A Heterogeneous Moran Process for the Analysis of Public Goods Games

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1 Introduction

Reducing global carbon emissions is one of the defining collective action problems of our time. Countries differ substantially in their economic capacity, historical emissions, and the costs they face when decarbonising, yet they share a mutual interest in the global public good of a stable climate. This structure maps onto a *public goods game* [22, 8]: each actor decides whether to incur a private cost to contribute to a shared resource whose total value is multiplied by some factor r and distributed equally. The temptation to free-ride means that purely selfish reasoning leads to a tragedy of the commons, even when collective cooperation would benefit everyone. Both theory [20] and controlled experiments [13] show that framing cooperation as a collective-risk problem, where failure to meet a cooperation threshold incurs a shared loss, can substantially raise cooperation rates. The challenge of overcoming free-riding is not merely theoretical. Nordhaus [15] proposed voluntary *climate clubs* in which member nations adopt a common carbon price and levy tariffs on non-members, turning the international climate problem into a coordination game with enforcement. The European Union's Carbon Border Adjustment Mechanism [4], which requires importers to purchase certificates reflecting the carbon content of goods produced under weaker environmental regulation, represents a real-world instance of this idea.

Evolutionary game theory offers a framework for studying how cooperation can emerge among self-interested actors without assuming rational foresight [8, 16]. In this approach, strategies that perform well spread through a population via one of several update rules, while poorly performing strategies decline. The conditions under which cooperative strategies can invade and stabilise have been extensively studied in two-player settings such as the iterated prisoner's dilemma [9, 1] and the snowdrift game [26]. The public goods game generalises these pairwise dilemmas to N players [6], making it well suited to studying multi-lateral cooperation such as climate agreements. In finite populations, evolutionary outcomes are stochastic: a rare mutant strategy spreads or vanishes with a probability that depends on the population size, payoff structure, and update rule [17, 25]. Computing these fixation and absorption probabilities exactly, rather than via simulation, is the approach taken here.

A feature of real-world collective action that standard models typically set aside is that actors are not identical. Players differ in wealth, productive capacity, and the costs they face, and this inequality can reshape the dynamics of cooperation [7]. In spatial and graphical settings, where players sit on a network and participate simultaneously in multiple games with their neighbours [18], heterogeneity generally promotes cooperation. Santos et al. [21] showed that social diversity in endowments promotes the emergence of cooperation in the public goods game directly: heterogeneous populations outperform their homogeneous counterparts over a wide range of parameters. This echoes earlier work showing that scale-free network topology, a form of structural heterogeneity, provides a unifying framework for the emergence of cooperation [19]. When contributions depend on dynamically accumulated reputation [12], high-reputation defectors receive reduced transfers, weakening defection's advantage. When heterogeneity arises from differences in network degree so that players allocate contributions according to the strength of their social ties [11], the payoff structure penalises high-degree defectors and protects cooperating clusters from invasion. The characteristic spatial mechanism is the formation of cooperating clusters that are more profitable than their defecting neighbours; provided r is sufficiently large, these clusters resist invasion and eventually spread. Heterogeneity does not, however, unconditionally favour cooperation. Flores et al. [5] show that when the contribution level is itself a strategic variable rather than a fixed attribute, low-value contributions act as a form of defection against the most collectively beneficial strategies, so that the emergence of cooperation does not guarantee the emergence of high-value cooperation.

The studies above are almost exclusively concerned with spatially structured populations, where network topology is central. Much less is known about fully heterogeneous, *well-mixed* populations, in which each individual is distinguished by a fixed attribute (such as their contribution capacity) independently of any graph structure. Furthermore, most existing analyses rely on simulation or mean-field approximations; exact treatments of heterogeneous evolutionary dynamics remain rare. How the choice of update rule interacts with population heterogeneity is also poorly understood. The Moran process [14], Fermi imitation dynamics [24], aspiration dynamics [**?**], and introspection dynamics [2, 3] all model strategy revision in distinct ways, and each may respond very differently to player-specific payoffs. In particular, *intrinsic* dynamics, in which a player evaluates a prospective strategy change based on their own counterfactual payoff, are better suited to heterogeneous populations than *extrinsic* dynamics, in which a player may copy a high-fitness neighbour without considering whether the same strategy would serve them as well. The introspection dynamics framework developed in [2] and extended to multiplayer asymmetric games in [3] formalises this distinction for homogeneous populations; extending it to the fully heterogeneous setting motivates the present work.

We make the following contributions. First, we establish a general framework for evolutionary dynamics on fully heterogeneous, well-mixed populations of N ordered individuals, in which each player may have a distinct payoff function and contribution level. Second, we derive exact transition matrices and, where applicable, exact absorption and steady-state distributions for four established dynamics (the Moran process, Fermi imitation dynamics, aspiration dynamics, and introspection dynamics) together with their mutation-augmented variants, applied to a heterogeneous public goods game. Third, we introduce a novel update rule, *Introspective Imitation Dynamics*, in which players select a role model proportionally to fitness (as in the Moran process) but evaluate whether to adopt that strategy by comparing their own current and counterfactual payoffs (as in introspection dynamics). This rule is well suited to heterogeneous populations, where copying a high-fitness individual is irrational if the same strategy yields a worse payoff for the copying player. Fourth, we show that purely extrinsic dynamics are subject to a ceiling on the probability of cooperation that intrinsic dynamics can exceed, and we characterise the conditions under which this advantage arises, with the ratio of the public goods multiplier r to population size N as the central governing parameter. We study two heterogeneous population structures: a *linear* population, in which players' capacities are graded uniformly across the group, and a *binomial* population, consisting of a low-capacity majority and a high-capacity minority.

The remainder of the paper is organised as follows. Section 2 introduces the population model, defines all five dynamics and their mutation variants, and specifies the heterogeneous public goods game. Section 3 presents our results, and supplementary material containing extended parameter sweeps is provided in Section 4.

2 The Model

In this section, similarly to [2, 3]we examine the underlying framework modelling dynamic processes on a fully heterogeneous population. We begin by defining a population of individuals and the actions which they may take, and then we see different methods by which we can model the evolution of a population over time.

2.1 An Initial System

- N **ordered** individuals who play actions from action sets $A_i = (a_{i,1}, a_{i,2}, ..., a_{i,k})$. In this paper, we shall assume that all action sets contain the same possible actions for each player.
- A state space S given by the set of ordered N-tuples with entries $\mathbf{a} = (a_1, a_2, a_3, \dots) \in A_1 \times A_2 \times A_3 \times ....$
- A fitness function $S \to \mathbb{R}^N$. We denote by π the payoff function for a given game. We denote the i^{th} entry of $\pi(\mathbf{a})$ by $\pi_i(\mathbf{a})$.

Now let $h(\mathbf{a},\ \mathbf{b})$ denote the Hamming distance between two states $\mathbf{a}, \mathbf{b} \in S$ [10]. The set of states $\mathbf{b}$ such that $h(\mathbf{a},\ \mathbf{b}) = 1$ is defined as $\mathrm{Neb}(\mathbf{a})$, the "Neighbourhood set of a state" [3]. Our system will be in the form of a Markov chain [23] which moves between states with a probability > 0 if and only if $h(\mathbf{a},\ \mathbf{b}) \leq 1$. We call the index at which $\mathbf{a}$ and $\mathbf{b}$ differ the *index of difference*, denoted $I(\mathbf{ab})$ [3]. We shall look at different population dynamics by which we may model such a Markov chain.

2.2 Population Dynamics

In this section, we give an overview of various population dynamics. This includes the following well-known dynamics, adapted to a fully heterogeneous population:

- The Moran process, first described in [14]
- Fermi Imitation Dynamics, a common choice of dynamic often adapted to the study of cooperation on a graph [12, 11, 24]
- Aspiration dynamics, a model first defined in [**?**]
- Introspection Dynamics, a more modern dynamic defined in [2]

We also introduce a new dynamic called "Introspective Imitation Dynamics". This is a novel dynamic particularly suited to the process of learning from others in a heterogeneous population.

2.2.1 Choice and Selection intensity

In Moran-process-type dynamics, which model biological evolution, we introduce a parameter ϵ referred to as *selection intensity* via the mapping

$$f_i(\mathbf{a}) = 1 + \epsilon\pi_i(\mathbf{a}).$$

Appropriate bounds on ϵ ensure that the fitness values f_i remain non-negative, which is necessary for the reproduction probabilities in the Moran process to be well-defined. Smaller values of ϵ correspond to weaker selection, placing less emphasis on payoff differences in the evolutionary update. In the limiting case $\epsilon = 0$, the process reduces to neutral drift, in which reproduction is independent of payoff. In contrast, population dynamics based on the Fermi update rule [24] employ a parameter β, which we refer to as the *choice intensity.* This parameter determines how sensitively individuals respond to payoff differences when making decisions. Depending on the specific dynamic under consideration, these payoff differences may arise from comparing an individual's current payoff with that of another individual in the population, or from comparing their current payoff with the payoff they would obtain by switching strategy.

2.2.2 The Moran Process

The Moran process [16] is a dynamic corresponding to the following algorithm:

1. A player is chosen to be duplicated with a probability proportional to their fitness in the population
2. A player is chosen with probability $\frac{1}{N}$ to be removed from the population

Let (a_{-j}) denote the state from the perspective of a_j, so that $\mathbf{a} = (a_j, a_{-j})$. Then, the entry $T_{\mathbf{ab}}$ of a transition matrix T in which each row and column correspond to a state in S is defined as the probability of a transition $\mathbf{a} = (a_1, a_2, \ldots, a_N) \to \mathbf{b} = (b_1, b_2, \ldots, b_N)$. For the Moran process, this is as follows:

$$T_{\mathbf{ab}} = \begin{cases} \frac{\sum_{a_i = b_{I(\mathbf{ab})}} f(a_i)}{N\sum_{a_j} f(a_j)} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \text{ differing at position } I(\mathbf{ab}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{1}$$

An important case which proceeds from (1) is that of a transition $\mathbf{a} \to \mathbf{b}$ where $\mathbf{b}$ contains an individual of a *type* not found in $\mathbf{a}$. For example, the transition $(0, 1) \to (0, 2)$. This would be forbidden by the intuition of a standard Moran process, as the new individual is the duplication of another individual in $\mathbf{a}$. However, we can see that the standard formula yields $p_{v,u} = 0$ because there does not exist $a_i \in \mathbf{a} : a_i = b_{i^*}$.

We can now discuss one of the key ideas in the Moran process. As no new action types can be introduced to the population once they are fully removed, we have a set of absorbing states $\mathbf{a}$ such that $a_1 = a_2 = ...$, for which we have $p_{\mathbf{a}\to\mathbf{a}} = 1$. We denote the set of absorbing states S^γ. We therefore have that the Moran process forms an absorbing Markov chain [23], and we can investigate the probability of the population being absorbed into each absorbing state.

Consider a game with two players, each playing one of two actions. We have a state space $\mathbf{a} = (0, 0), \mathbf{b} = (0, 1), \mathbf{c} = (1, 0), \mathbf{d} = (1, 1)$, and our transition matrix T would take the form

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 1 - \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} - \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 0 & \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} \\ \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & 0 & 1 - \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} - \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

Our transition matrix represents the probability of transitioning from the state represented by the row to the state represented by the column. Here our columns represent **a**,**b**, **c** and **d** from left to right, and the rows represent these states in the same order from top to bottom.

Now, by taking the fundamental matrix of T and performing a right multiplication by R, the matrix with entries which represent transitions from transitive states to absorbing states[23], we can obtain the absorption matrix for this system.

For the above game, we find the following equation:

$$\left(I - \begin{bmatrix} 1 - \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} - \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 0 \\ 0 & 1 - \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} - \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \end{bmatrix}\right)^{-1} \begin{bmatrix} \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} \\ \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \end{bmatrix} \tag{3}$$

And solve to find our absorption matrix:

$$\begin{bmatrix} \frac{f_1(\mathbf{b})}{\left(\frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} + \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}\right)(2f_1(\mathbf{b})+2f_2(\mathbf{b}))} & \frac{f_2(\mathbf{b})}{\left(\frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} + \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}\right)(2f_1(\mathbf{b})+2f_2(\mathbf{b}))} \\ \frac{f_2(\mathbf{c})}{\left(\frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} + \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}\right)(2f_1(\mathbf{c})+2f_2(\mathbf{c}))} & \frac{f_1(\mathbf{c})}{\left(\frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} + \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}\right)(2f_1(\mathbf{c})+2f_2(\mathbf{c}))} \end{bmatrix} \tag{4}$$

In this matrix, the rows represent our initial transitive state (in our system, row 1 represents $\mathbf{b}$ and row 2 represents $\mathbf{c}$), and our columns represent which state the system is absorbed into (with $\mathbf{a}$ on the left, and $\mathbf{d}$ on the right). It is immediately clear from this result that the probability of the system ending up in a particular absorbing state does not rely on the fitness of individuals in the absorbing state, but rather their fitness in the transitive states. Thus, players can remain in a state which could be bettered by a single change, however they will not make such a change if no individual of another type exists in the population.

2.2.3 Fermi Imitation Dynamics

The Fermi Imitation Dynamic is another method of modelling evolution in a population. It models the following process:

- Two players, i and j are selected at random. Unlike [24], the player j does not have to be the neighbour of player i.
- Player i copies the action type of the player j with a probability according to the Fermi function $\phi(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})) = \frac{1}{1+e^{(\beta(\pi_i(\mathbf{a})-\pi_j(\mathbf{a})))}}$

Where β is the choice intensity of the system, representing how sensitive our players' choice is to the difference between their current payoff, and the payoff which they are considering. This method has been commonly used in studies of evolutionary games, for example [12, 11, 8, 24].

Unlike a Moran Process, we can see that the players are not selected with a probability proportional to their fitness. However, Fermi Imitation Dynamics compared the fitness after the selection of players. This means that in Fermi Imitation Dynamics, there is not necessarily a player changing action in each timestep, whereas in a Moran process, there is always a player changing action type (although it may be a change to the same action type that they are currently performing) We now define our transition matrix according to the Fermi function:

$$T_{\mathbf{ab}} = \begin{cases} \left(\frac{1}{N(N-1)} \sum_{a_j = b_{I(\mathbf{ab})}} \phi(\pi_{\mathrm{I}(\mathbf{ab})}(a) - \pi_j(\mathbf{a}))\right) & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{5}$$

Here we see how the Fermi function can be used to define the transition probability between states in a Markov chain. Importantly, we see that the first case of equation 5 is always in the interval $(0, 1)$, as the Fermi function always takes a value on $(0, 1)$, and the summation term is at most $N - 1$ values of said function, and thus takes a value $< N - 1$.

Now, let us look once more at our state space $\mathbf{a} = (0, 0), \mathbf{b} = (0, 1), \mathbf{c} = (1, 0), \mathbf{d} = (1, 1)$. We can define a transition matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2}\phi(\pi_2(\mathbf{b}) - \pi_1(\mathbf{b})) & \frac{1}{2} & 0 & \frac{1}{2}\phi(\pi_1(\mathbf{b}) - \pi_2(\mathbf{b})) \\ \frac{1}{2}\phi(\pi_1(\mathbf{c}) - \pi_2(\mathbf{c})) & 0 & \frac{1}{2} & \frac{1}{2}\phi(\pi_2(\mathbf{c}) - \pi_1(\mathbf{c})) \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}$$

and using our previous method, we can acquire our absorption matrix:

$$\begin{bmatrix} \phi(\pi_2(\mathbf{b}) - \pi_1(\mathbf{b})) & \phi(\pi_1(\mathbf{b}) - \pi_2(\mathbf{b})) \\ \phi(\pi_1(\mathbf{c}) - \pi_2(\mathbf{c})) & \phi(\pi_2(\mathbf{c}) - \pi_1(\mathbf{c})) \end{bmatrix} \tag{7}$$

This shows that, similarly to the Moran process, our absorption probabilities rely on the fitness of players in transitive states and not those of players in the absorbing states. Thus, this dynamic is still not fully suited for modelling a heterogeneous system as a player can commonly copy a high-fitness strategy without considering the impact on their own fitness, and in some cases may enter an absorbing state with a transition which worsens their position with a high probability.

2.2.4 Aspiration Dynamics

Instead of considering the difference between two players' fitness, aspiration dynamics models an update rule based on how far away a player's performance in a game is from the fitness which they desire. It follows the algorithm:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. This player changes strategy with a probability $\phi(\pi_i(\mathbf{a}) - A)$

Here, A is player i's *aspiration*, meaning the payoff which they desire to make from a given game. A player will accept a new strategy with a higher than random probability if their current payoff is lower than their aspired payoff. This dynamic is almost always only defined for two actions - traditionally the actions to either cooperate or defect. The related transition matrix can be seen below:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N}\,\phi(\pi_{I(\mathbf{ab})}(\mathbf{a}) - A_{I(\mathbf{ab})}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{ab}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a})\text{and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{8}$$

Now, consider the transition matrix for our general four-state system:

$$\begin{bmatrix} 1 - \frac{1}{2(1+e^{-\beta(A_2 - f_2(a))})} - \frac{1}{2(1+e^{-\beta(A_1 - f_1(a))})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta f_2(a)})} & \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta f_1(a)})} & 0 \\ \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta f_2(b)})} & 1 - \frac{1}{2(1+e^{-\beta(A_2 - f_2(b))})} - \frac{1}{2(1+e^{-\beta(A_1 - f_1(b))})} & 0 & \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta f_1(b)})} \\ \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta f_1(c)})} & 0 & 1 - \frac{1}{2(1+e^{-\beta(A_2 - f_2(c))})} - \frac{1}{2(1+e^{-\beta(A_1 - f_1(c))})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta f_2(c)})} \\ 0 & \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta f_1(d)})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta f_2(d)})} & 1 - \frac{1}{2(1+e^{-\beta(A_2 - f_2(d))})} - \frac{1}{2(1+e^{-\beta(A_1 - f_1(d))})} \end{bmatrix} \tag{9}$$

We can see that as our players no longer consider the actions of the rest of the population when considering their decision, we no longer have an absorbing Markov chain. Thus, we no longer consider an absorption matrix, but instead we consider the steady state of our Markov chain [23] This is the proportion of time spent in each state when the Markov chain reaches a stable distribution across the states. It is well proven that any irreducible, finite Markov chain eventually reaches it's unique stationary distribution. This is also independent of the starting state of the distribution. [**?**]. For our four state system, we acquire the following result for a stationary distribution:

2.2.5 Introspection Dynamics

In introspection dynamics, the following process take place [2]

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. Player i chooses another possible strategy, $a_{i,l}$, from the set of all possible strategies.
3. The chosen player replaces their strategy with probability $p_{a_{I(\mathbf{ab})} \to b_{I(\mathbf{ab})}}(a_{-I(\mathbf{ab})}) = \phi(\Delta\pi_{I(\mathbf{ab})})$, where $\Delta\pi_{I(\mathbf{ab})} = \pi_{I(a,b)}(\mathbf{a}) - \pi_{I(a,b)}(\mathbf{b})$ is the difference between the possible fitness of the new strategy and the fitness of the player's current strategy.

[2] considers this process for the two player case, and it is expanded to multiple players in [3].

This process is well suited to a heterogeneous population. This is because in some cases using imitation dynamics, a player may copy a strategy due it's fitness when played by another individual in the population, despite the fact that such a strategy actually has a lower payoff for the focal player than their current strategy. In introspection dynamics, players do not consider the fitness of other individuals, and so cannot be fooled by high fitness strategies with attributes different to that of our focal player.

The transition matrix for the introspection dynamics process [3] is given by:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N(m_j - 1)}\, \phi(\Delta\pi_{I(\mathbf{ab})}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{ab}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{10}$$

where m_j is the number of actions available to individual j. Since we assume that all individuals have the same action set with size k, we can replace $m_j - 1$ with $k - 1$.

Now, Couto also calculated in [2] the explicit strategy distribution for an asymmetric interaction between two players. We can adapt these results to define our transition matrix for our example state space **a,b,c,d**:

$$\begin{bmatrix} 1 - \frac{1}{2\left(e^{\beta(\pi_2(a)-\pi_2(b))}+1\right)} - \frac{1}{2\left(e^{\beta(\pi_1(a)-\pi_1(c))}+1\right)} & \frac{1}{2\left(e^{\beta(\pi_2(a)-\pi_2(b))}+1\right)} & \frac{1}{2\left(e^{\beta(\pi_1(a)-\pi_1(c))}+1\right)} & 0 \\ \frac{1}{2\left(e^{\beta(-\pi_2(a)+\pi_2(b))}+1\right)} & 1 - \frac{1}{2\left(e^{\beta(\pi_1(b)-\pi_1(d))}+1\right)} - \frac{1}{2\left(1+e^{-\beta(\pi_2(a)-\pi_2(b))}\right)} & 0 & \frac{1}{2\left(e^{\beta(\pi_1(b)-\pi_1(d))}+1\right)} \\ \frac{1}{2\left(e^{\beta(-\pi_1(a)+\pi_1(c))}+1\right)} & 0 & 1 - \frac{1}{2\left(e^{\beta(\pi_2(c)-\pi_2(d))}+1\right)} - \frac{1}{2\left(1+e^{-\beta(\pi_1(a)-\pi_1(c))}\right)} & \frac{1}{2\left(e^{\beta(\pi_2(c)-\pi_2(d))}+1\right)} \\ 0 & \frac{1}{2\left(e^{\beta(-\pi_1(b)+\pi_1(d))}+1\right)} & \frac{1}{2\left(e^{\beta(-\pi_2(c)+\pi_2(d))}+1\right)} & 1 - \frac{1}{2\left(1+e^{-\beta(\pi_2(c)-\pi_2(d))}\right)} - \frac{1}{2\left(1+e^{-\beta(\pi_1(b)-\pi_1(d))}\right)} \end{bmatrix} \tag{11}$$

We then consider the steady state vectors for this system. For this, we take the left eigenvector of the transition matrix, which is given by:

Unlike in the case of imitation dynamics, there are no absorbing states for a Markov chain transitioning according to imitation dynamics. Therefore, we define a steady state vector $\mathbf{u}$ according to the method Couto uses in [2].

2.2.6 Introspective Imitation Dynamics

Introspection dynamics allows players to learn based on their own fitness, resulting in a reduced probability of state changes that worsen the focal individual's fitness in a heterogeneous population. Now we consider a population dynamic that utilises this property of introspection dynamics but within the context of an imitative process. Consider a population of N players with the same action set. We define the introspective imitation process as follows:

1. A focal player is chosen at random to reconsider their strategy
2. Another player a_j in the population is chosen with a probability proportional to their fitness within the population to have their strategy considered
3. The focal player changes their strategy with probability $\phi(\Delta(f_{I(\mathbf{ab})}))$

We can define a transition matrix T of an introspective imitation process as follows:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N} \frac{\sum_{a_j = b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{\sum_k f_k(\mathbf{a})} \phi(\Delta(f_{I(\mathbf{ab})})) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{12}$$

We see that players can only copy the actions present in the state, however they determine whether or not to copy a player based on their own fitness rather than by considering the fitness of another player. This models a behaviour where individuals learn from each other, but do not blindly trust that the strategy will work for themself just because it worked for another player. Instead, players look to others to see what action they *could* play, but judge based on their own fitness to decide what action they *should* play.

Note that Introspective Imitation Dynamics takes into account both selection intensity and choice intensity. This means that both the level to which a player's fitness is considered when their strategy is being chosen for consideration, and also how much it is considered when choosing whether to accept the new strategy, is present in the dynamic.

Let us now see this model applied to our ongoing example in the state space $S = \{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}\}$. Let $\eta_{i,j}(\mathbf{x,y}) = 1 + e^{\beta((f_i(\mathbf{x}) - f_j(\mathbf{y})))}$ be the denominator of the Fermi function. We first see our transition matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{f_1(b)}{2\eta_{2,2}(\mathbf{b,a})(f_1(b)+f_2(b))} & 1 - \frac{f_2(b)}{\eta_{1,1}(\mathbf{b,d})(2f_1(b)+2f_2(b))} - \frac{f_1(b)}{\eta_{2,2}(\mathbf{b,a})(2f_1(b)+2f_2(b))} & 0 & \frac{f_2(b)}{2\eta_{1,1}(\mathbf{b,d})(f_1(b)+f_2(b))} \\ \frac{f_2(c)}{2\eta_{1,1}(\mathbf{c,a})(f_1(c)+f_2(c))} & 0 & 1 - \frac{f_1(c)}{\eta_{2,2}(\mathbf{c,d})(2f_1(c)+2f_2(c))} - \frac{f_2(c)}{\eta_{1,1}(\mathbf{c,a})(2f_1(c)+2f_2(c))} & \frac{f_1(c)}{2\eta_{2,2}(\mathbf{c,d})(f_1(c)+f_2(c))} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{13}$$

And, using the same method as in the other methods of imitation dynamics, we obtain the absorption matrix:

$$\begin{bmatrix} \frac{\eta_{1,1}(\mathbf{b,d})f_1(b)}{\eta_{1,1}(\mathbf{b,d})f_1(b)+\eta_{2,2}(\mathbf{b,a})f_2(b)} & \frac{\eta_{2,2}(\mathbf{b,a})f_2(b)}{\eta_{1,1}(\mathbf{b,d})f_1(b)+\eta_{2,2}(\mathbf{b,a})f_2(b)} \\ \frac{\eta_{2,2}(\mathbf{c,d})f_2(c)}{\eta_{1,1}(\mathbf{c,a})f_1(c)+\eta_{2,2}(\mathbf{c,d})f_2(c)} & \frac{\eta_{1,1}(\mathbf{c,a})f_1(c)}{\eta_{1,1}(\mathbf{c,a})f_1(c)+\eta_{2,2}(\mathbf{c,d})f_2(c)} \end{bmatrix} \tag{14}$$

Unlike other imitation dynamics, we can see that the absorption probability in an introspective imitation process relies on the fitness that at least one individual will possess in the absorbing state. This means that players will enter absorbing states with a higher probability if it improves their own fitness, despite any heterogeneity between payoffs for the same action played by different players. This solves the problem of players making worsening moves based on the fitness of another individual in the population, while still retaining the property of imitation dynamics that new strategies cannot be introduced to the system.

2.3 Mutation

We can also model how individuals may mutate within the system. This is the process by which after a state transition, the chosen player will instead mutate to a different action type with some small probability. We then have two possible feasible transitions:

- If $h(\mathbf{ab}) = 0$, then the only way in which the transition can be achieved is through the transition of a player $i \to i$ and then no mutation occuring.
- If $h(\mathbf{ab}) = 1$, then there are two ways in which the transition can occur. Either, the player i which is different between $\mathbf{a}$ and $\mathbf{b}$ transitions by the standard process, with no mutation occuring, or no transition occurs, but player i mutates into the type which they possess in $\mathbf{b}$

We let $\mu_{i,j}$ be the probability that individual i mutates to type j. Thus, given a transition matrix T, we can produce the related transition matrix with mutation as follows:

$$T^{(m)}_{\mathbf{ab}} = \begin{cases} T_{\mathbf{ab}}(1 - \sum_k \mu_{I(\mathbf{ab}),k}) + (\frac{\mu_{i,\mathbf{b}_{I(\mathbf{ab})}}}{N}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{15}$$

In all cases, this forms an ergodic Markov chain rather than an absorbing chain. Thus, we do not obtain absorption matrices for models which use mutation, but instead we obtain a steady state, which no longer depends on our starting state.

We will now see each of our processes for the case where mutation is considered.

2.3.1 The Moran Process

In the case of the Moran process, our transition matrix becomes:

$$T^{(m)}_{\mathbf{ab}} = \begin{cases} (\frac{\sum_{a_i = b_{I(\mathbf{ab})}} f(a_i)}{N \sum_{a_j} f(a_j)})(1 - \sum_k \mu_{I(\mathbf{ab}),k}) + (\frac{\mu_{i,\mathbf{b}_{I(\mathbf{ab})}}}{N}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \text{ differing at position } I(\mathbf{ab}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{16}$$

For the case of the two player game, we now have the transition matrix:

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & \frac{\mu_{12}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} - \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} + 1 & 0 & \frac{\mu_{12}}{2} + \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} \\ \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} - \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} + 1 & \frac{\mu_{22}}{2} + \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} \\ 0 & \frac{\mu_{11}}{2} & \frac{\mu_{21}}{2} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix}$$

and the steady state:

2.3.2 Fermi Imitation Dynamics

When we introduce mutation to Fermi Imitation Dynamics, we obtain the following transition matrix:

$$T^{(m)}_{\mathbf{ab}} = \begin{cases} \left(\frac{1}{N(N-1)} \sum_{a_j = b_{I(\mathbf{ab})}} \phi(\pi_{\mathrm{I}(\mathbf{ab})}(a) - \pi_j(\mathbf{a}))\right)(1 - \sum_k \mu_{I(\mathbf{ab}),k}) + \left(\frac{\mu_{i,\mathbf{b}_I(\mathbf{ab})}}{N}\right) & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{17}$$

In the case of the two player game, we obtain the transition matrix:

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & \frac{\mu_{12}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(-f_1(b)+f_2(b))}+1\right)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(f_1(b)-f_2(b))}+1\right)} - \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(-f_1(b)+f_2(b))}+1\right)} & 0 & \frac{\mu_{12}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(f_1(b)-f_2(b))}+1\right)} \\ \frac{\mu_{11}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(f_1(c)-f_2(c))}+1\right)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(f_1(c)-f_2(c))}+1\right)} - \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(-f_1(c)+f_2(c))}+1\right)} & \frac{\mu_{22}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(-f_1(c)+f_2(c))}+1\right)} \\ 0 & \frac{\mu_{11}}{2} & \frac{\mu_{21}}{2} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix}$$

And the steady state:

2.3.3 Aspiration Dynamics

As aspiration and introspection dynamics do not form absorbing Markov chains, we do not need mutation to be applied in order to form an ergodic chain. However, we will consider these cases for completeness, and thus we obtain the following transition matrix:

$$T^{(m)}_{\mathbf{ab}} = \begin{cases} \left(\frac{1}{N} \phi(\pi_{I(\mathbf{ab})} - A_i)\right)(1 - \sum_k \mu_{I(\mathbf{ab}),k}) + \left(\frac{\mu_{i,\mathbf{b}_I(\mathbf{ab})}}{N}\right) & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{ab}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases}$$

In the case of the general 2-player game, we get the following:

$$\begin{bmatrix} 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(a))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(a))}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(a)}} & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(a)}} & 0 \\ \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(b)}} & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(b))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(b))}} & 0 & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(b)}} \\ \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(c)}} & 0 & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(c))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(c))}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(c)}} \\ 0 & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(d)}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(d)}} & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(d))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(d))}} \end{bmatrix}$$

2.3.4 Introspection Dynamics

As introspection dynamics does not form an absorbing Markov chain, we do not need mutation to form an ergodic chain. However, we will consider the following transition matrix for the case with mutation:

$$T^{(m)}_{\mathbf{ab}} = \begin{cases} \left(\frac{1}{N(m_j - 1)} \phi(\Delta\pi_{I(\mathbf{ab})})\right)(1 - \sum_k \mu_{I(\mathbf{ab}),k}) + \left(\frac{\mu_{i,\mathbf{b}_I(\mathbf{ab})}}{N}\right) & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{ab}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a}) \text{and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases}$$

which, in the case of the two player game, gives us the following:

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(f_2(a)-f_2(b))}+1\right)} - \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(f_1(a)-f_1(c))}+1\right)} & \frac{\mu_{22}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(f_2(a)-f_2(b))}+1\right)} & \frac{\mu_{12}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(f_1(a)-f_1(c))}+1\right)} & 0 \\ \frac{\mu_{21}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(-f_2(a)+f_2(b))}+1\right)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(-f_2(a)+f_2(b))}+1\right)} - \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(f_1(b)-f_1(d))}+1\right)} & 0 & \frac{\mu_{12}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(f_1(b)-f_1(d))}+1\right)} \\ \frac{\mu_{11}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(-f_1(a)+f_1(c))}+1\right)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(f_2(c)-f_2(d))}+1\right)} - \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(-f_1(a)+f_1(c))}+1\right)} & \frac{\mu_{22}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(f_2(c)-f_2(d))}+1\right)} \\ 0 & \frac{\mu_{11}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(-f_1(b)+f_1(d))}+1\right)} & \frac{\mu_{21}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(-f_2(c)+f_2(d))}+1\right)} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2\left(e^{\beta(-f_2(c)+f_2(d))}+1\right)} - \frac{-\mu_{11}-\mu_{12}+1}{2\left(e^{\beta(-f_1(b)+f_1(d))}+1\right)} \end{bmatrix}$$

2.3.5 Introspective Imitation Dynamics

Finally, for the case of introspective imitation dynamics, we get the following transition matrix with mutation:

$$T^{(m)}_{\mathbf{ab}} = \begin{cases} \left(\frac{1}{N}\frac{\sum_{a_j=b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{\sum_k f_k(\mathbf{a})}\phi(\Delta(f_{I(\mathbf{ab})}))\right)(1-\sum_k \mu_{I(\mathbf{ab}),k}) + \left(\frac{\mu_{i,\mathbf{b}_{I(\mathbf{ab})}}}{N}\right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{18}$$

For the two player game, we obtain the transition matrix:

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & \frac{\mu_{12}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2\left(e^{\beta(-f_2(a)+f_2(b))}+1\right)(\epsilon f_1(b)+\epsilon f_2(b)+2)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2\left(e^{\beta(-f_2(a)+f_2(b))}+1\right)(\epsilon f_1(b)+\epsilon f_2(b)+2)} - \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2\left(e^{\beta(f_1(b)-f_1(d))}+1\right)(\epsilon f_1(b)+\epsilon f_2(b)+2)} + 1 & 0 & \frac{\mu_{12}}{2} + \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2\left(e^{\beta(f_1(b)-f_1(d))}+1\right)(\epsilon f_1(b)+\epsilon f_2(b)+2)} \\ \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2\left(e^{\beta(-f_1(a)+f_1(c))}+1\right)(\epsilon f_1(c)+\epsilon f_2(c)+2)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2\left(e^{\beta(f_2(c)-f_2(d))}+1\right)(\epsilon f_1(c)+\epsilon f_2(c)+2)} - \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2\left(e^{\beta(-f_1(a)+f_1(c))}+1\right)(\epsilon f_1(c)+\epsilon f_2(c)+2)} + 1 & \frac{\mu_{22}}{2} + \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2\left(e^{\beta(f_2(c)-f_2(d))}+1\right)(\epsilon f_1(c)+\epsilon f_2(c)+2)} \\ 0 & \frac{\mu_{11}}{2} & \frac{\mu_{21}}{2} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix}$$

which gives us a steady state:

2.4 Classification of Population Dynamics

We will now classify our population dynamics into one of two categories. The first are population dynamics which accept strategies from the population based on the fitness of said strategies when played by other individuals. These dynamics are poorly suited to a heterogeneous population, as individuals do not consider how the change will affect them. However, these do offer a good model for biological processes such as the evolution of traits in nature, as in these cases the fitness of an individual causes them to pass on their traits. The second are population dynamics which consider their own change in payoff in the case that they accept a new strategy. These dynamics are more well suited to a heterogeneous population, modelling state changes where individuals make an active decision to change state in order to improve their own fitness.

Below, we see a diagram showing the classification of the population dynamics shown in this section:

2.5 The Public Goods Game

In this section, we shall see how different population dynamics can be applied to the public goods game. [22, 8] In this game, individuals choose whether or not to contribute to a public resource pool. In the end, the total resource is multiplied by some factor $r > 1$ and distributed equally between each player. This model often encourages a behaviour known as "free-riding" , where players refuse to contribute to the pool and simply take the benefit provided by other individuals' contribution. The classic problem of a public goods game is to provide an environment where such free-riding is less profitable than contributing to the public good.

We can model a public goods game by providing the following payoff function:

$$\pi_i(\mathbf{a}) = \frac{r\sum_{j=1}^{N}\alpha_j}{N} - \alpha_i \tag{19}$$

(where α_j is the contribution by individual j.) In the homogeneous public goods game, we have that α_j can only take one of two values: some constant α if individual j cooperates, and 0 if not. However, the heterogeneous game will require a more tailored α_j. Thus, we consider two cases of contribution distributions. Let M be the total possible contribution when all players cooperate. The first case, we consider a linear wealth distribution, under which a player i contributes $\frac{2Mi}{N(N+1)}$. In this population, each player contributes more than the last. The second population we call the binomial population. In this population, the first $n < N$ players contribute some low amount α_L, and the final $N - n$ players contribute some greater amount α_H.

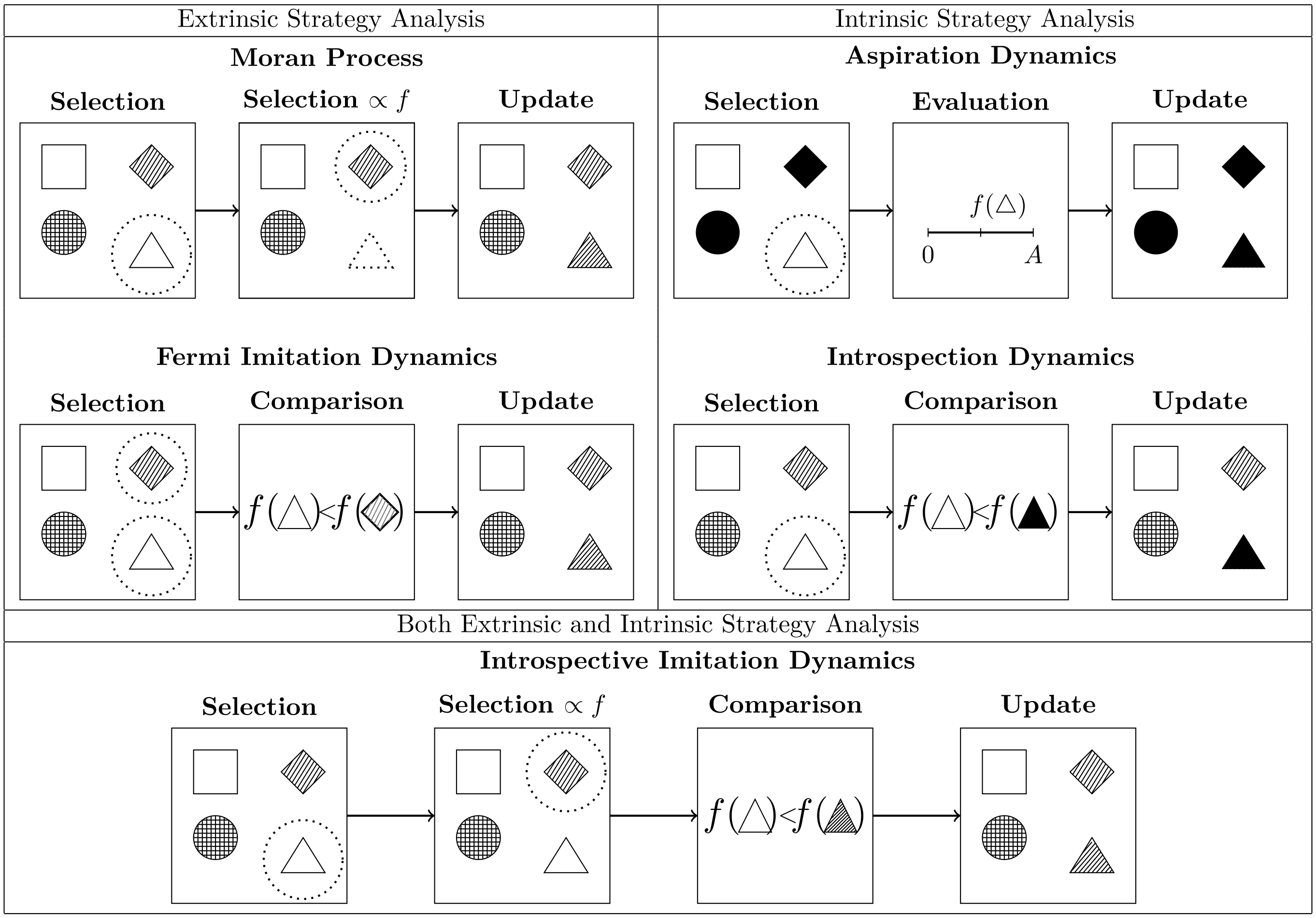
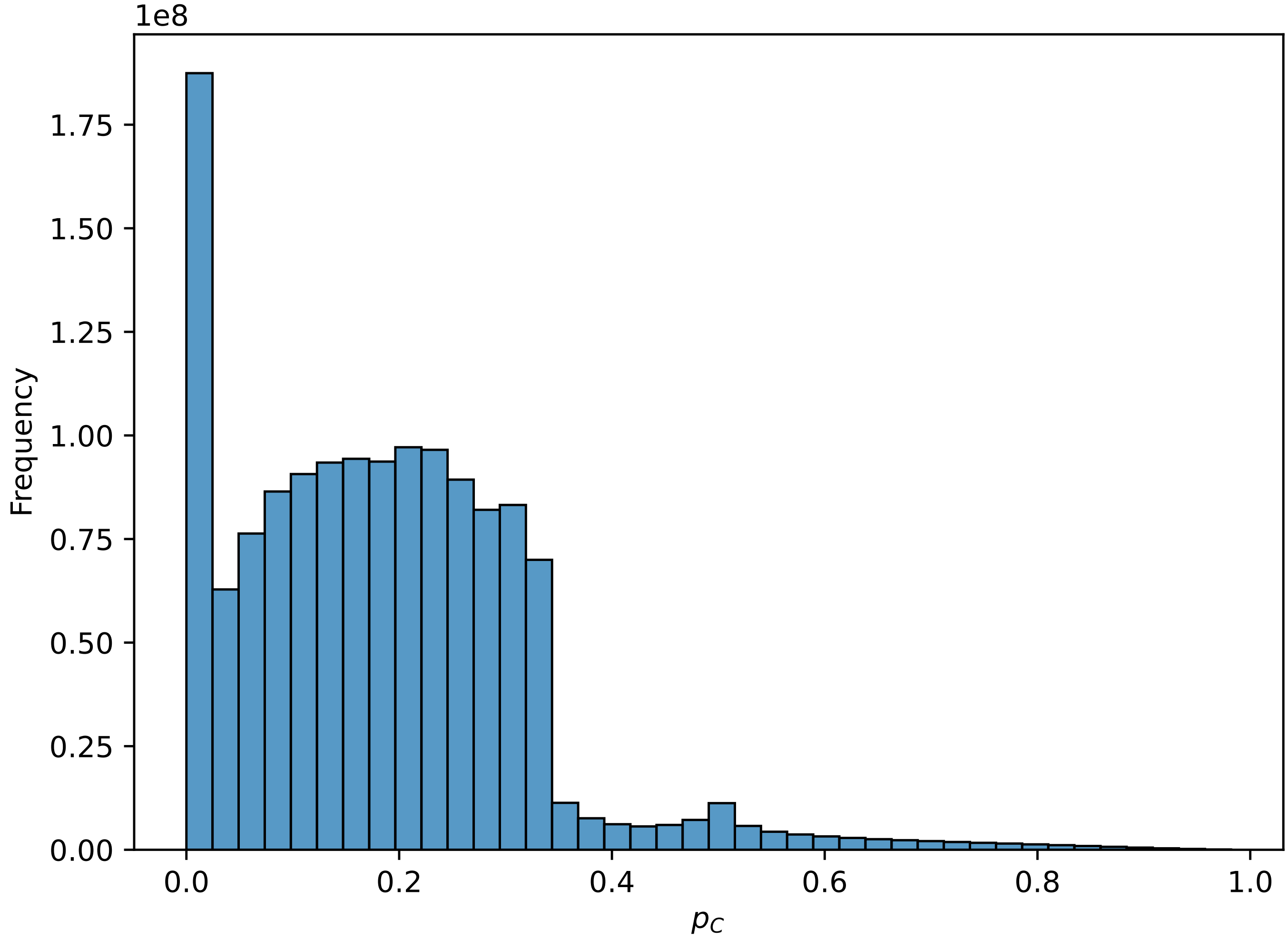


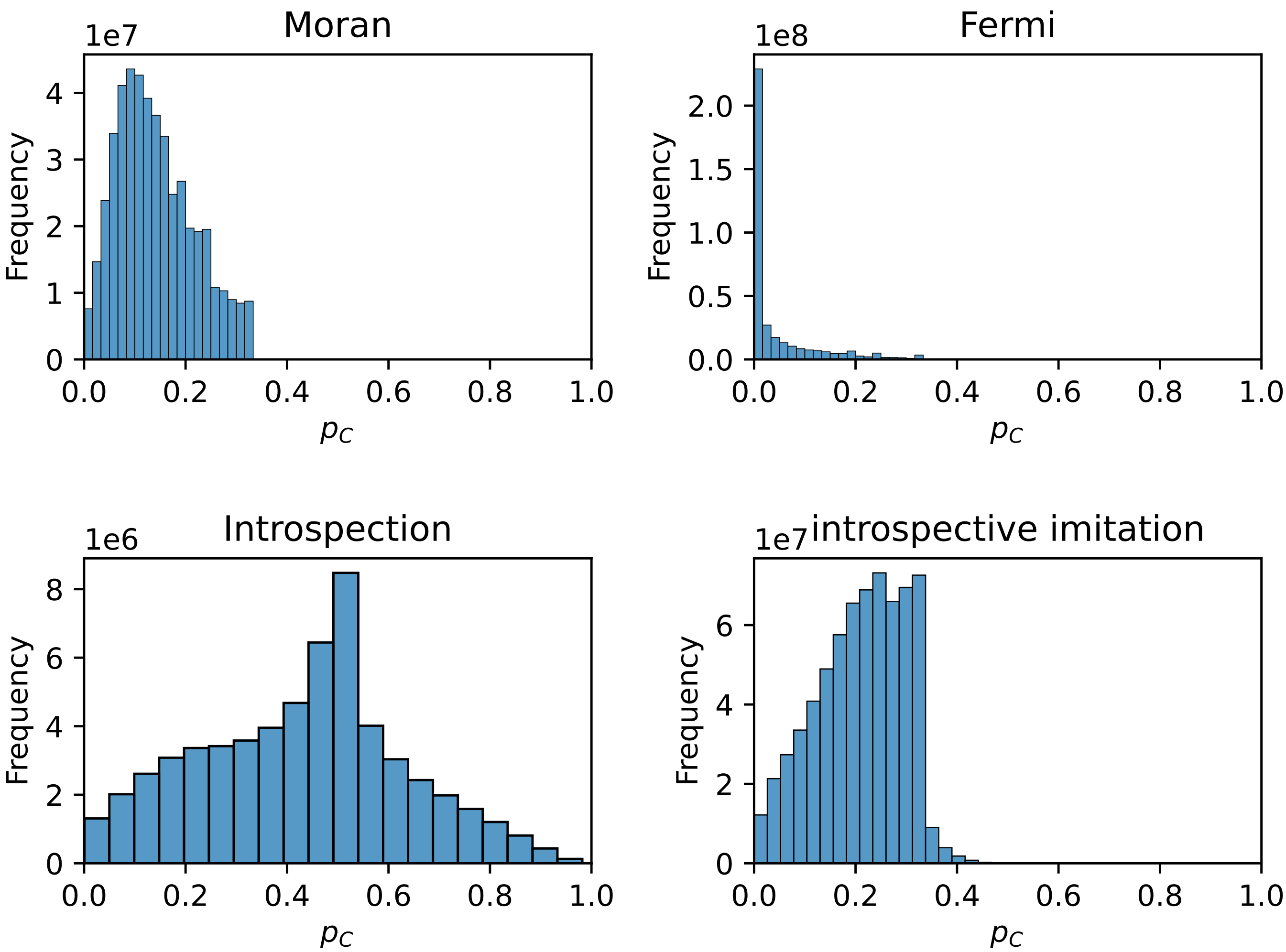
Table 1: A comparison of different population dynamics. We assume in all cases that the triangle is chosen for replacement with a probability $\frac{1}{N}$ and the diamond is chosen with a varying probability depending on the process. In the case of Introspection dynamics, we assume that the strategy represented by the shape being filled in is chosen with probability $\frac{1}{k-1}$, and for aspiration dynamics there is only one other strategy to accept.

3 Results

First, let us see the overall distribution of p_C across all models.

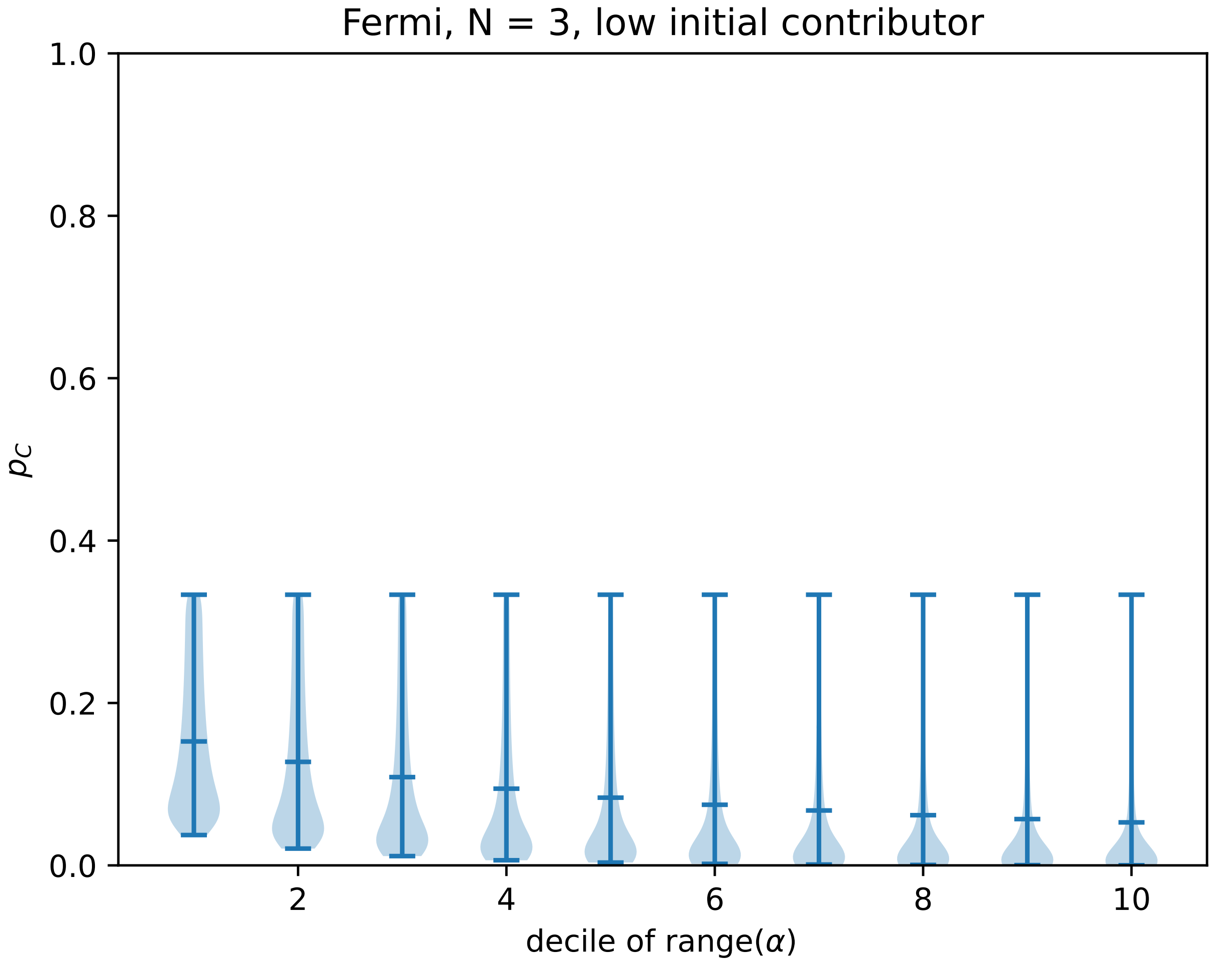


We see that there most systems do not produce a high value of p_C, however we can encourage cooperation under some conditions. Let us begin to investigate these conditions by seeing the distribution of p_C across our population dynamics.

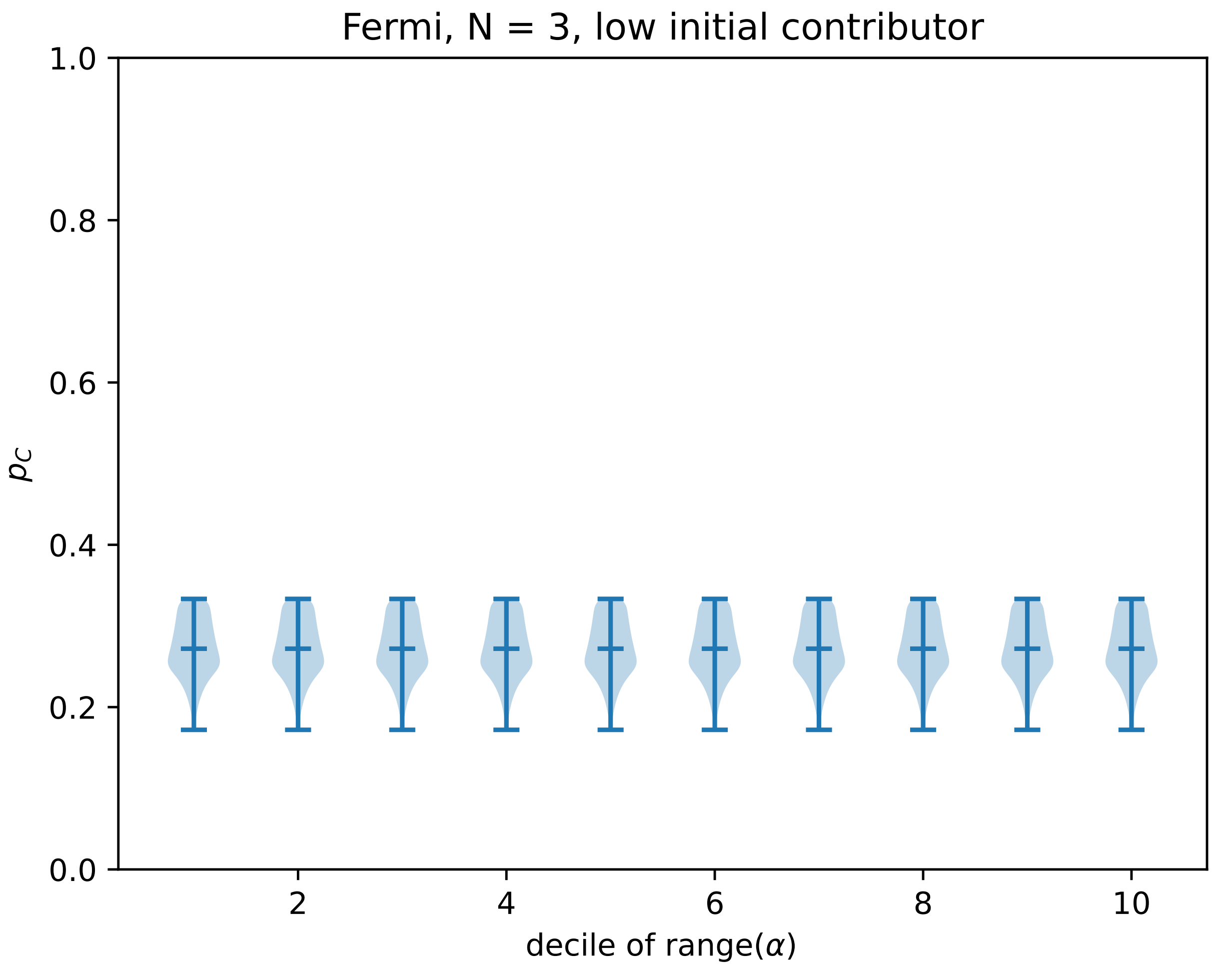


There is clearly a large difference in p_C values in different population dynamics, with Fermi imitation dynamics producing a very low value of p_C, and introspective imitation dynamics giving the highest values of any absorbing Markov chain. We now see how different parameters affect p_C.

First, we shall see how the heterogeneity of α effects the probability of cooperation with a low-contributing mutant in a linear population, and we shall ask the question "Is there a process which maximises p_C?"

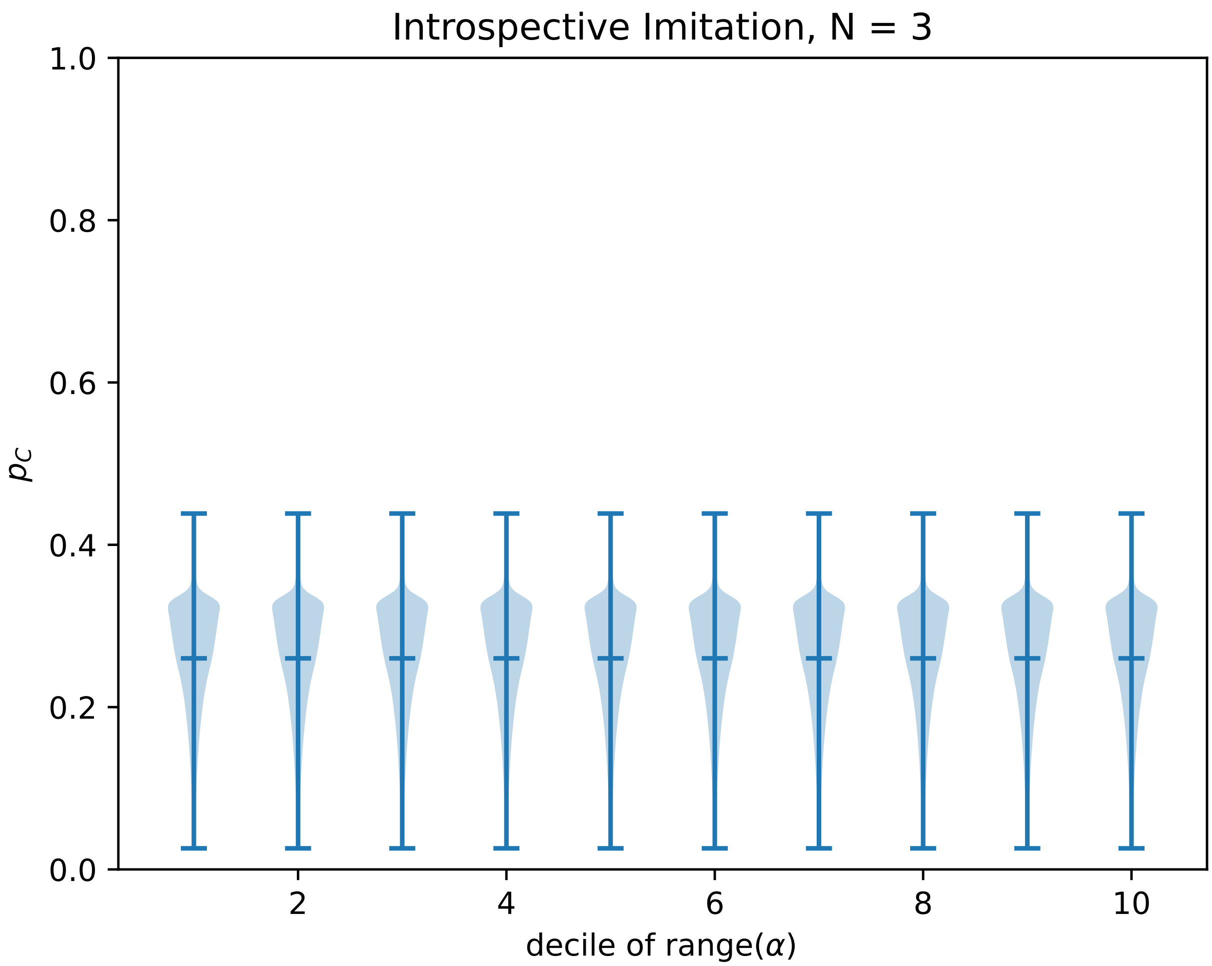


In Fermi imitation dynamics, we see that an increase in heterogeneity actually decreases the probability of cooperation. This is because with a higher heterogeneity, high contributing individuals are much harder to maintain, since the difference between their payoff and the payoff for a defector in the same state will be greater. This is because the high values of α_i are greater, and thus the probability of a defector copying the high contributors is lower, and the probability of a high contributor copying a defector is greater.

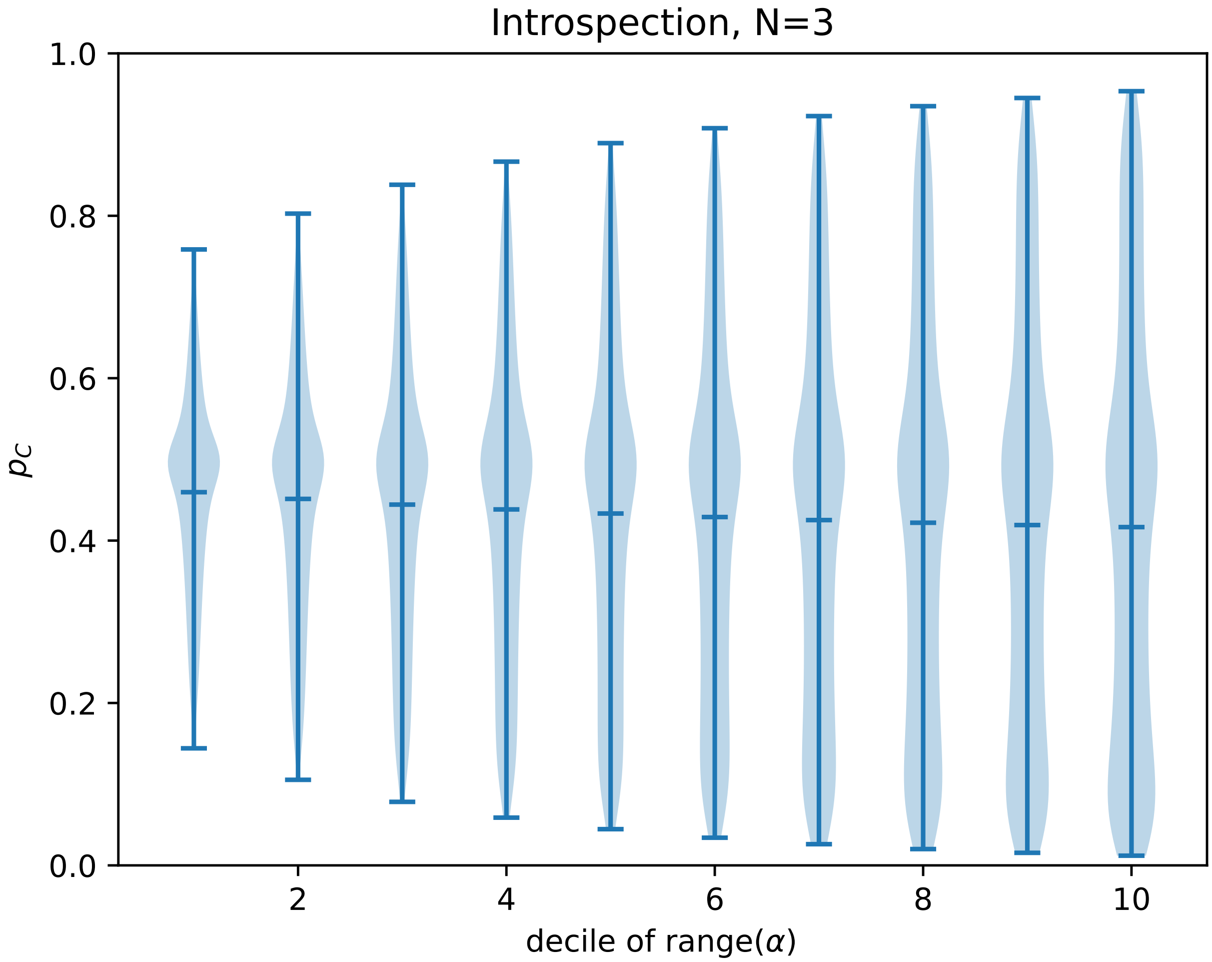


In the Moran process, we see no effect from the heterogeneity of α_i.

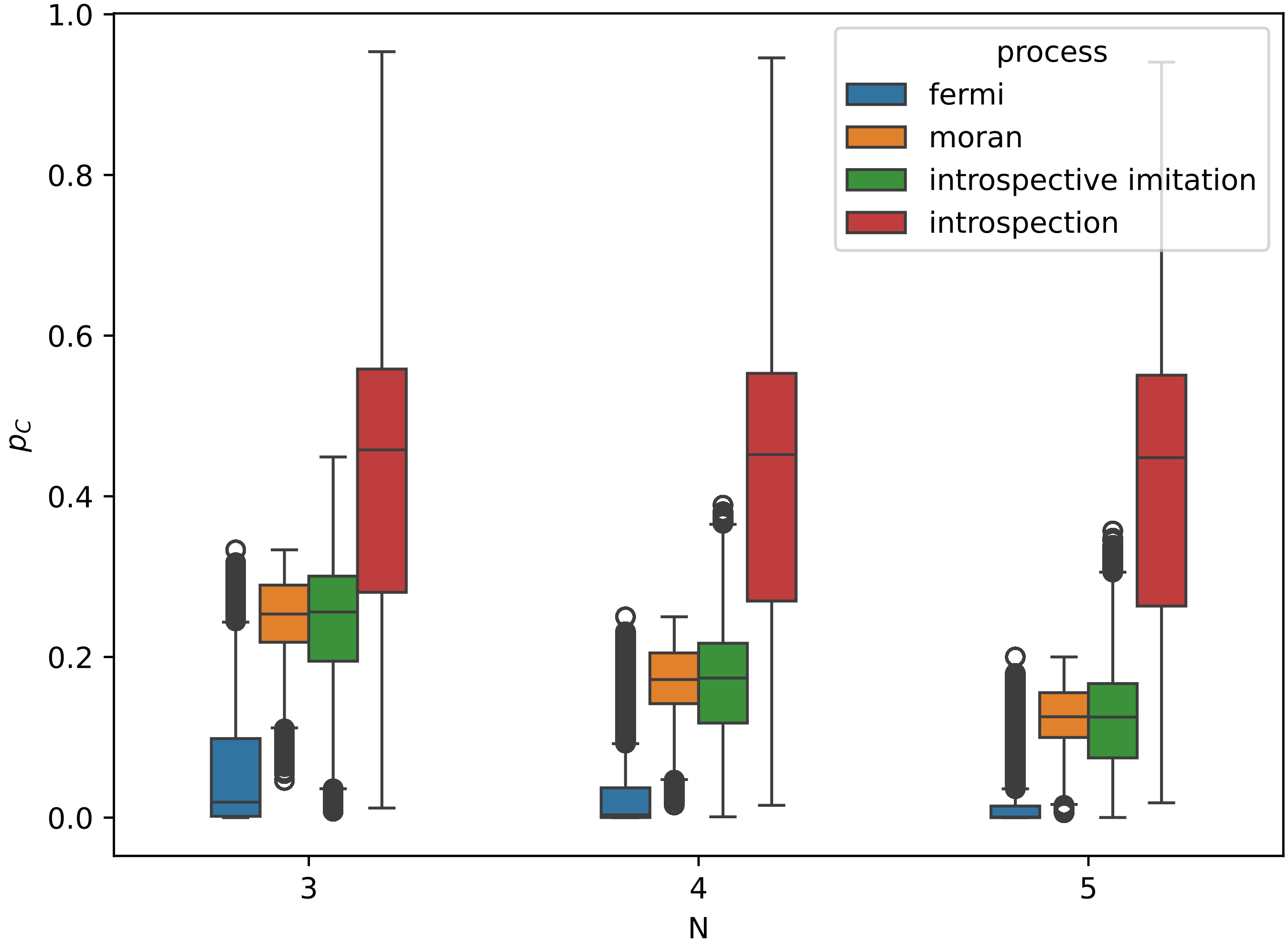
However, we do notice that in both of the above dynamics, there is a limit of $p_C = \frac{1}{3}$. This is because it is impossible for a state to have a contributor with a higher payoff than a defector if $\alpha_i > 0$, and thus in a purely extrinsic dynamic, the probability of a player accepting a defecting strategy will always be greater than the probability of a player accepting a cooperative strategy. Thus, we cannot achieve a value of p_C greater than that where players decide on strategies randomly, which would give us $p_C \leq \frac{1}{N}$



Here we see that introspective imitation is able to exceed the limit imposed on purely extrinsic processes. The introspective step is able to have a higher probability of accepting cooperation than defection if a player will have a higher payoff by cooperating than by defecting, even if in a given state the defectors will always have a higher payoff. Thus, it is possible to obtain higher values of p_C in introspective imitation dynamics than in purely extrinsic dynamics. It is also important to note that there is no result to show that the maximum value of p_C could not be improved from those shown in this data.

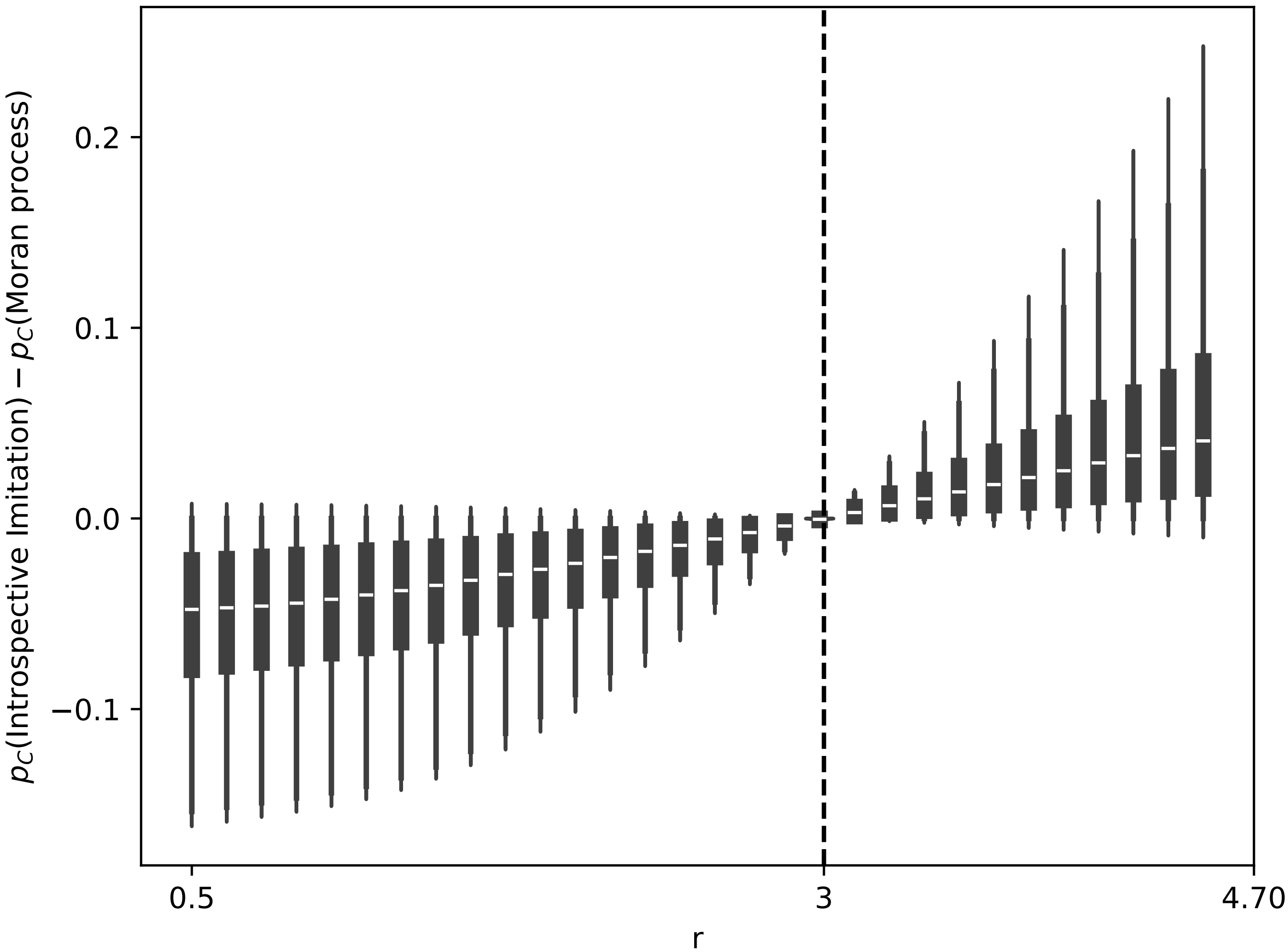


We see that this result also exists for introspection dynamics, though due to the different interpretation of p_C, this must be taken lightly.

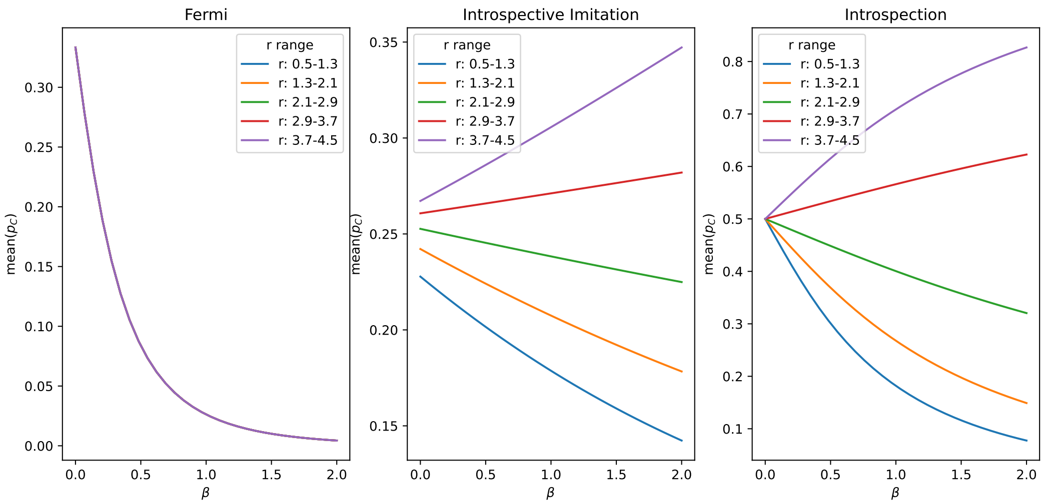


This result is not just true for the $N = 3$ case, but is also clearly true accross multiple values of N. This result is also true across different populations and even higher values of N, as can be seen in the supplementary information.

So to answer our previous question, intrinsic processes can in fact provide us with a greater probability of cooperation than purely extrinsic processes. In the next section, we will see under which conditions an intrinsic process begins to produce higher values of p_C than a purely extrinsic process. To do this, we compare the values of p_C in the Moran process and introspective imitation dynamics. This is because introspective imitation dynamics takes the form of the Moran process with an additional introspective step.



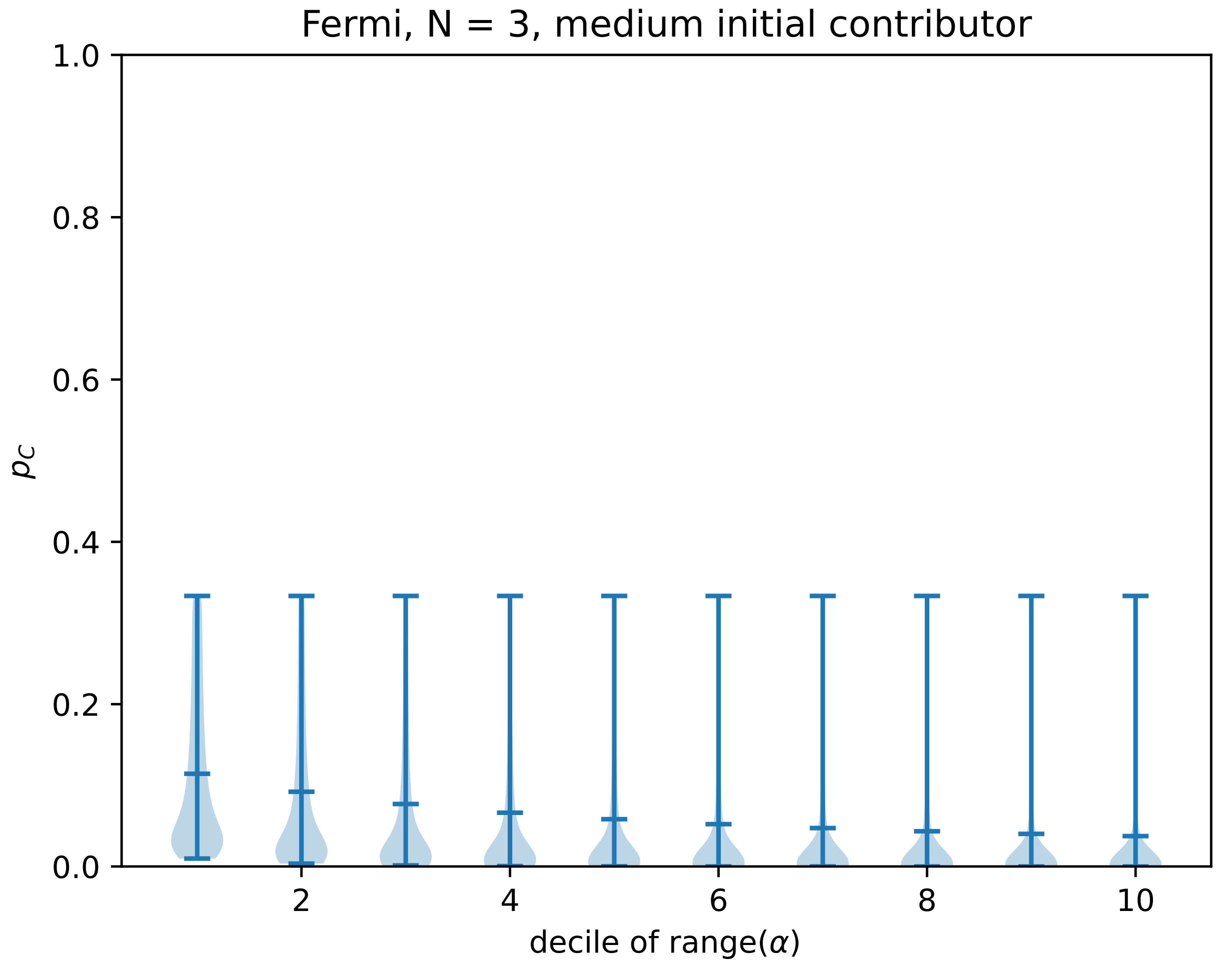
We can see that the parameter $r = N$ provides an exact point where the intrinsic processes begin to provide high p_C values. This is because when $r > N$, all players will gain a higher payoff by contributing than by defecting, regardless of the actions of the rest of the state, as we have $\frac{r\alpha_i}{N} - \alpha_i > 0$ for any value of α_i. Thus, this is the point at which the intrinsic step of introspective imitation dynamics begins to favour cooperation. For $r < N$, the intrinsic step favours defection, and thus the intrinsic step actually reduces p_C. We see in the supplementary information that this is indeed the point at which intrinsic processes begin to dominate.

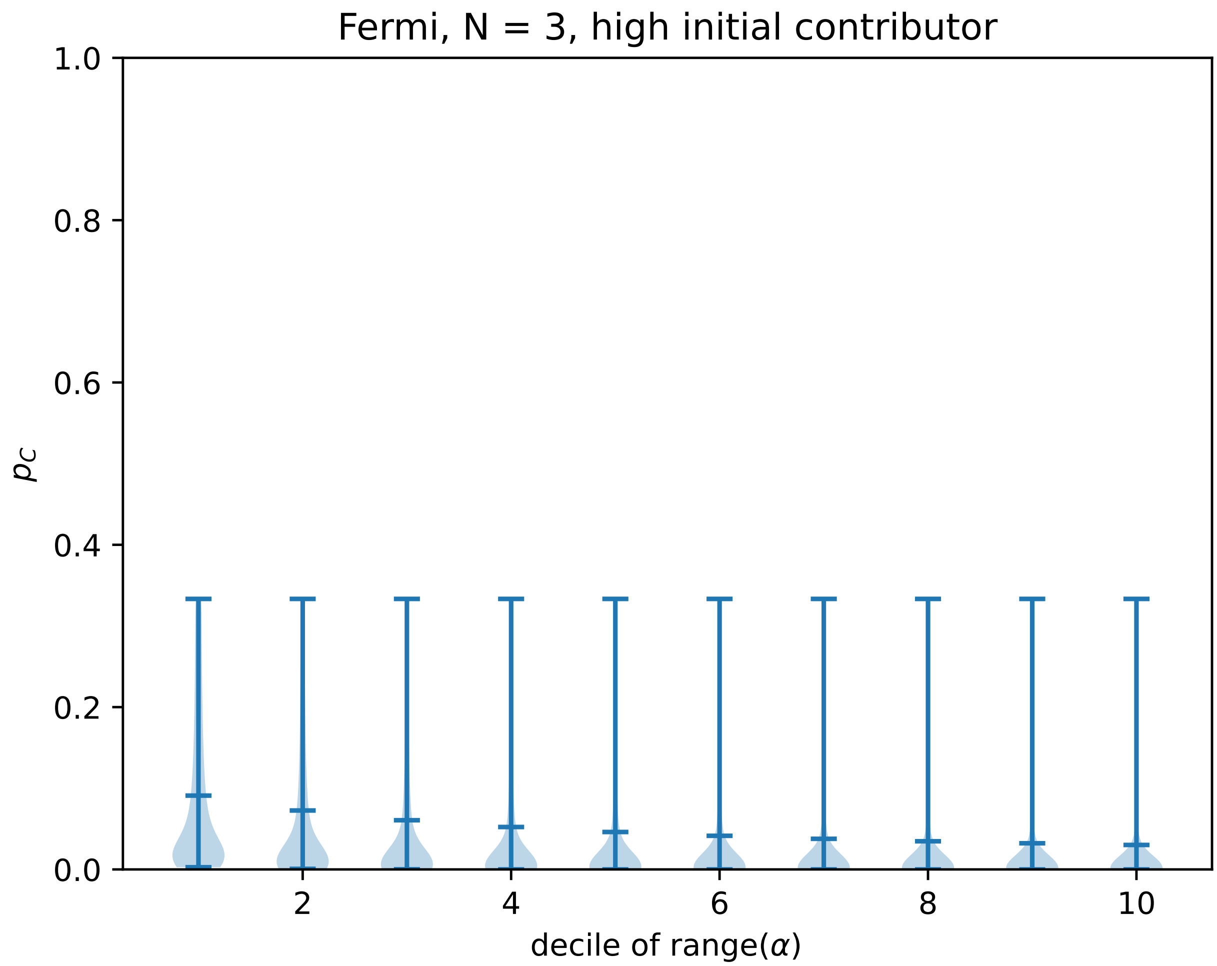


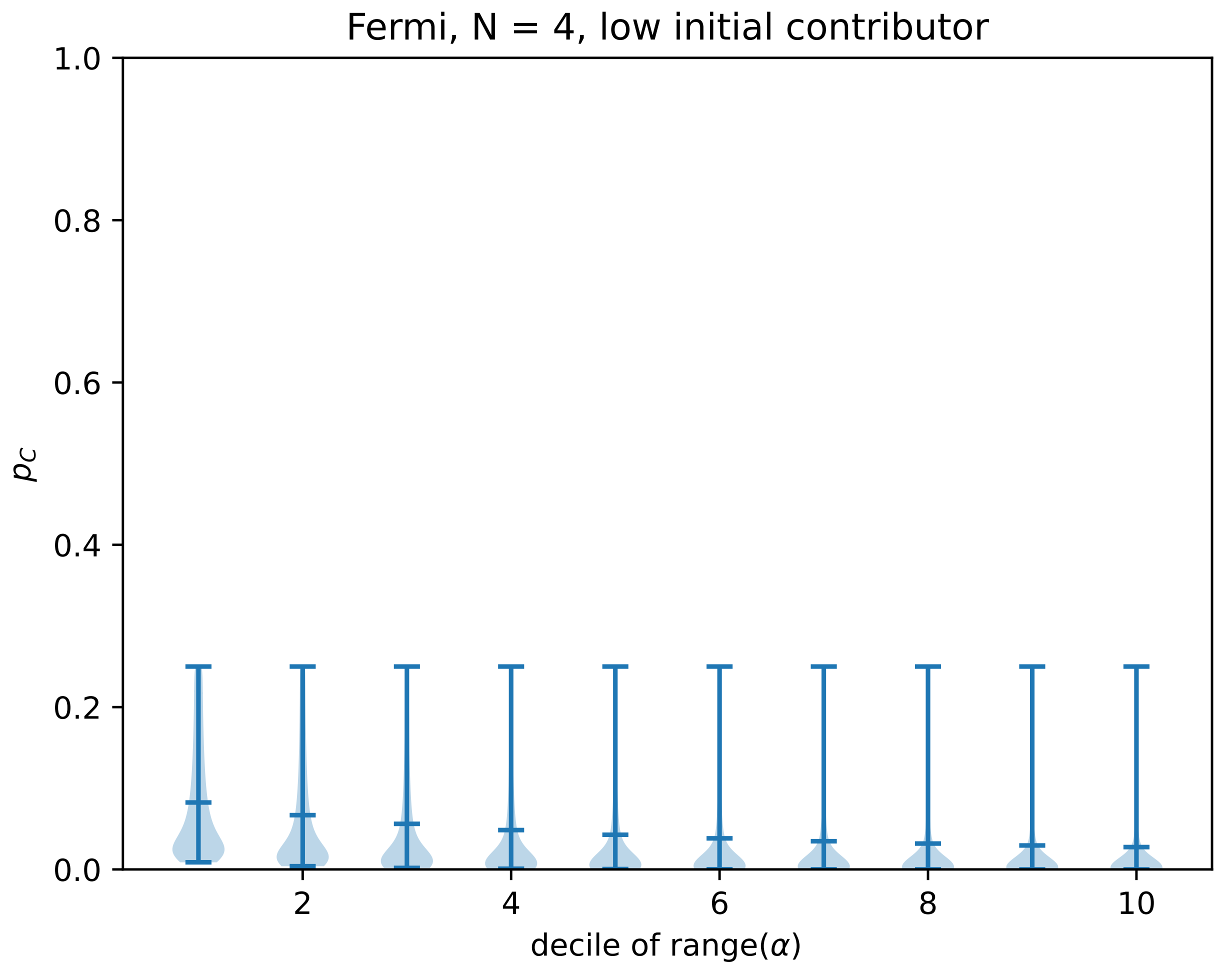
We also see that as the choice intensity β of a population dynamic increases, an intrinsic dynamic is able to facillitate an increase in p_C, whereas even for high r values, purely extrinsic processes will always have a negative correlation between β and p_C. This is because a purely extrinsic process will always see a defector as performing better than a cooperator, so a high choice intensity will always favour defection. However, in an intrinsic process, we can see that under certain conditions (specifically, the condition $r > N$ as before) the comparison between a cooperator and a defector may favour cooperation by comparing a defector to their own possible payoff by cooperating, and thus a higher p_C may be obtained by a higher value of β in an intrinsic process.

4 supplementary information

4.1 Higher N and other populations




Fermi, N = 3, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 4, low initial contributor
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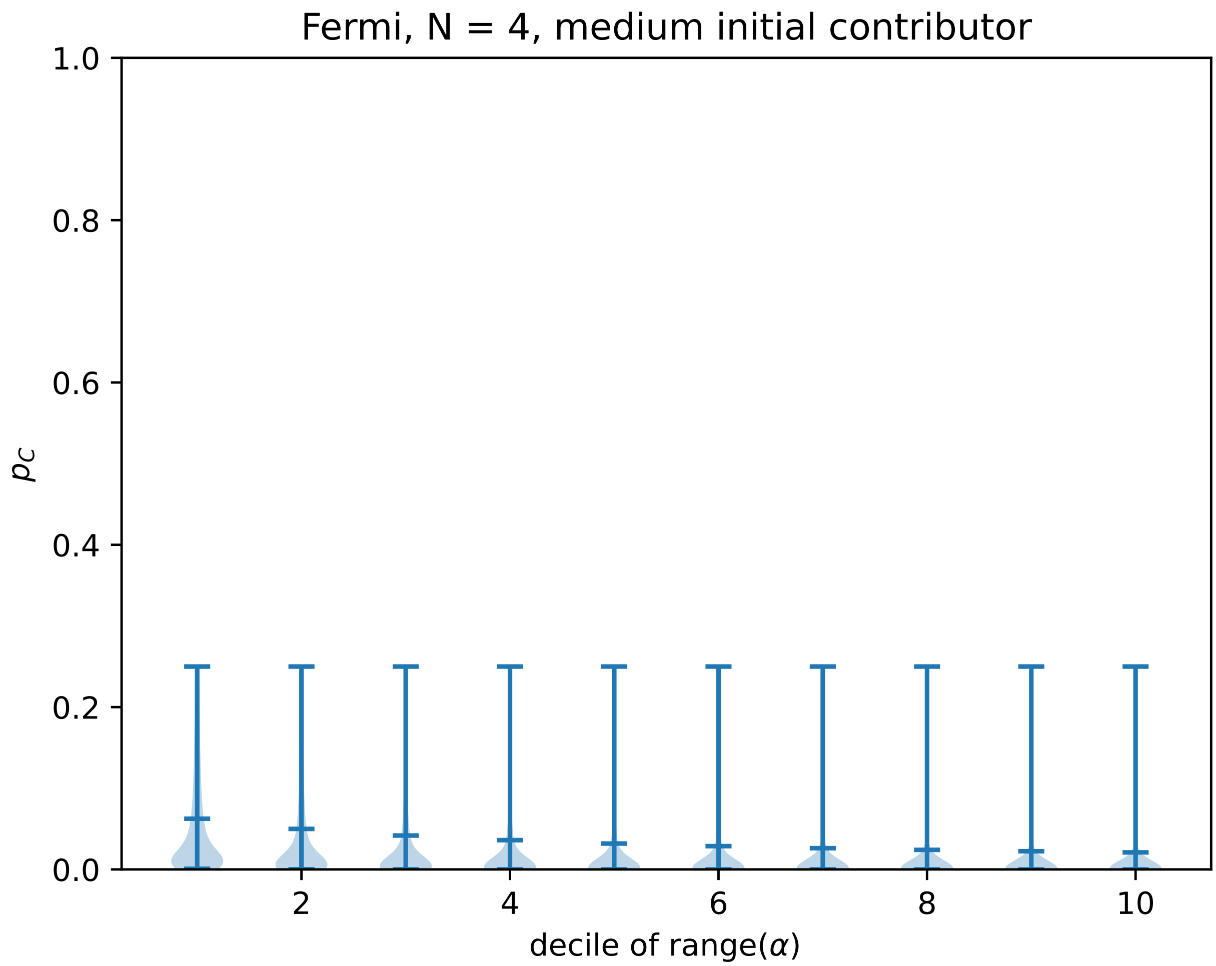
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 4, medium initial contributor
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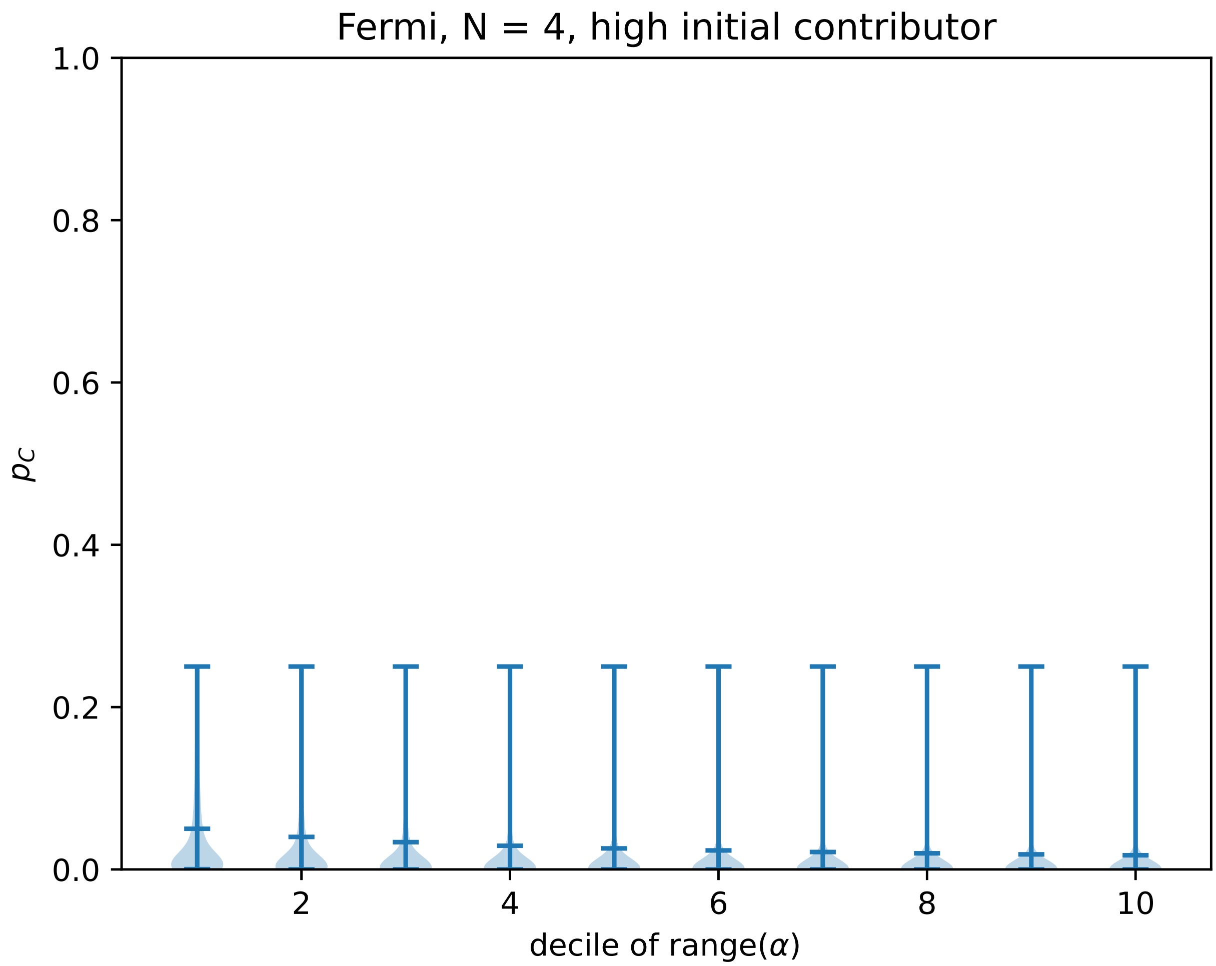
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 4, high initial contributor
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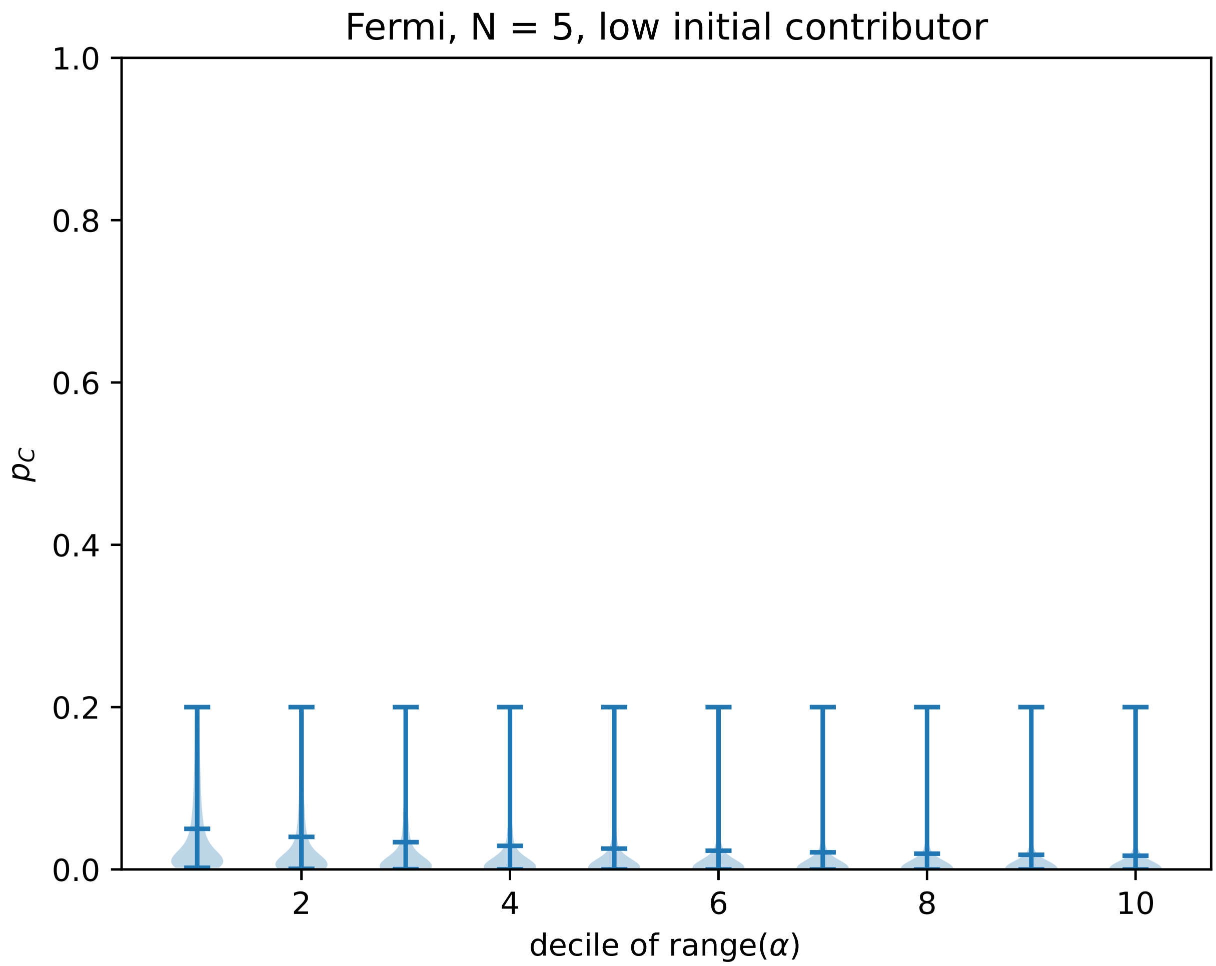
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 5, low initial contributor
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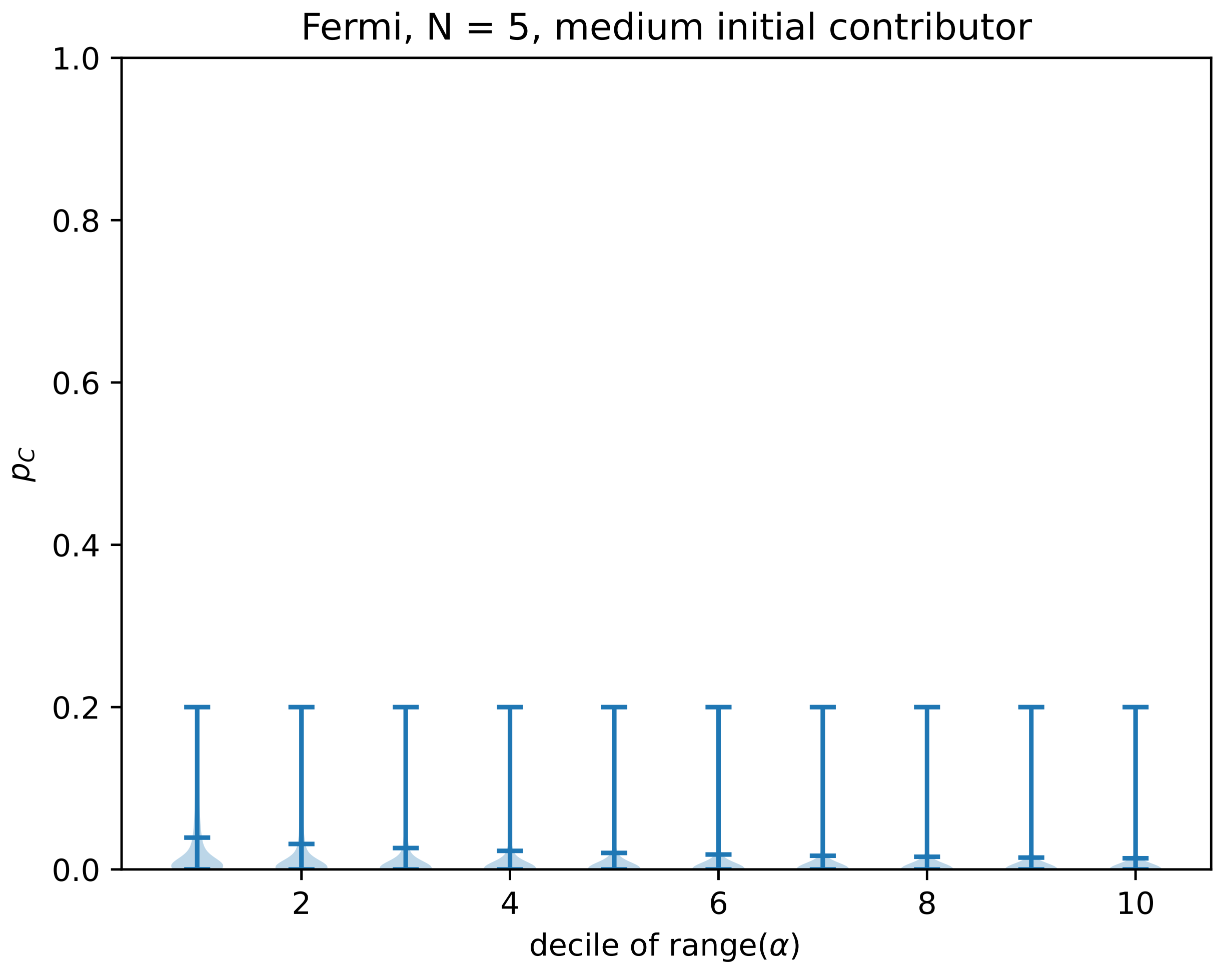
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 5, medium initial contributor
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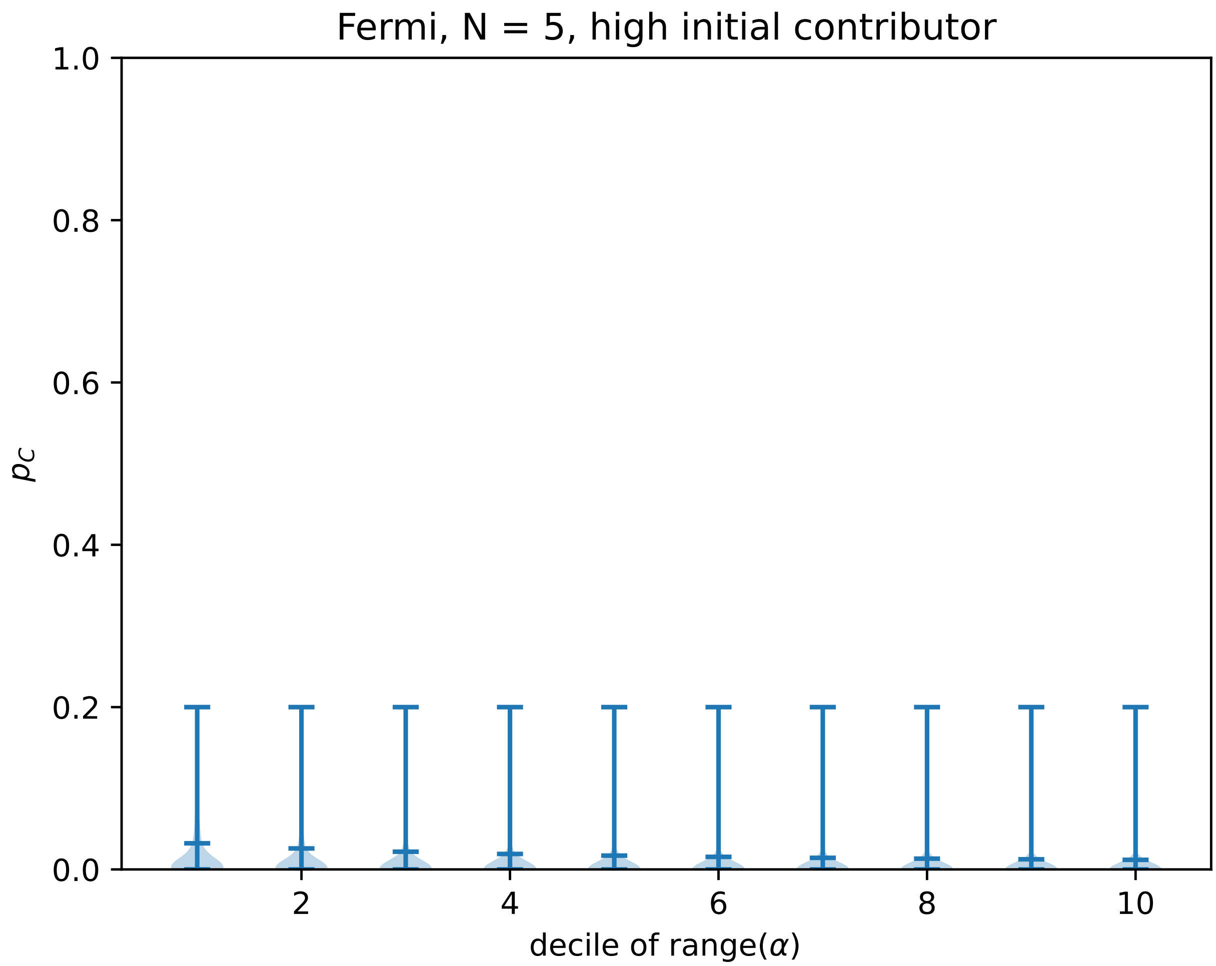
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 5, high initial contributor
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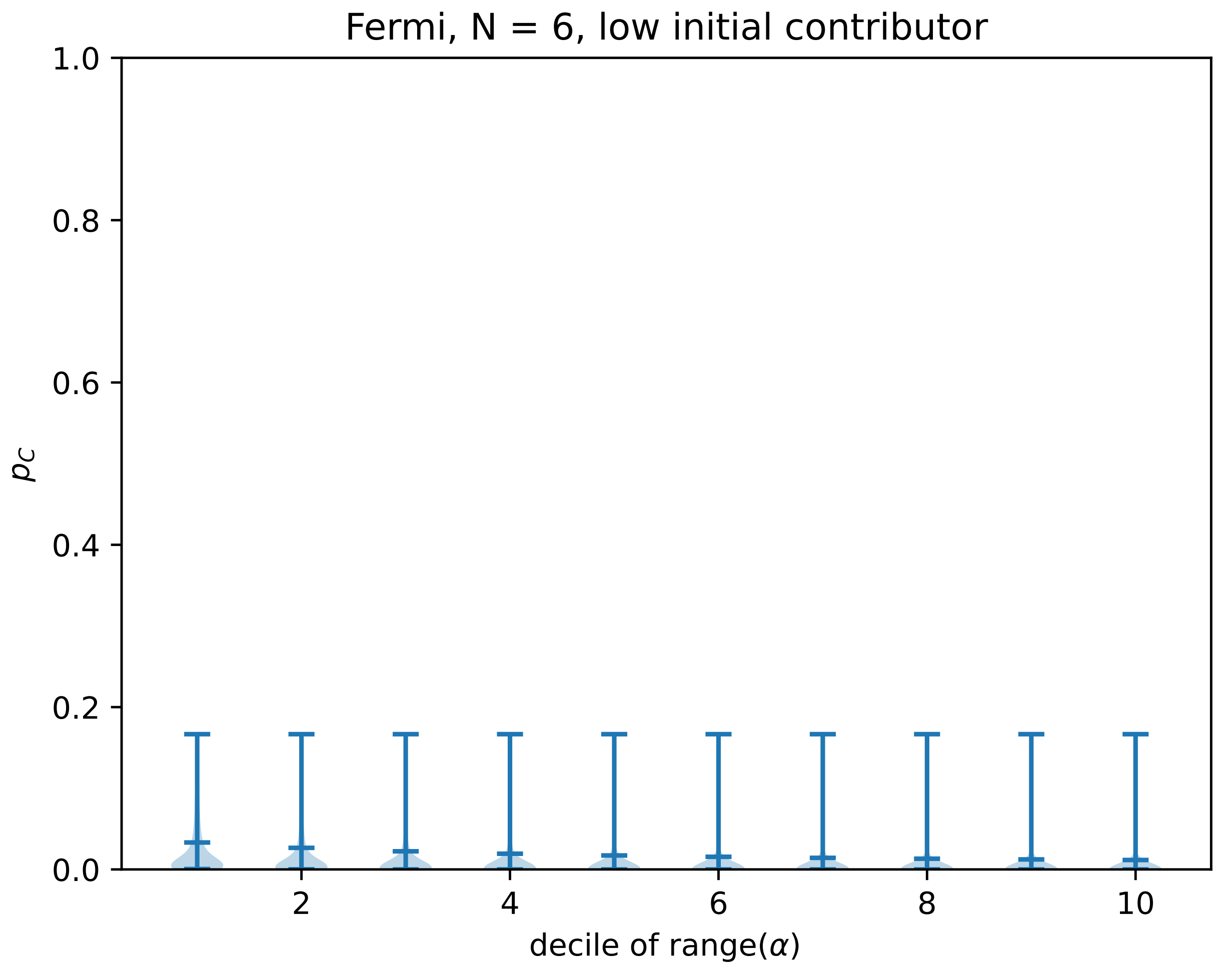
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 6, low initial contributor
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0.8
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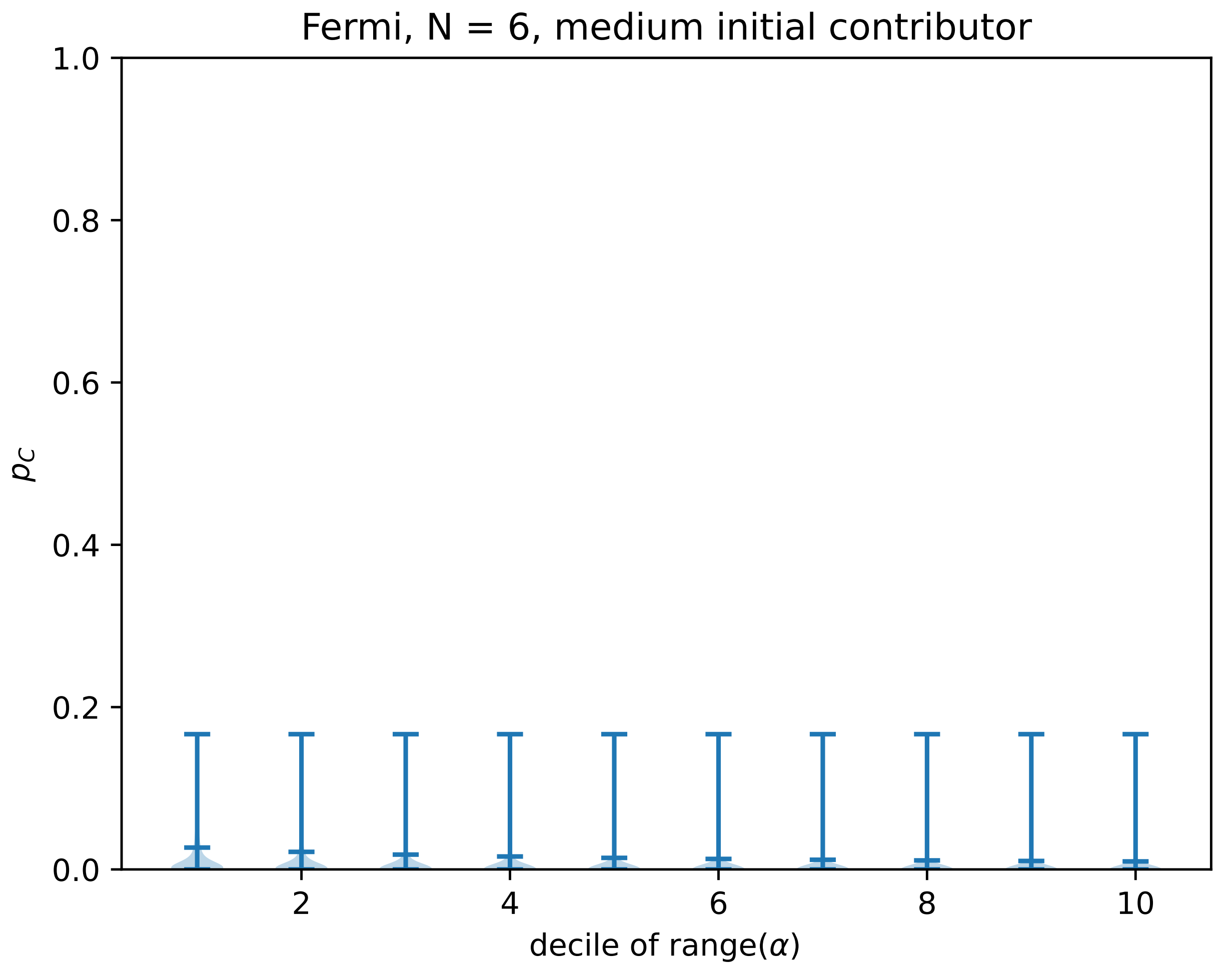
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 6, medium initial contributor
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0.8
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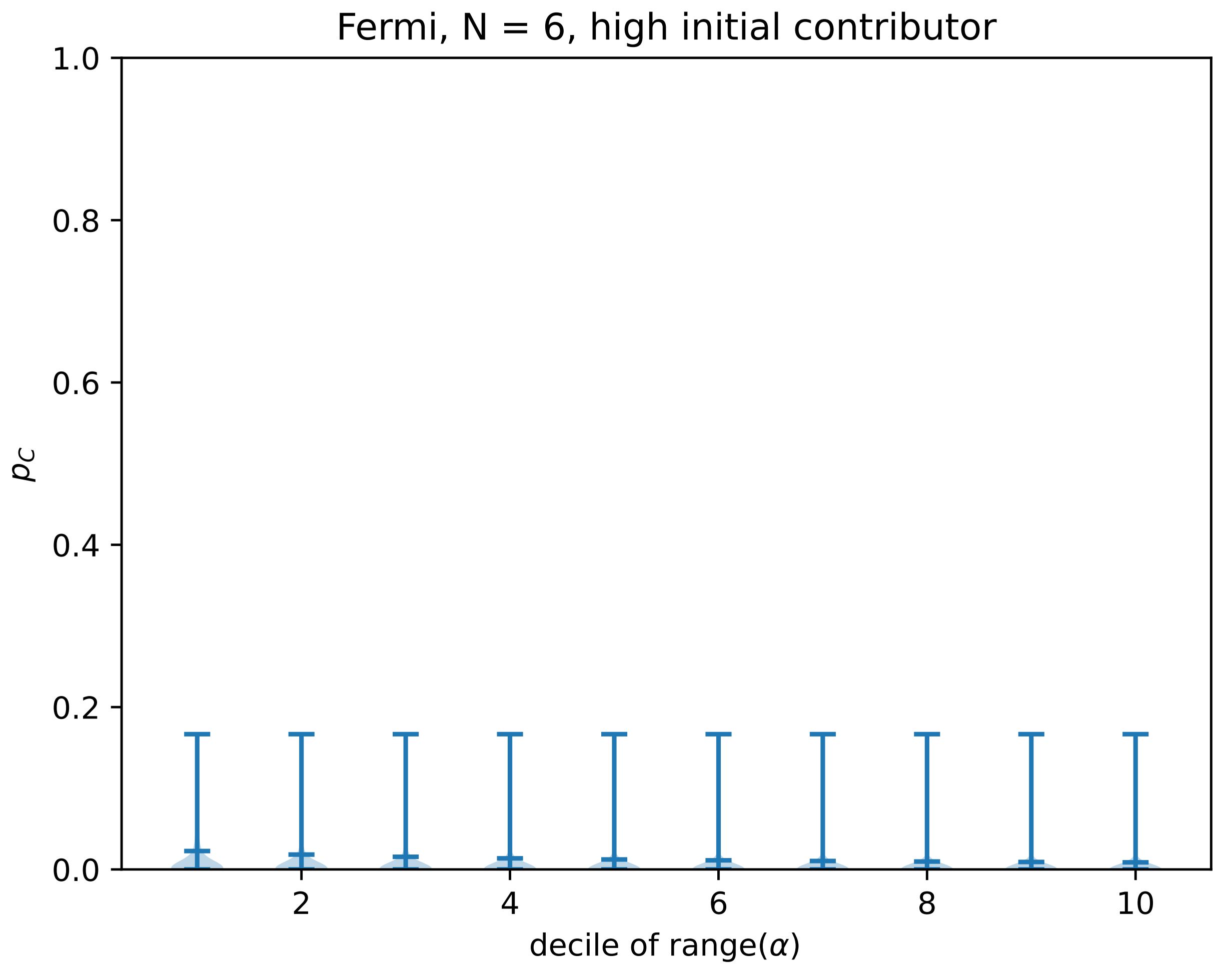
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 6, high initial contributor
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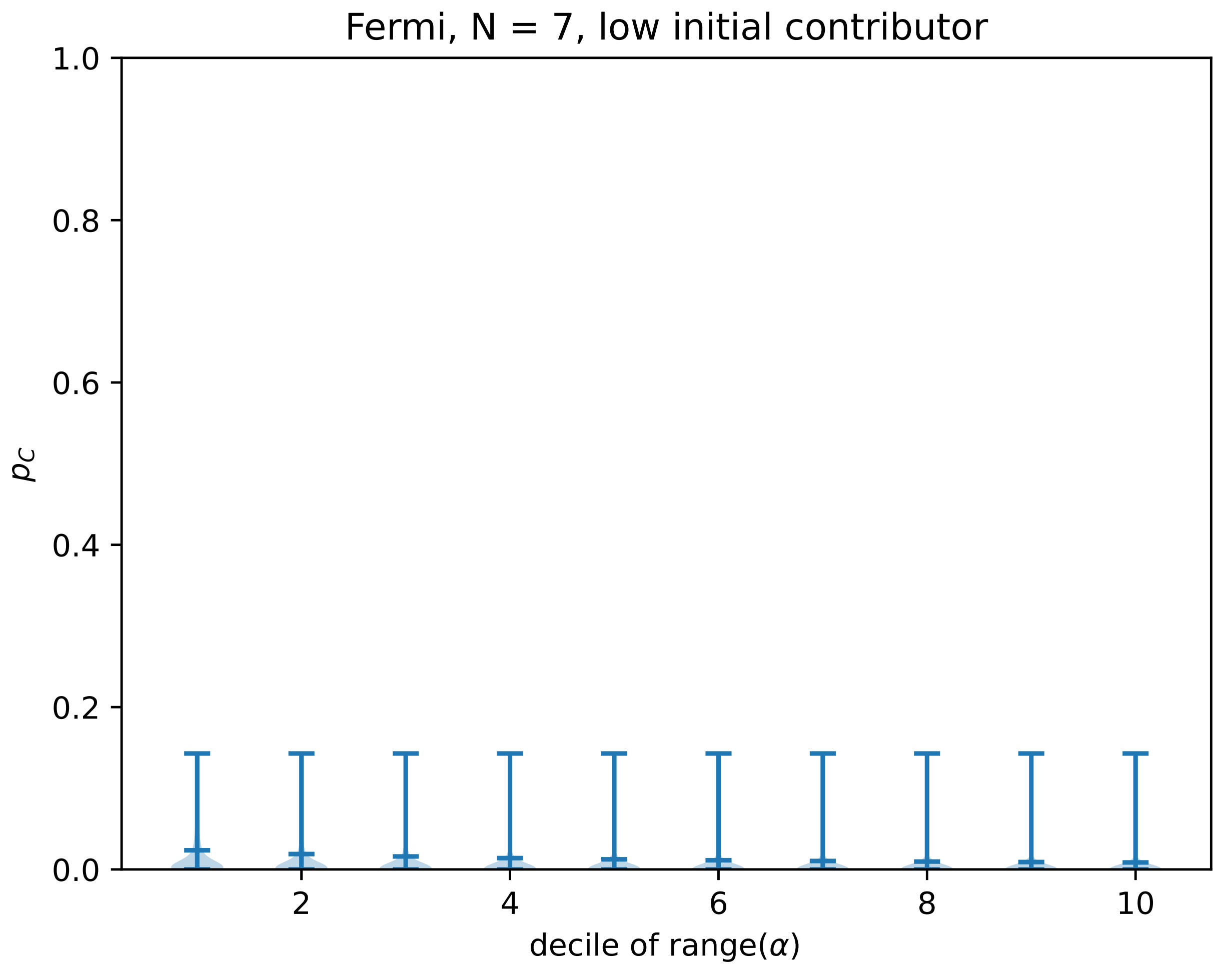
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 7, low initial contributor
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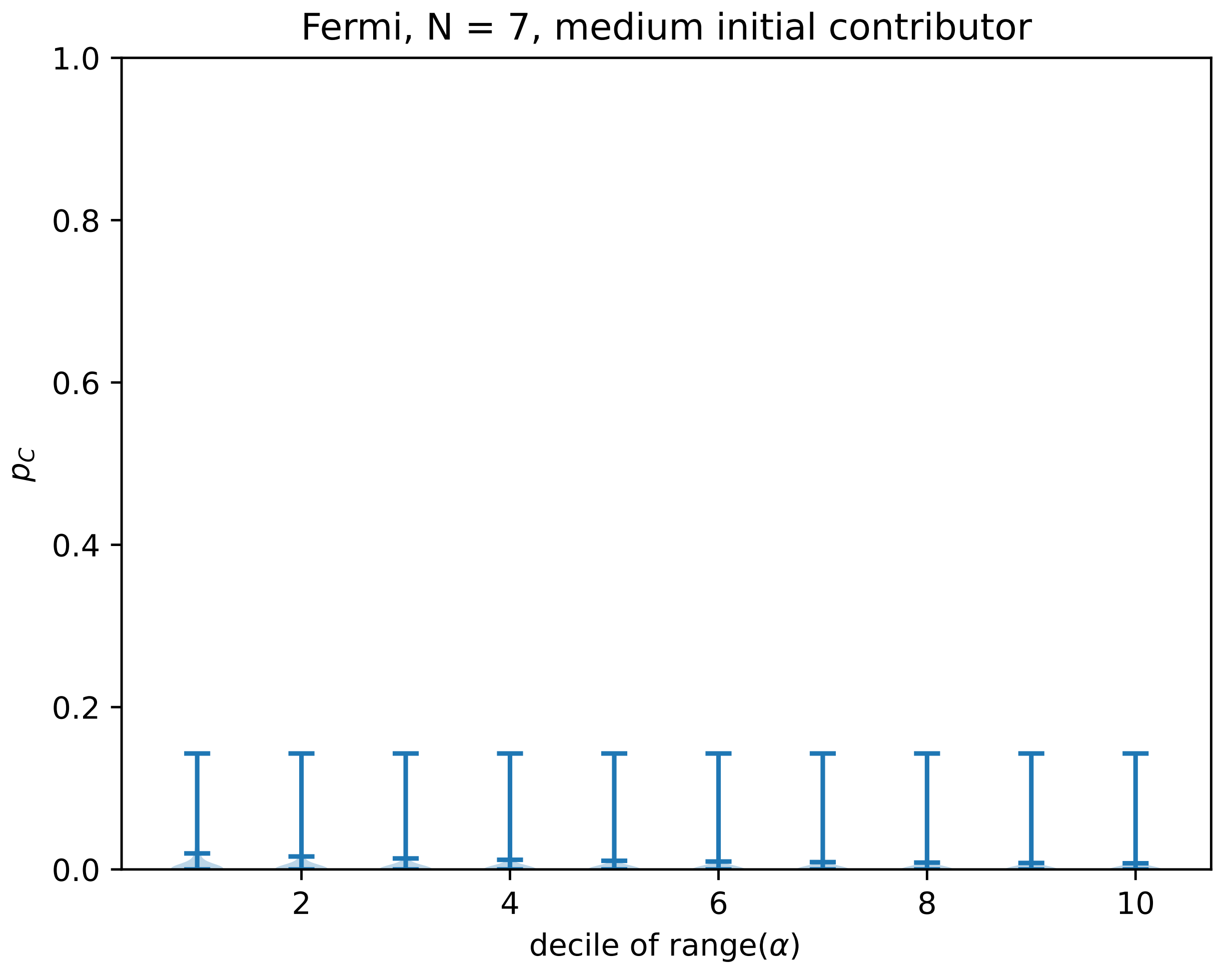
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 7, medium initial contributor
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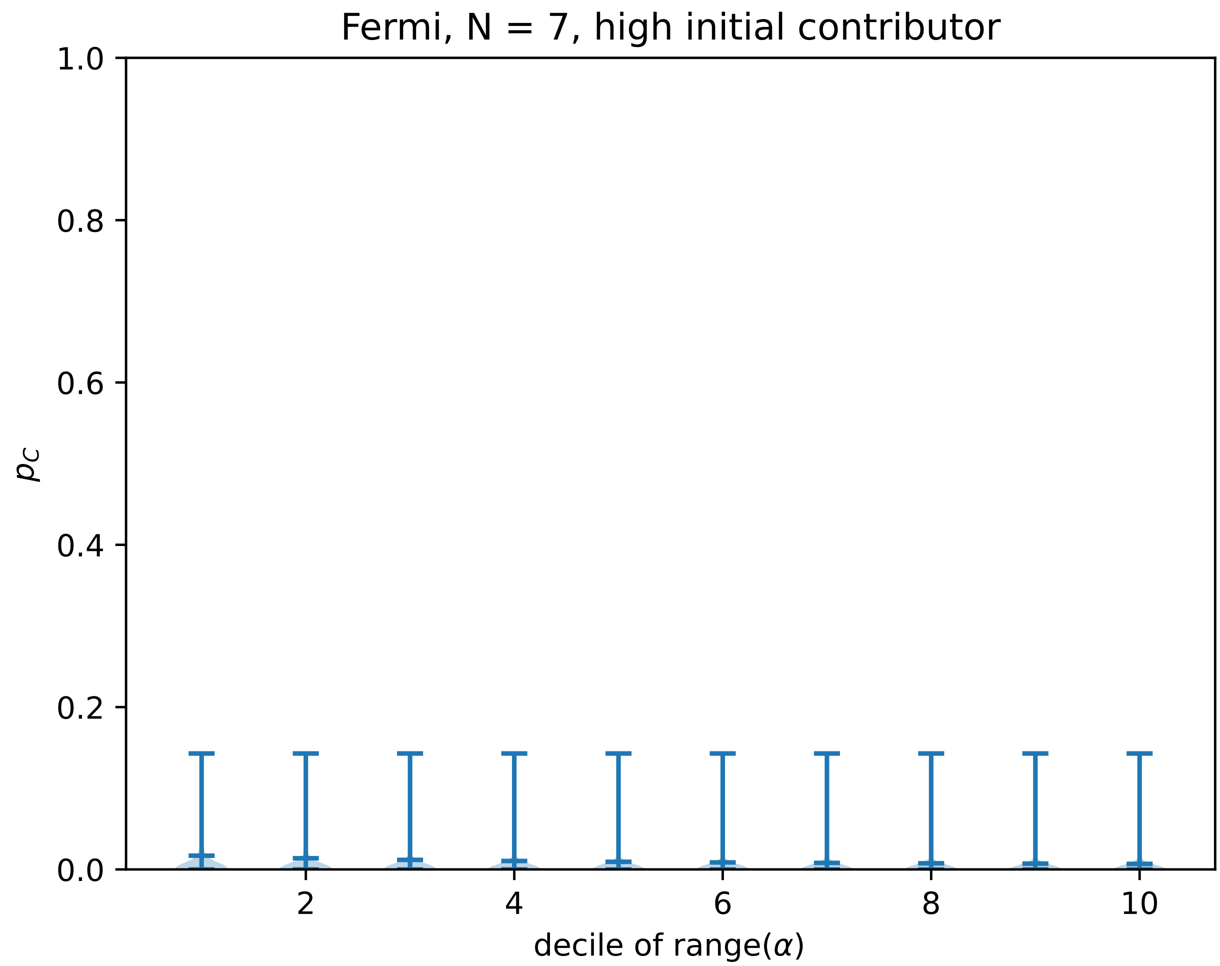
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 7, high initial contributor
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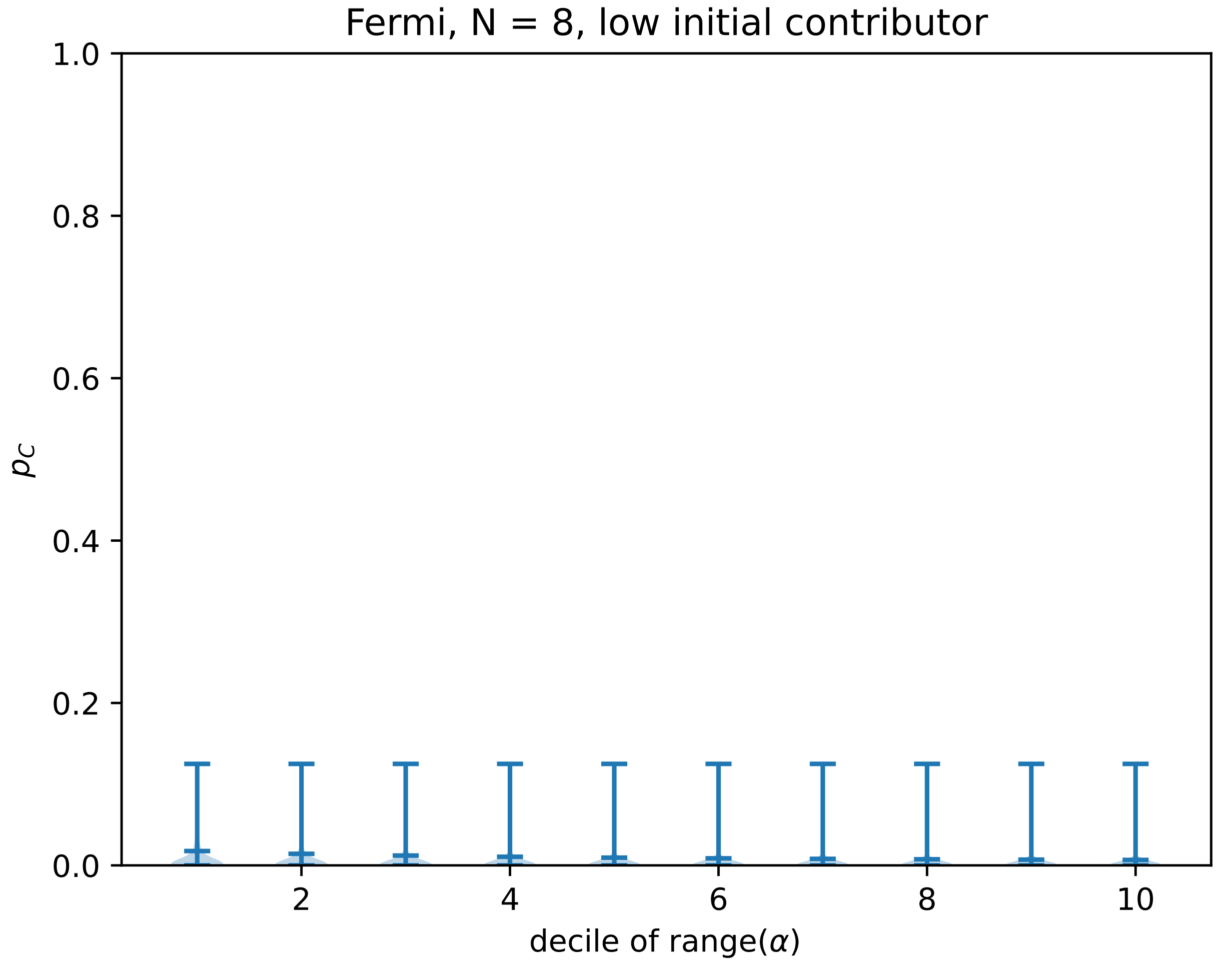
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 8, low initial contributor
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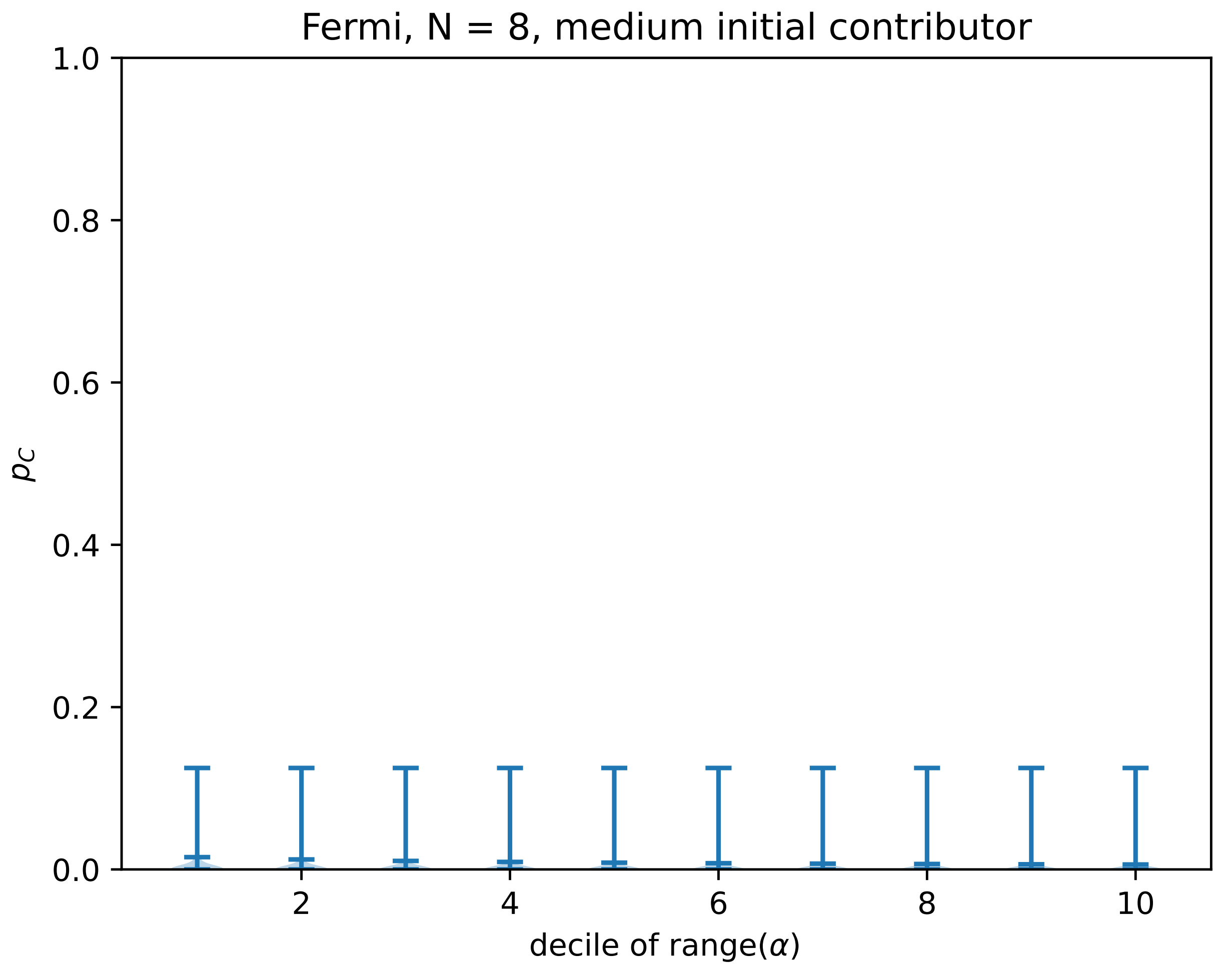
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 8, medium initial contributor
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0.6
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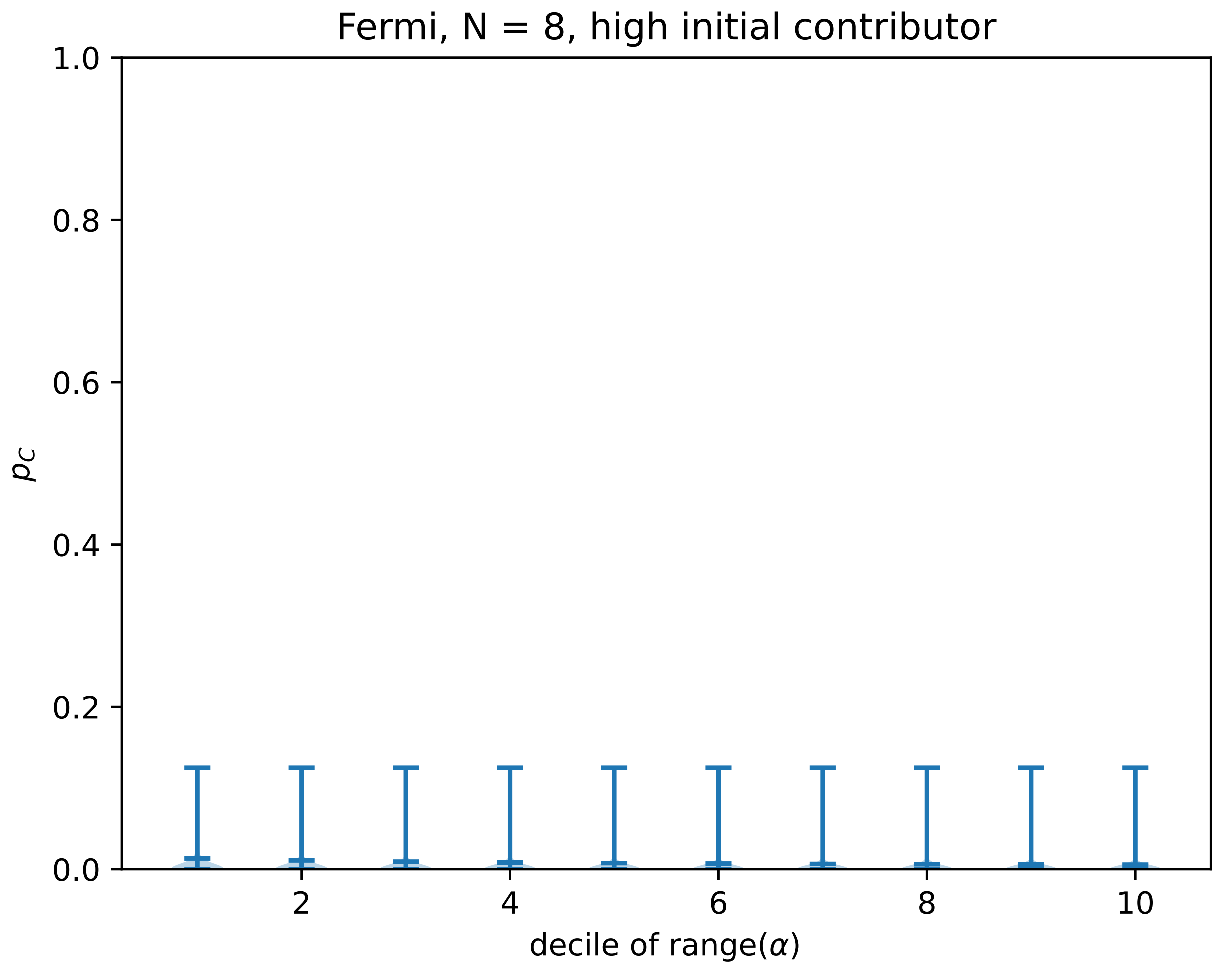
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 8, high initial contributor
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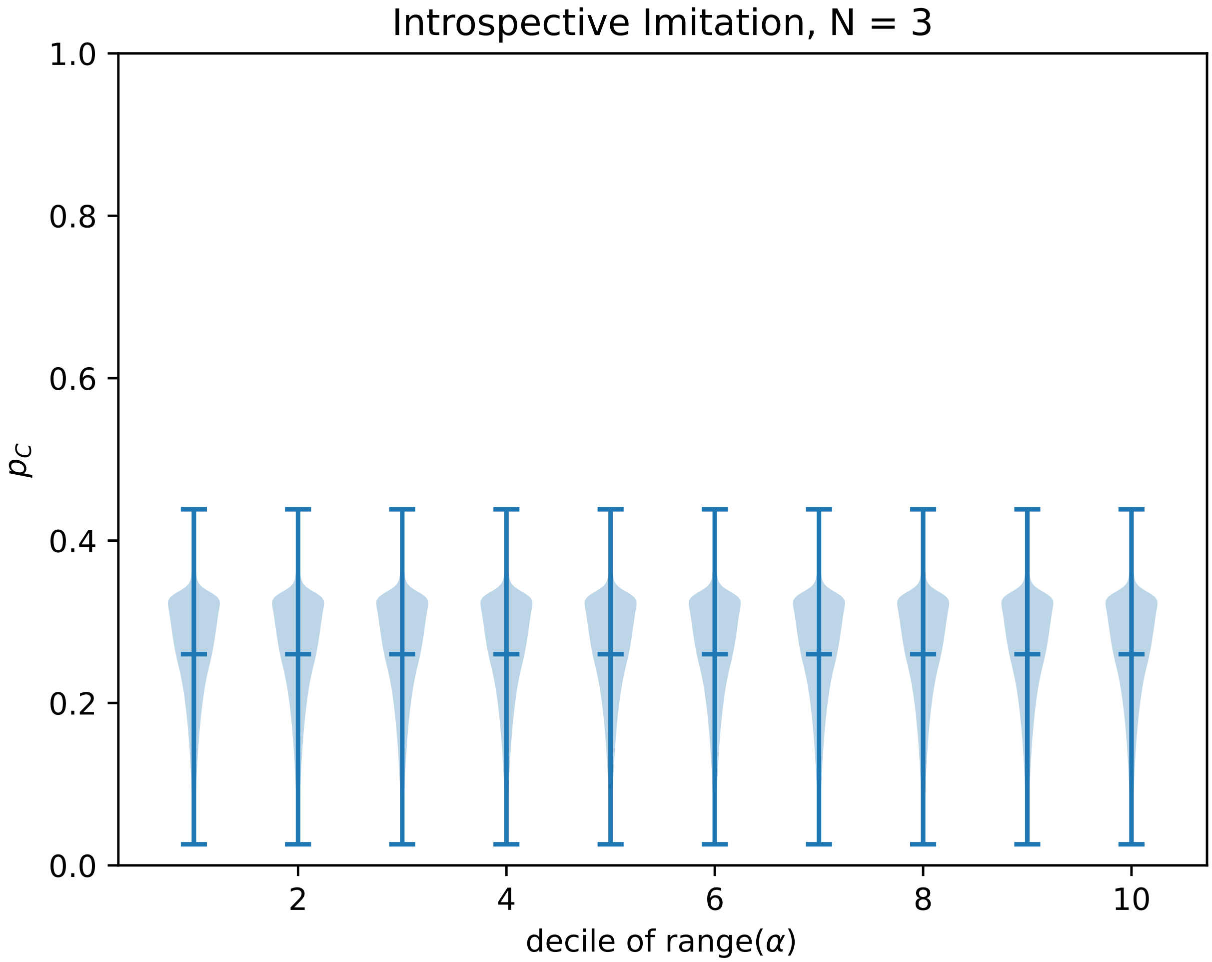
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Introspective Imitation, N = 3
p_C
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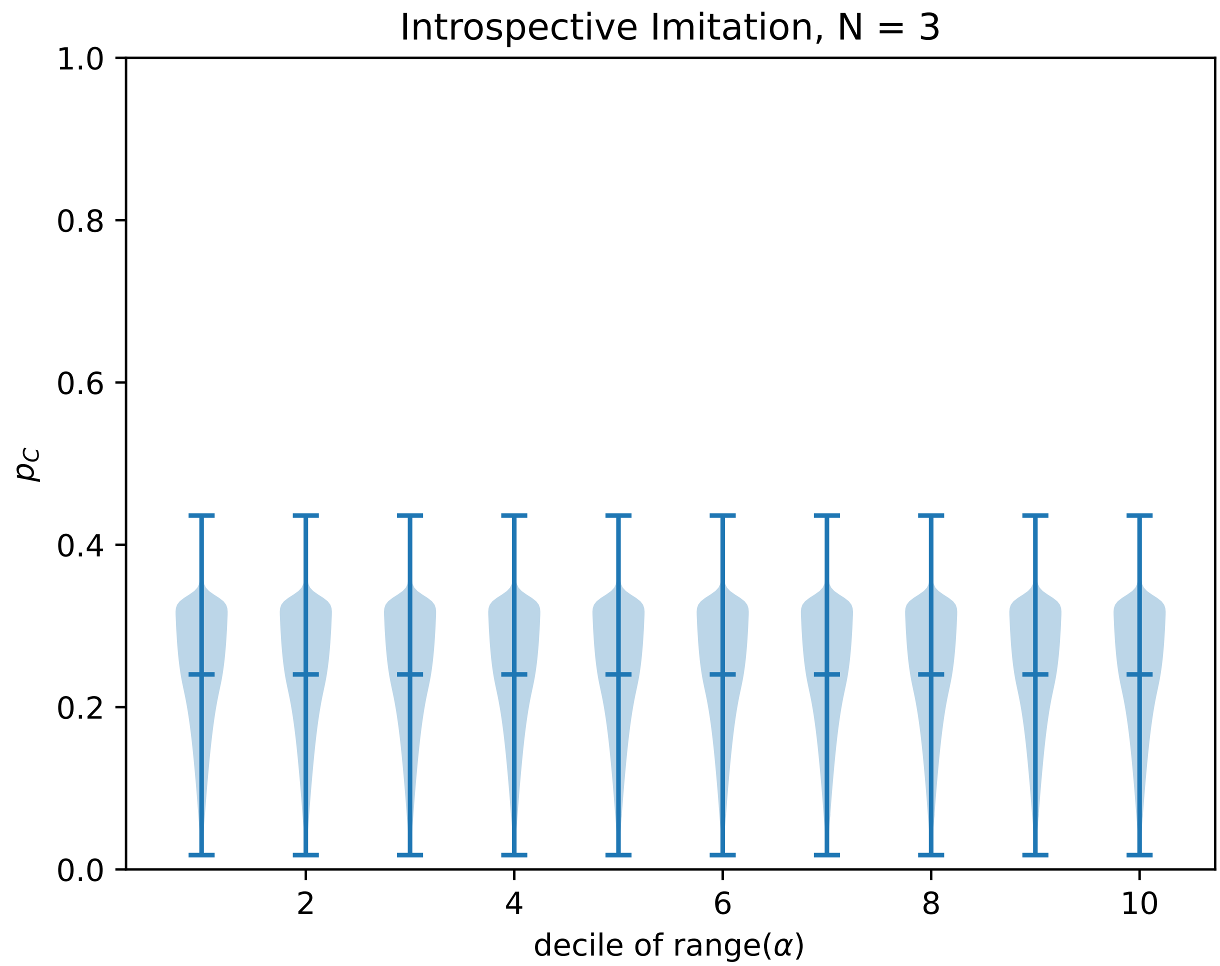
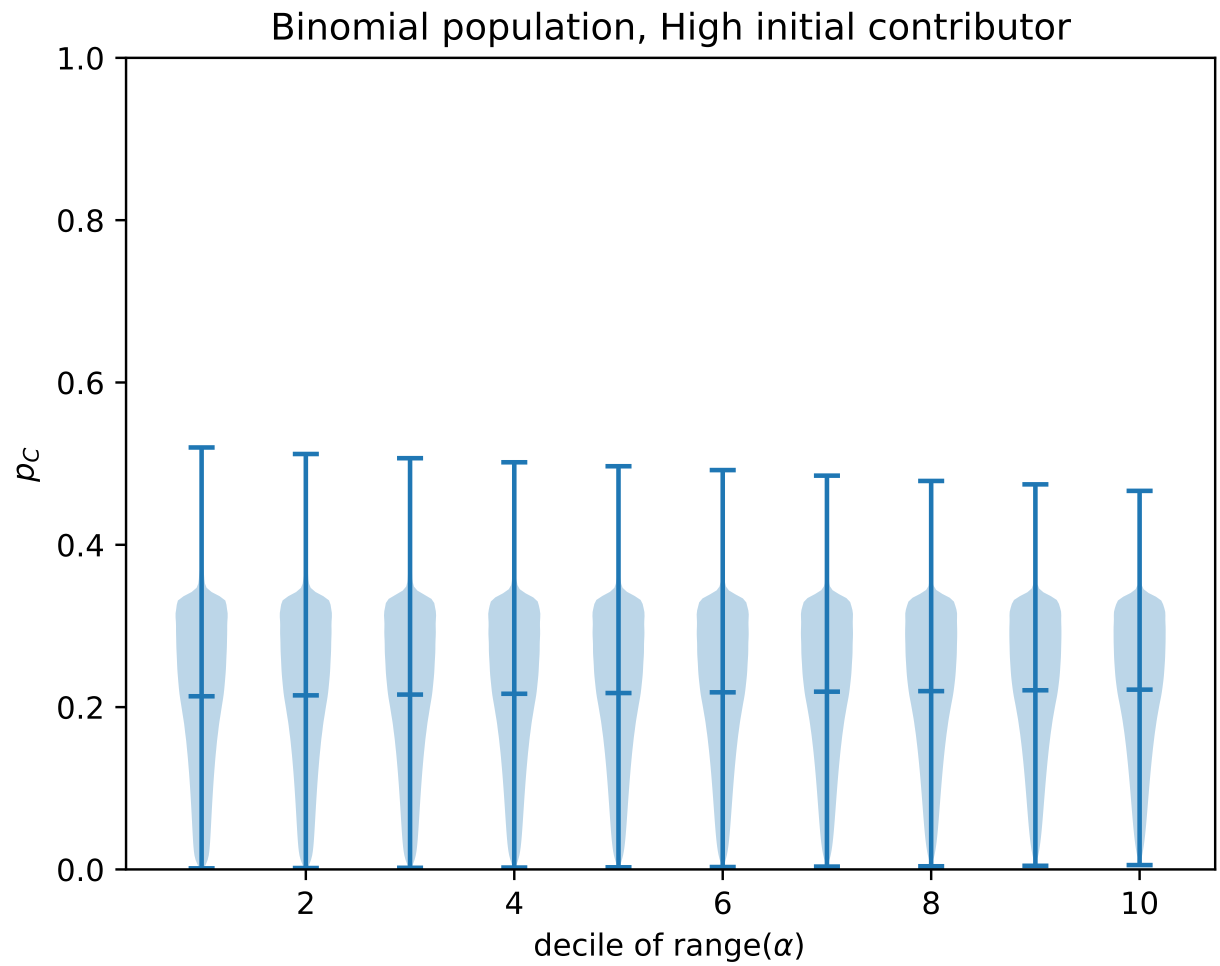
0.6
0.4
0.2
0.0
2
4
6
8
10
decile of range(α)


Introspective Imitation, N = 3
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Binomial population, High initial contributor
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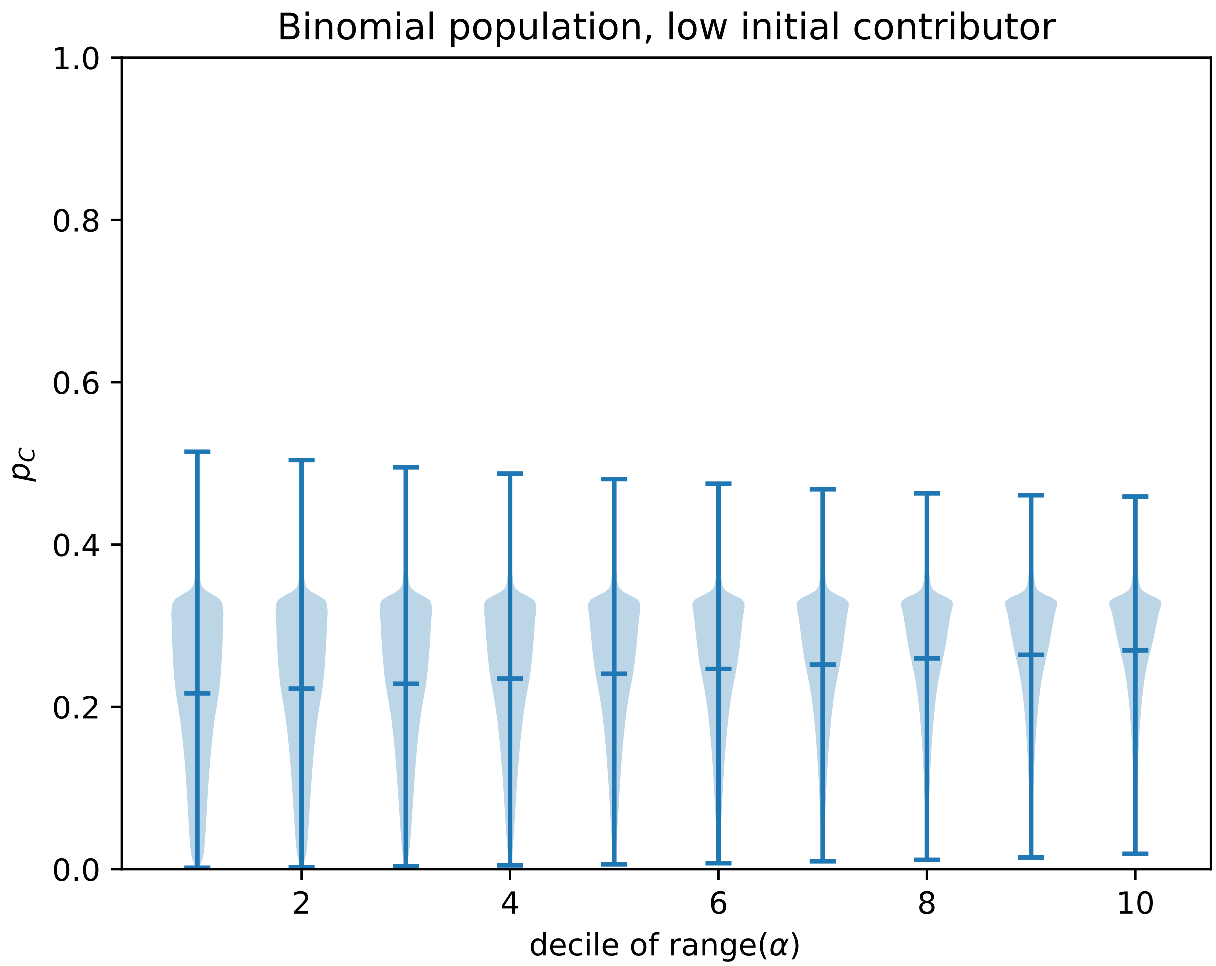
p_C
2
4
6
8
10
decile of range(α)


Binomial population, low initial contributor
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0.6
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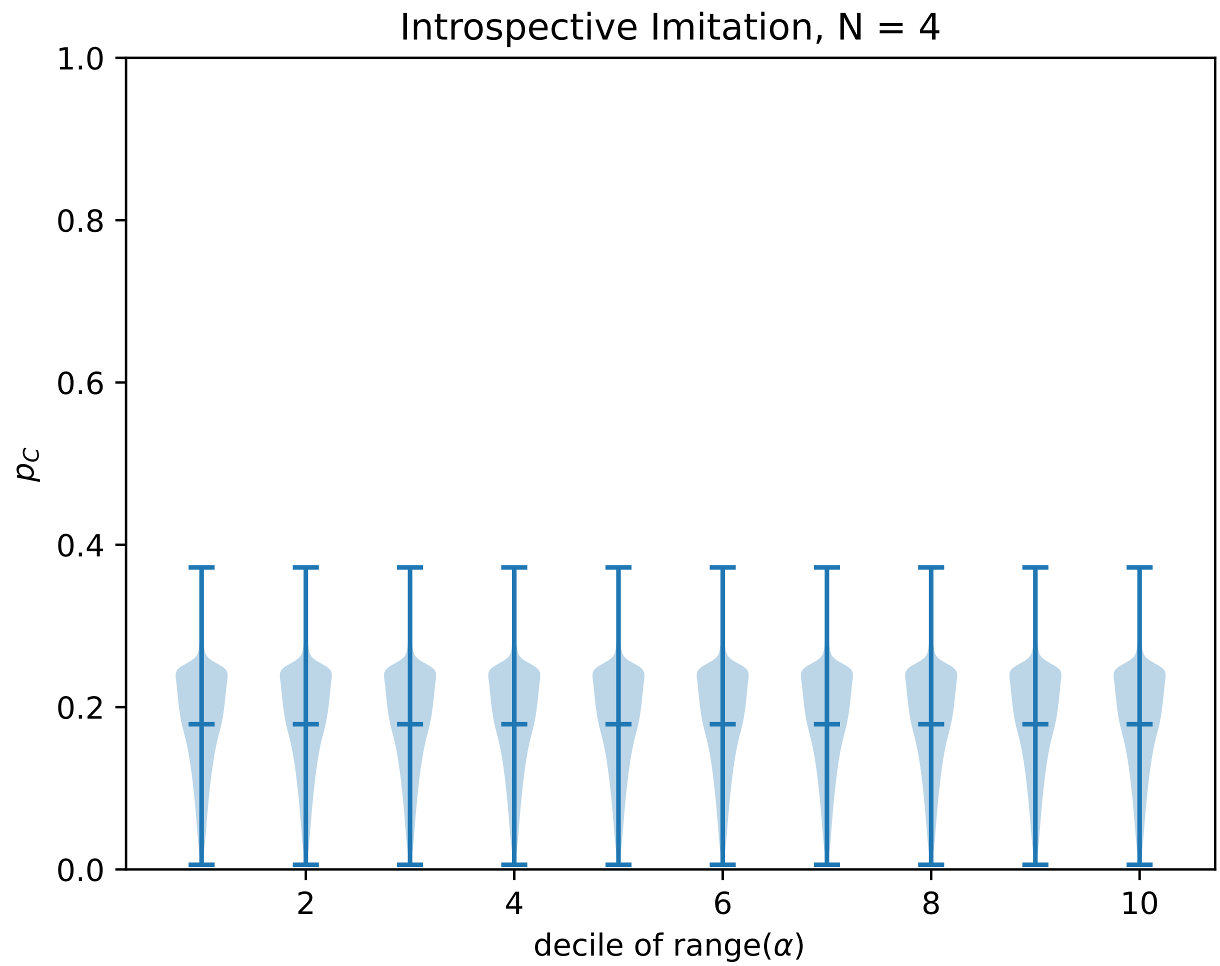
p_C
2
4
6
8
10
decile of range(α)



Introspective Imitation, N = 4
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p_C
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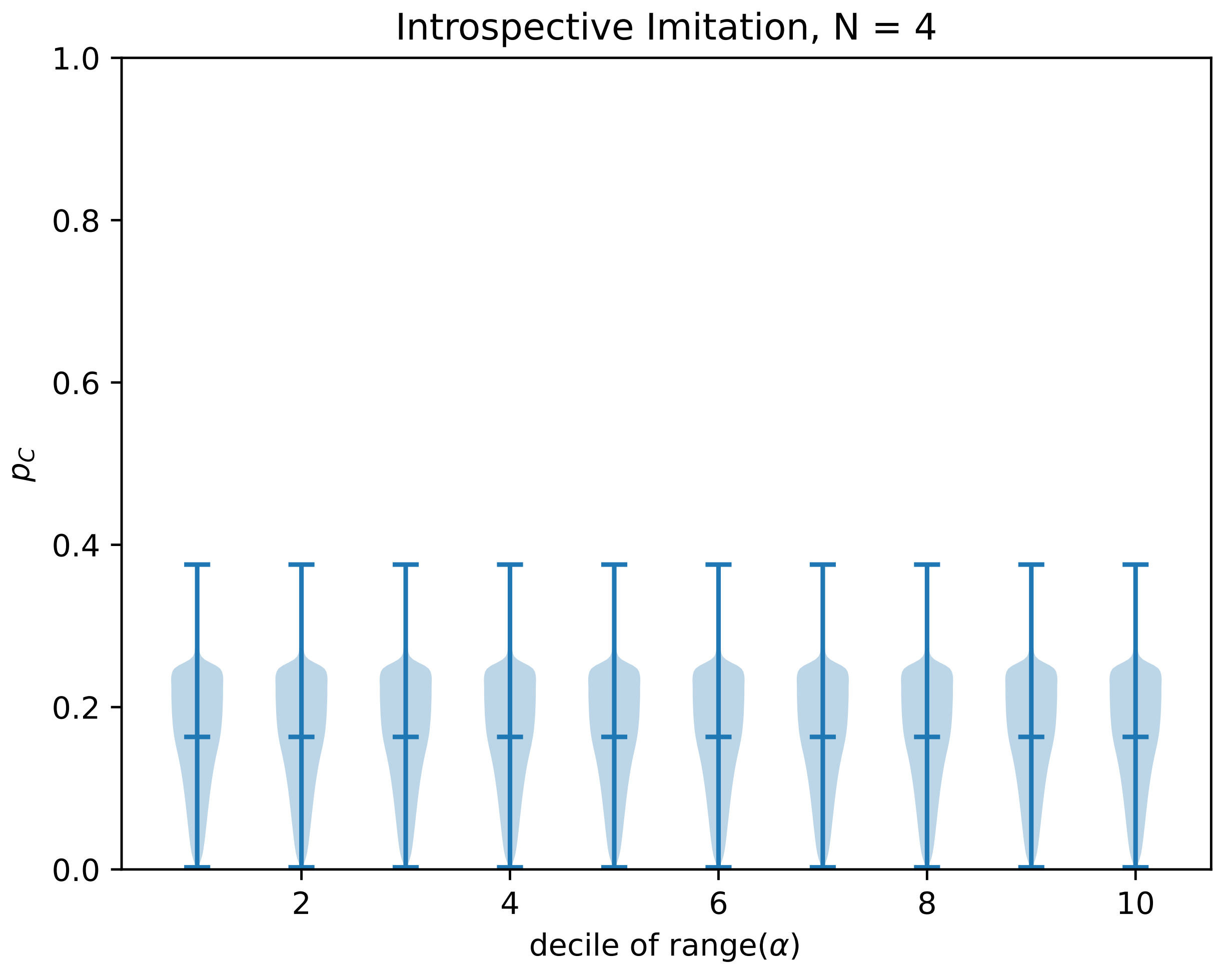
4
6
8
10
decile of range(α)


Introspective Imitation, N = 4
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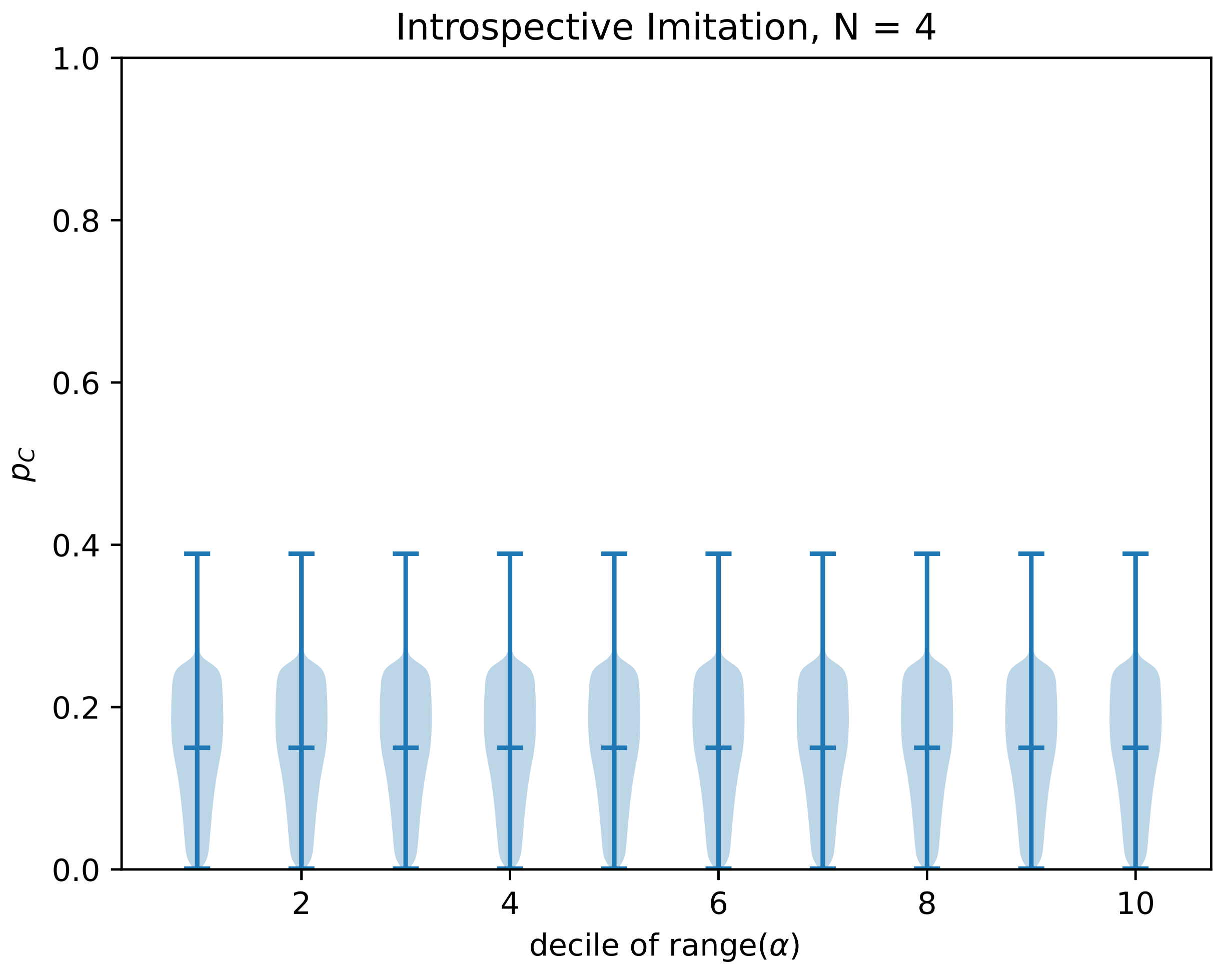
0.0
p_C
2
4
6
8
10
decile of range(α)


Introspective Imitation, N = 4
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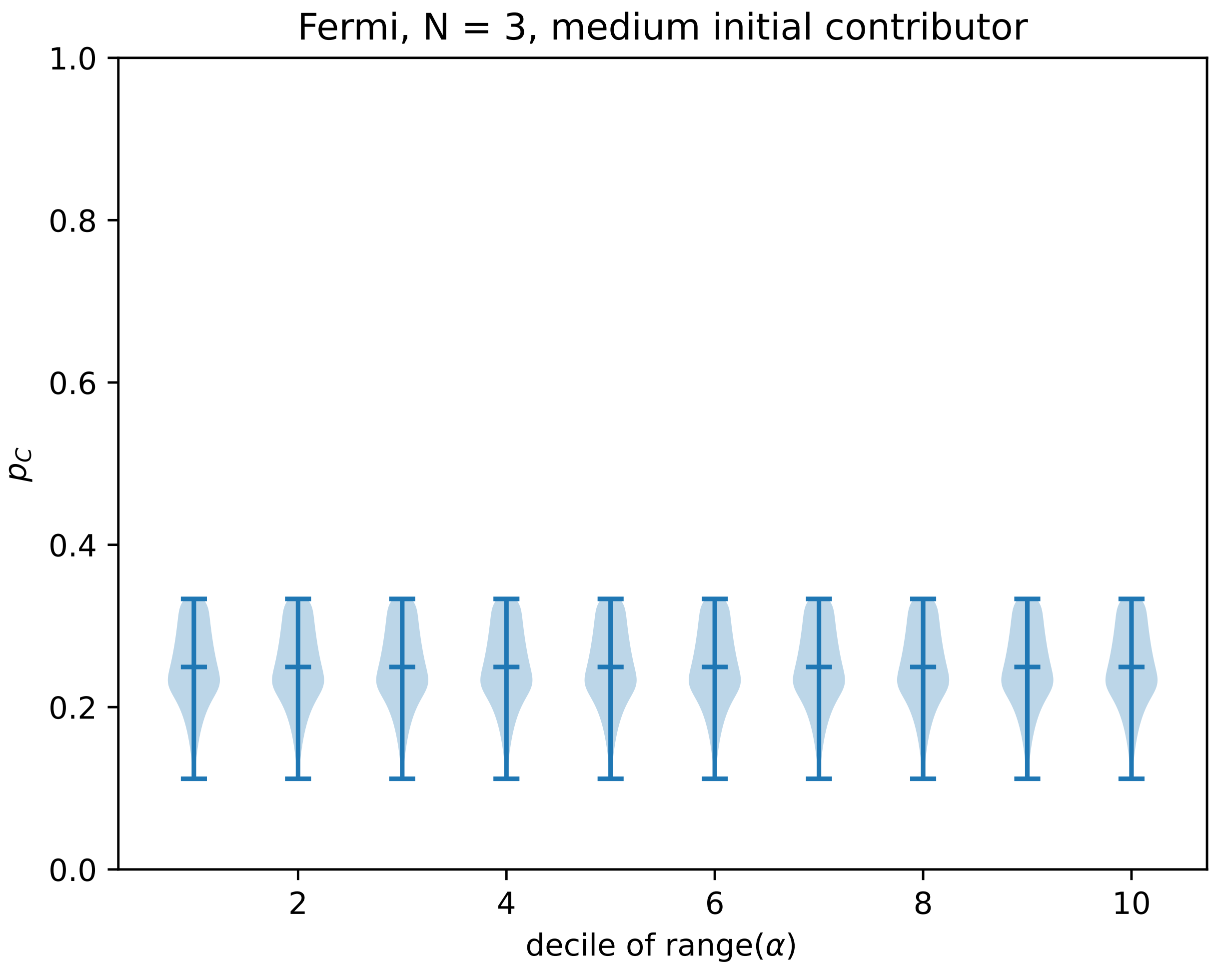
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 3, medium initial contributor
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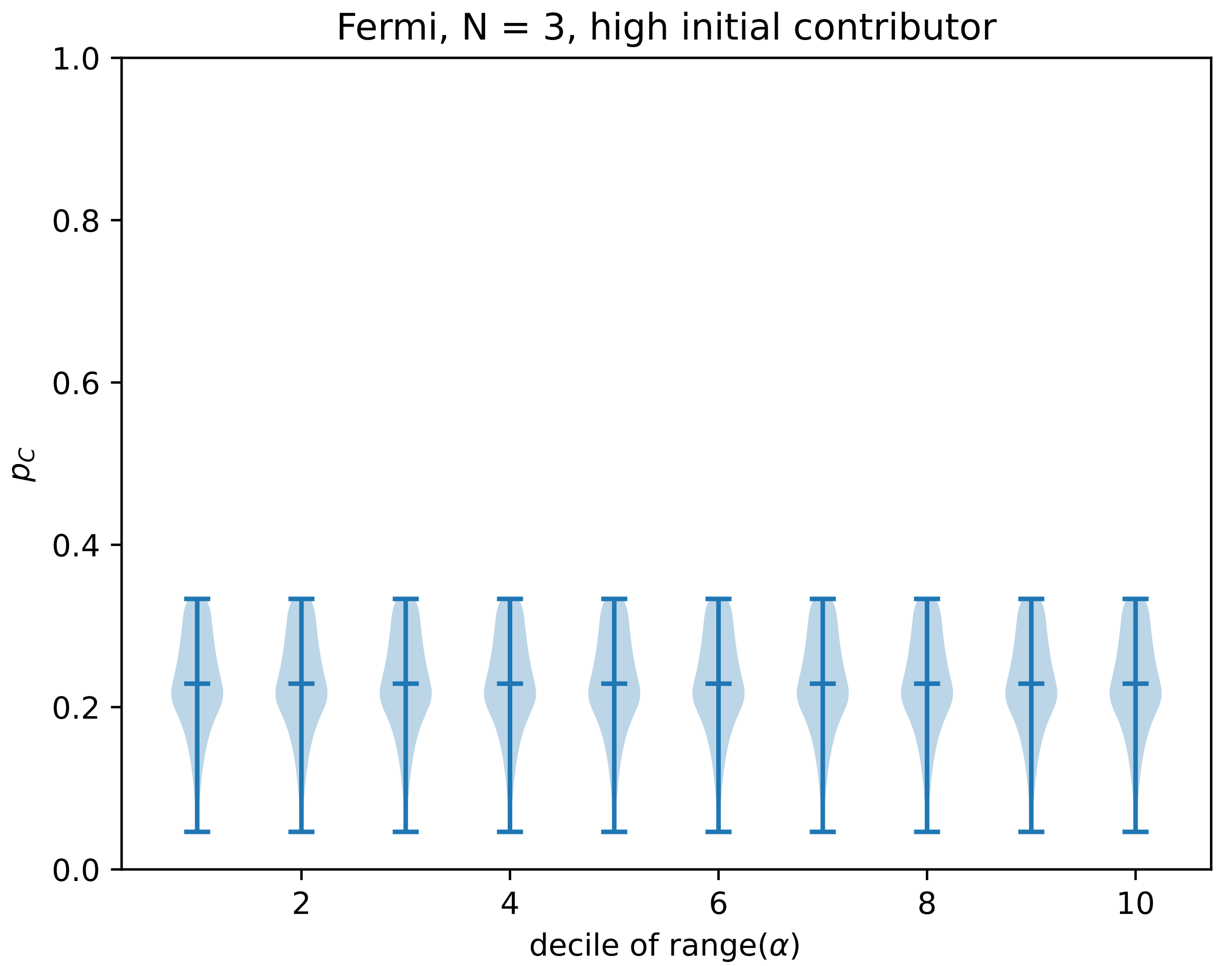
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 3, high initial contributor
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0.0
p_C
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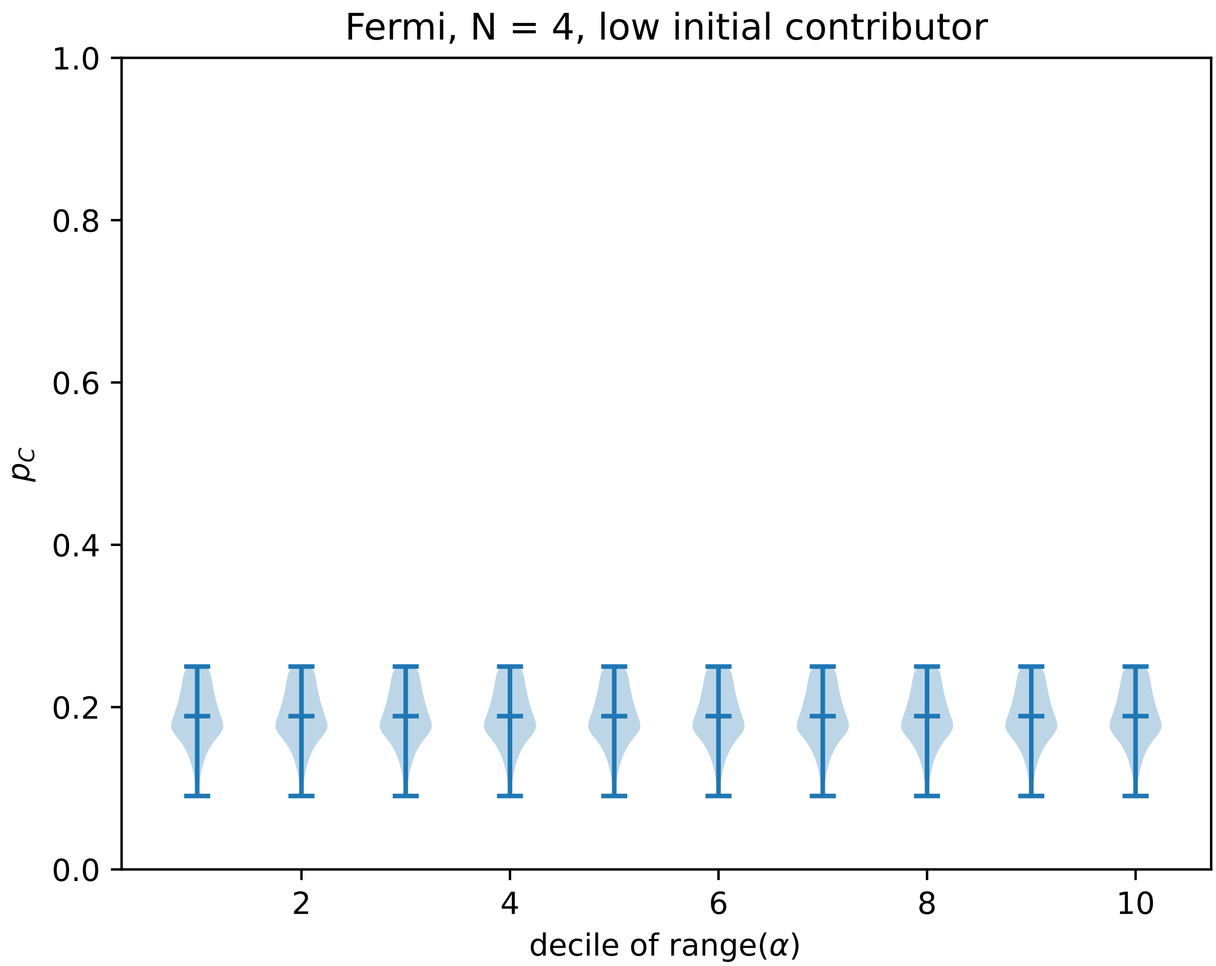
4
6
8
10
decile of range(α)



Fermi, N = 4, low initial contributor
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0.6
0.4
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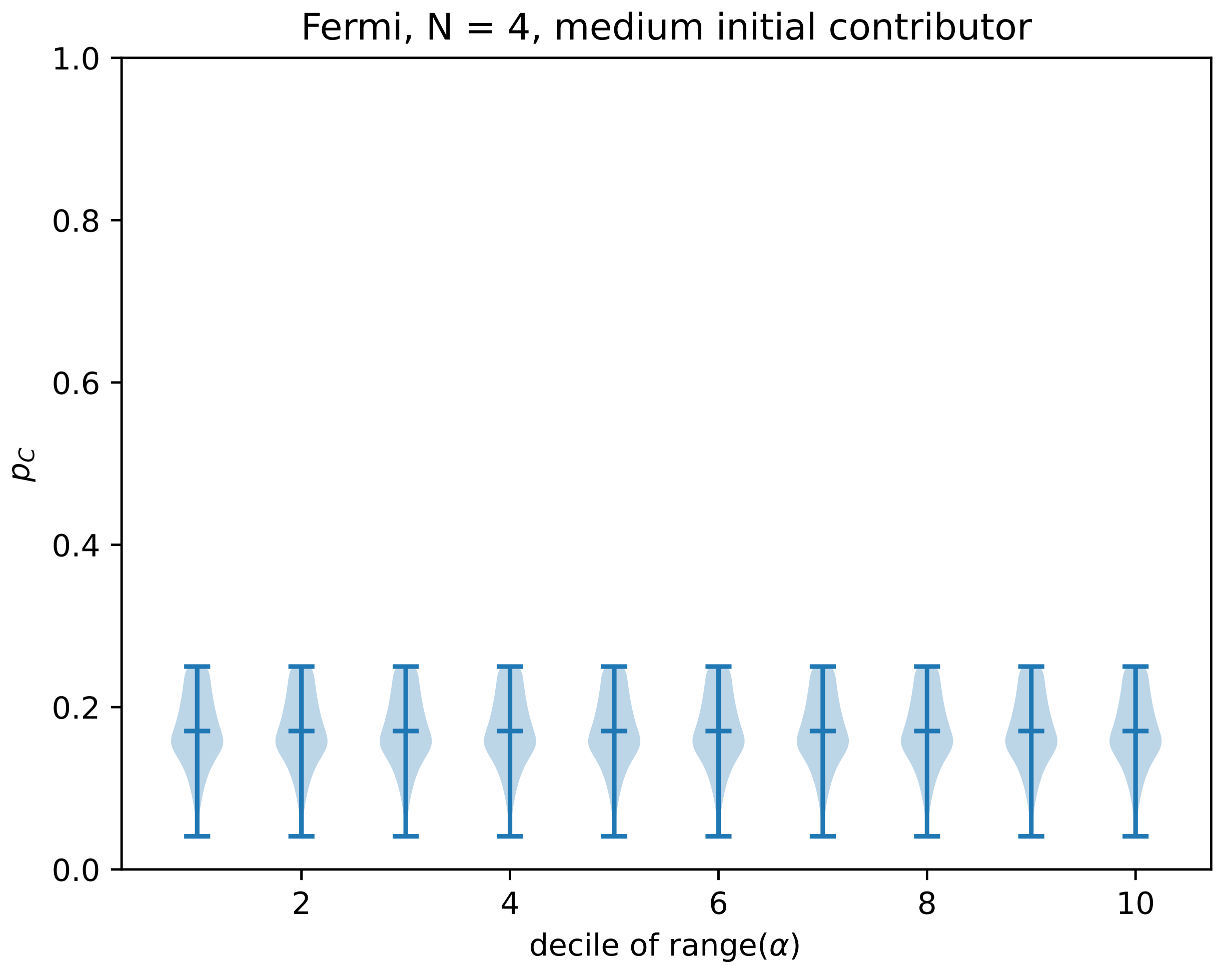
0.0
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 4, medium initial contributor
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0.6
0.4
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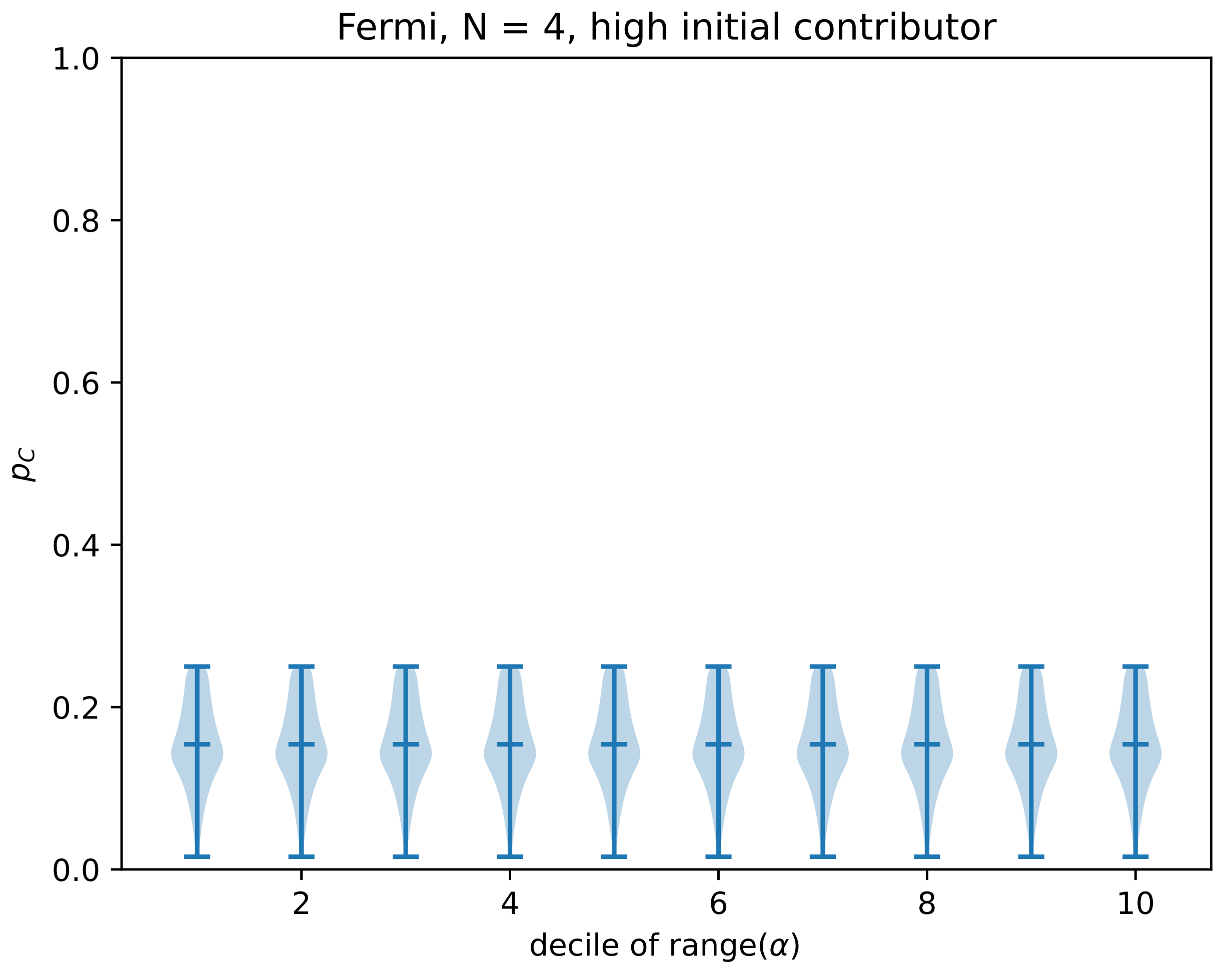
0.0
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 4, high initial contributor
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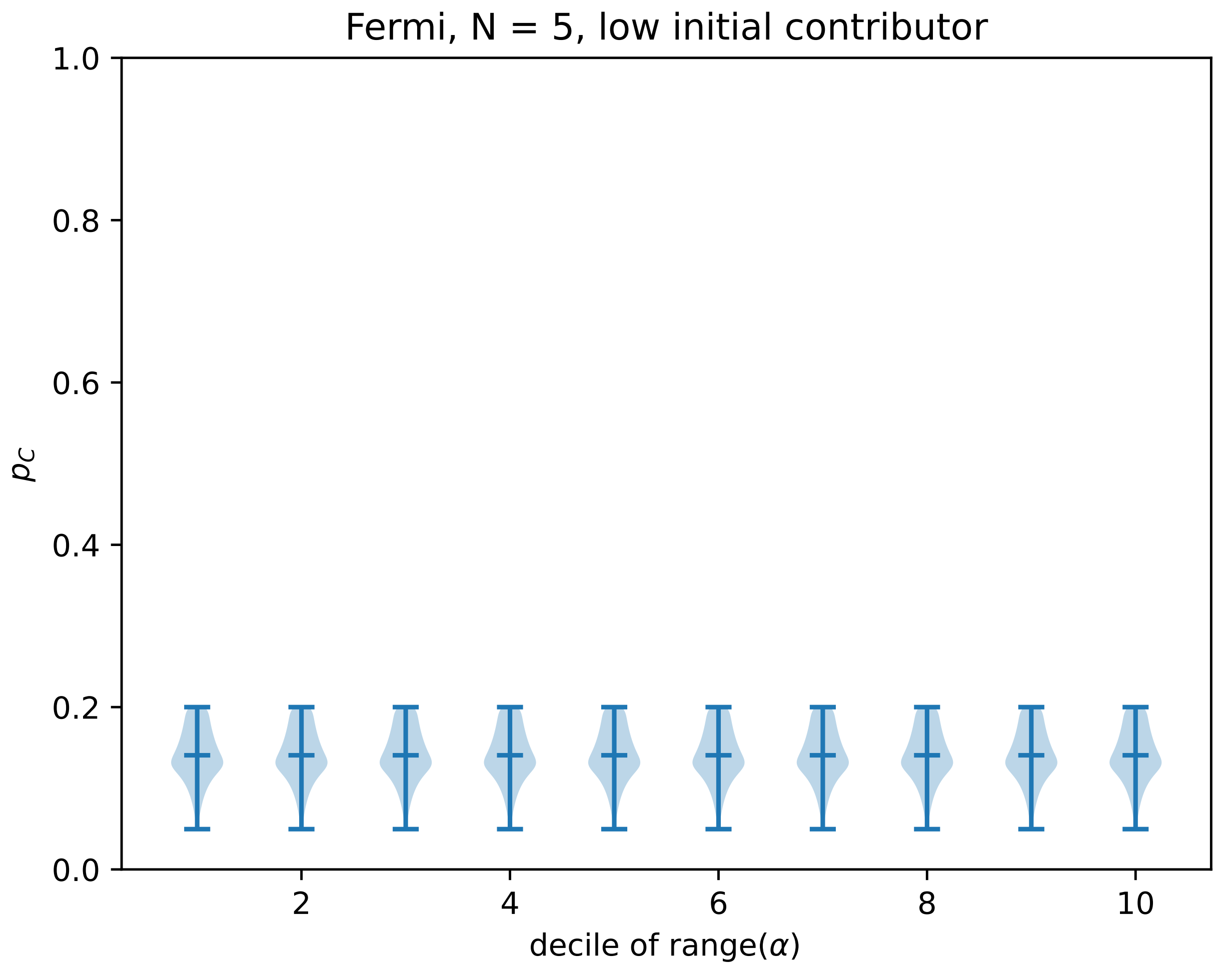
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 5, low initial contributor
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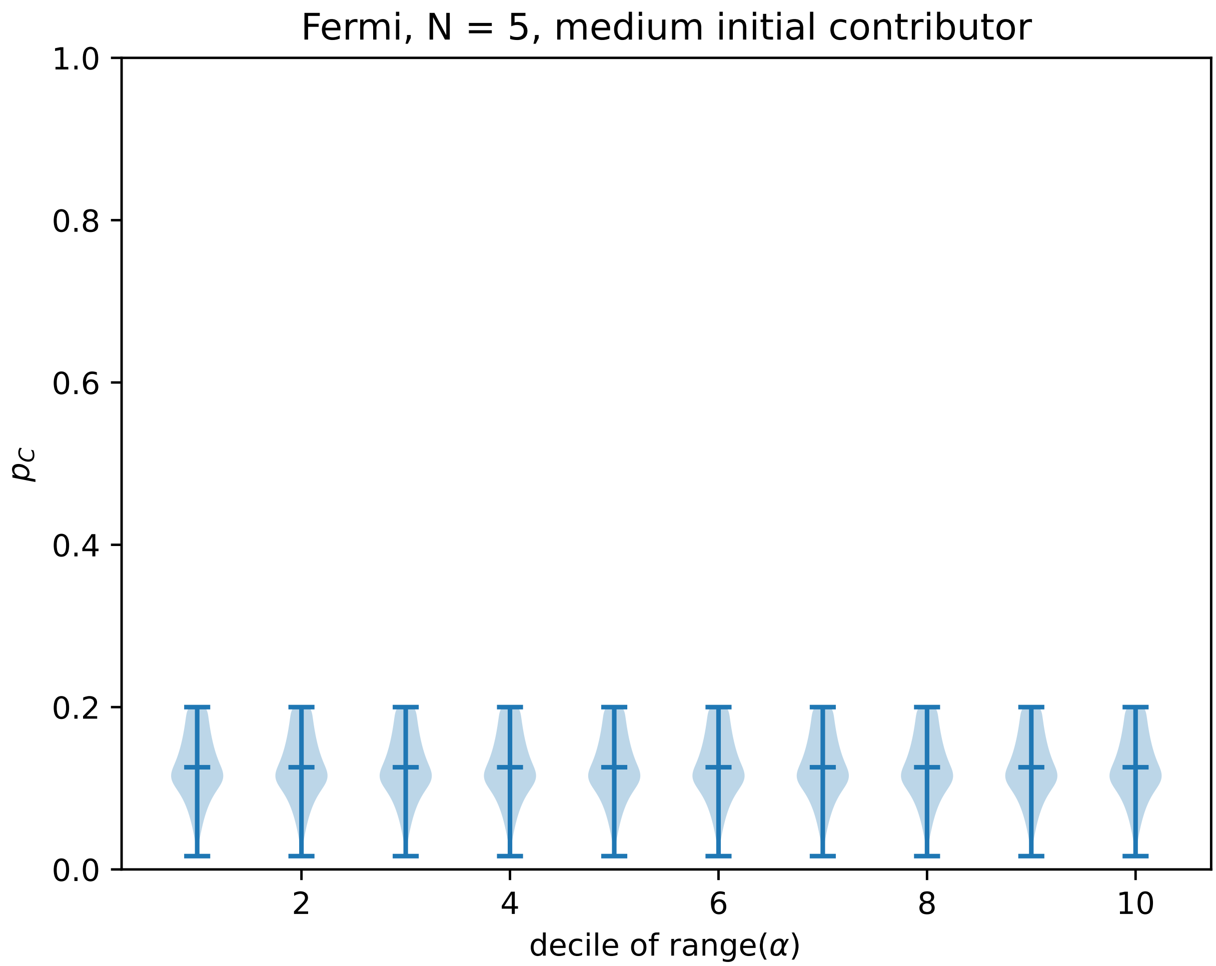
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 5, medium initial contributor
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0.8
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0.4
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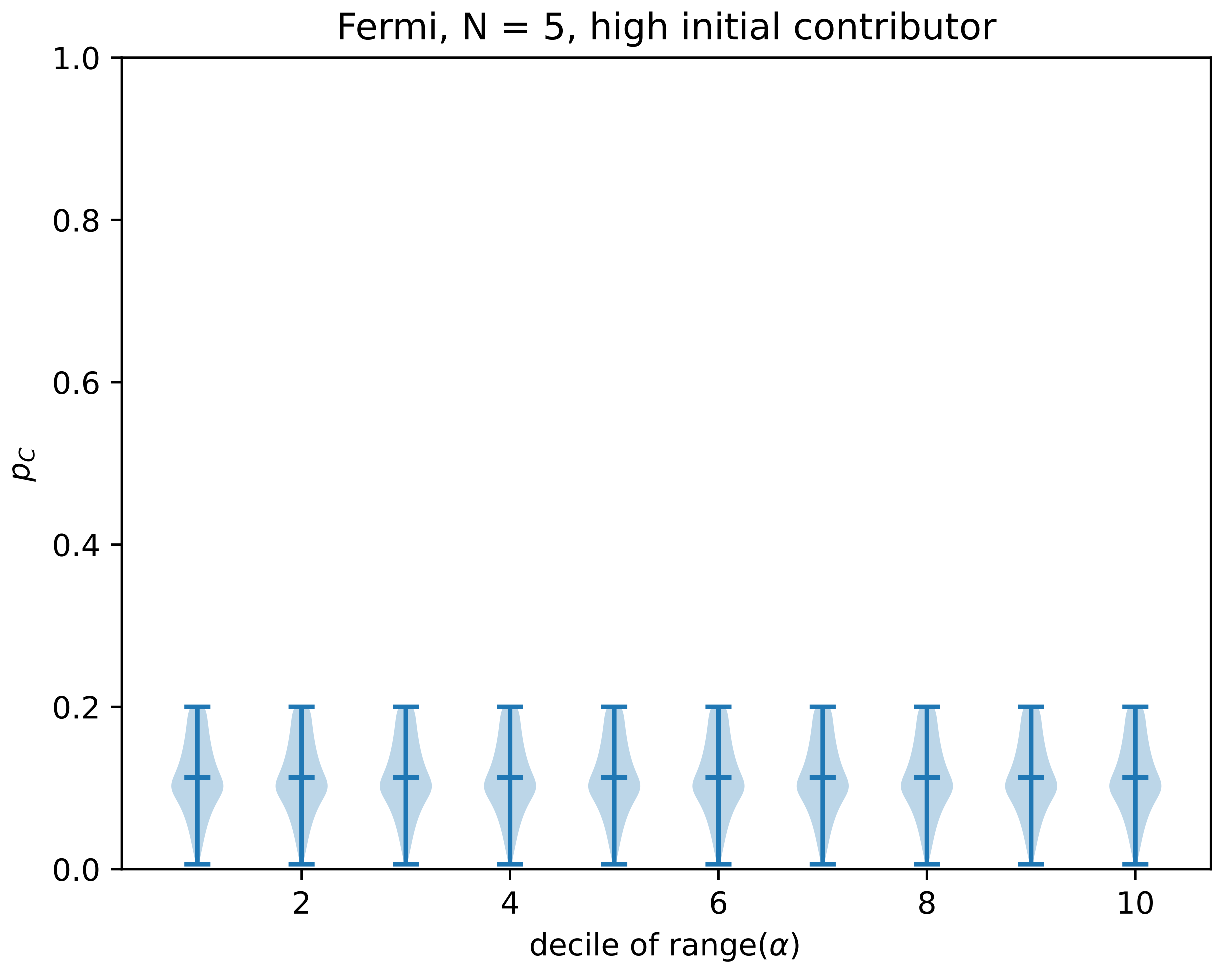
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 5, high initial contributor
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0.4
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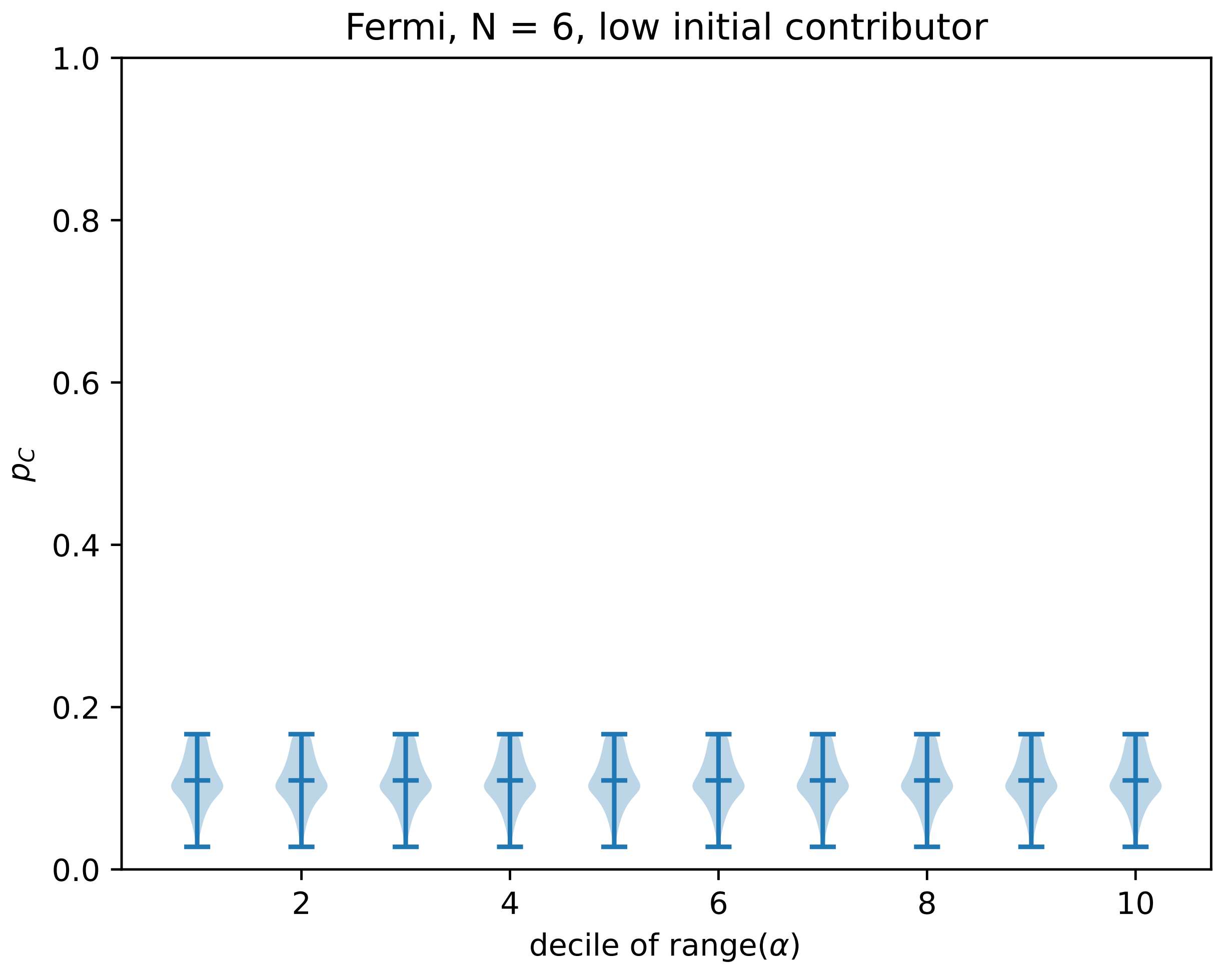
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 6, low initial contributor
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p_C
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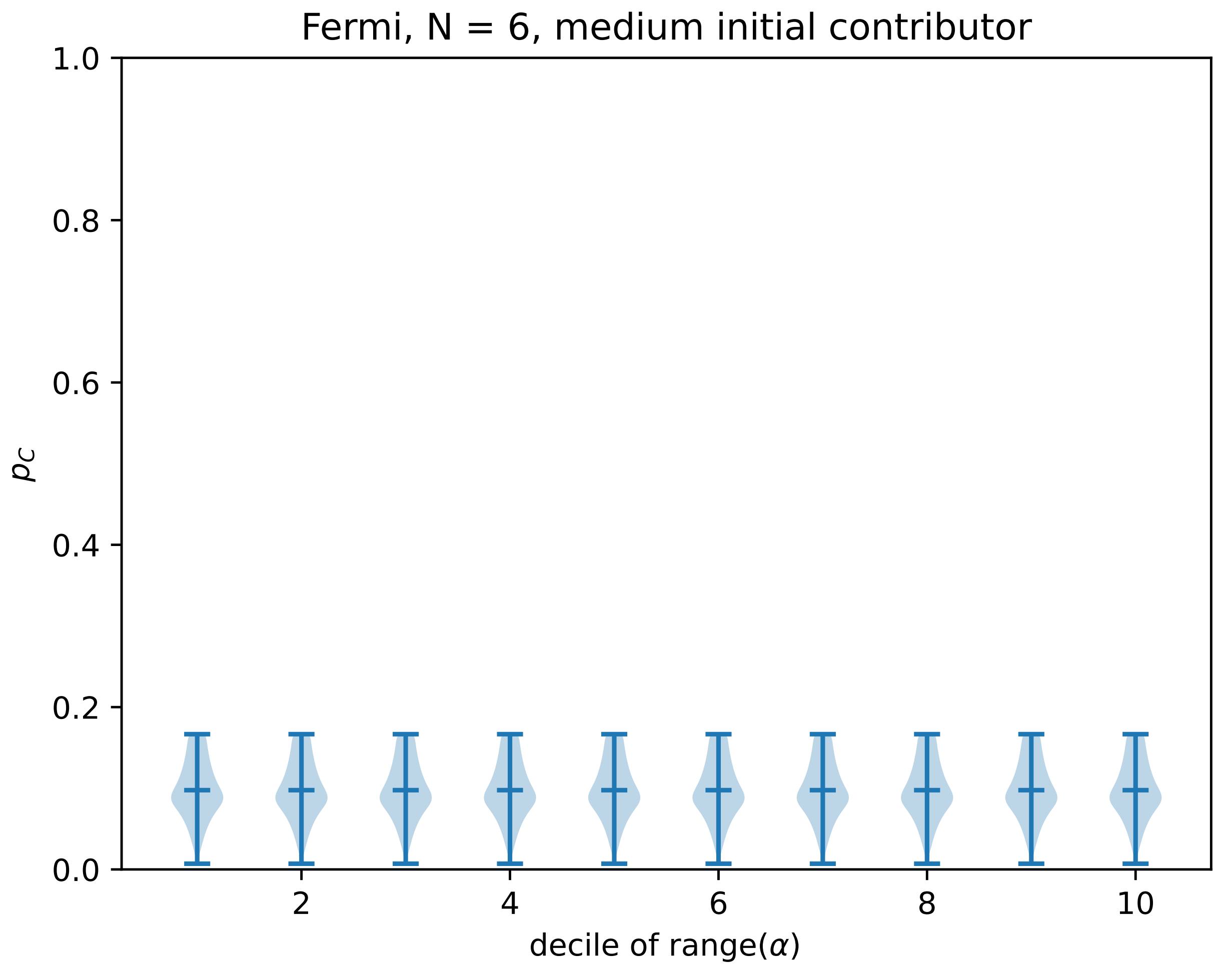
4
6
8
10
decile of range(α)


Fermi, N = 6, medium initial contributor
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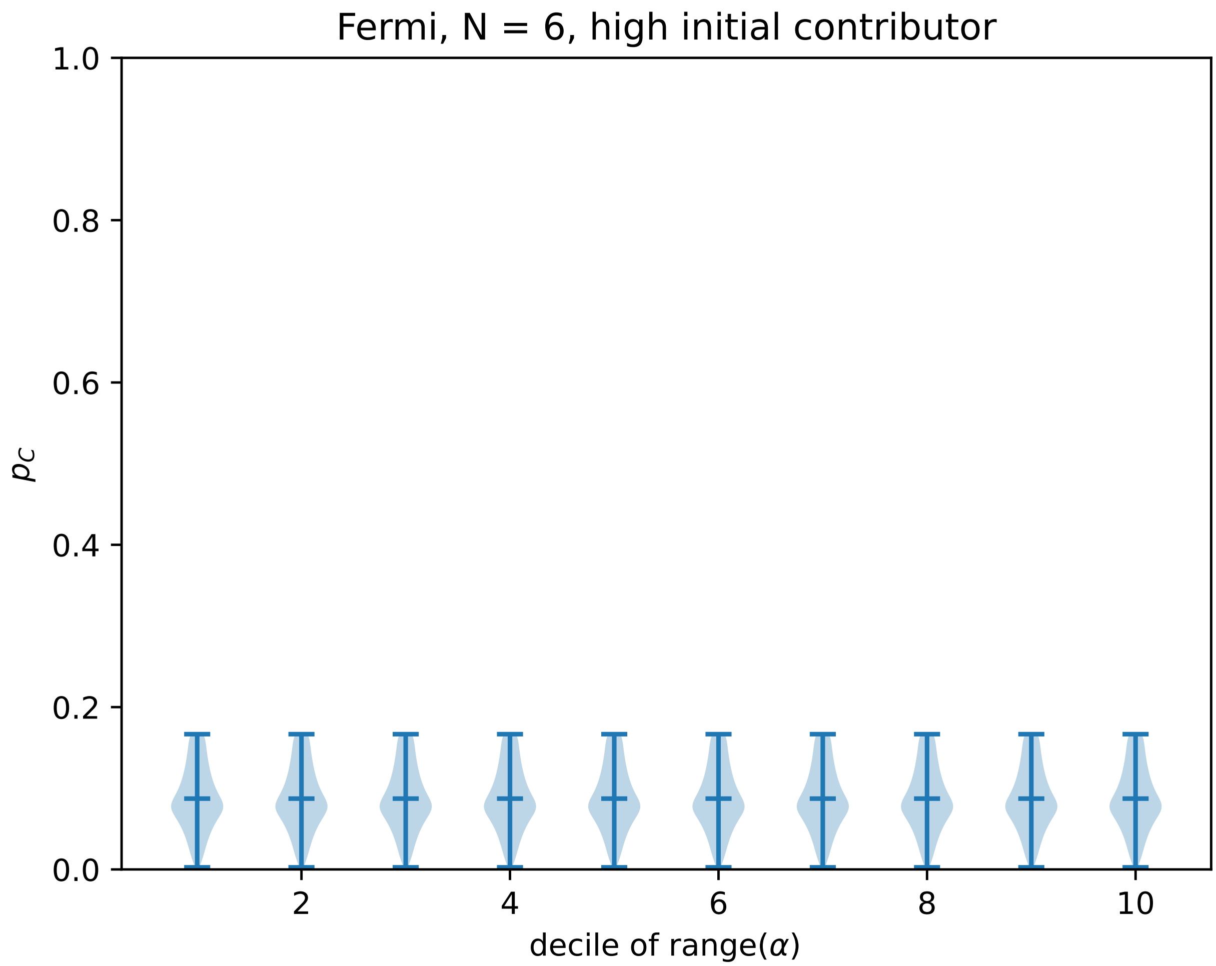
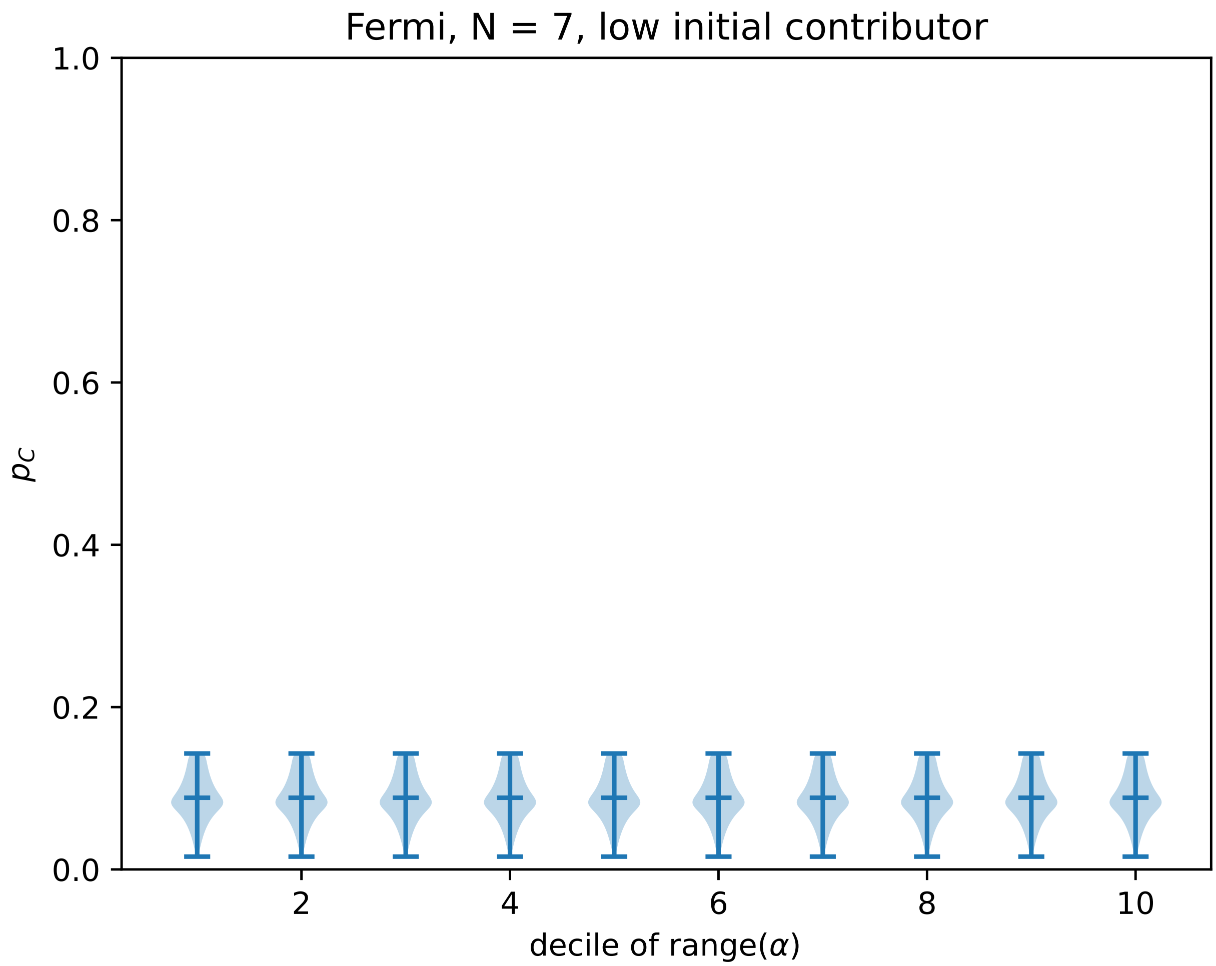
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 6, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 7, low initial contributor
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0.6
0.4
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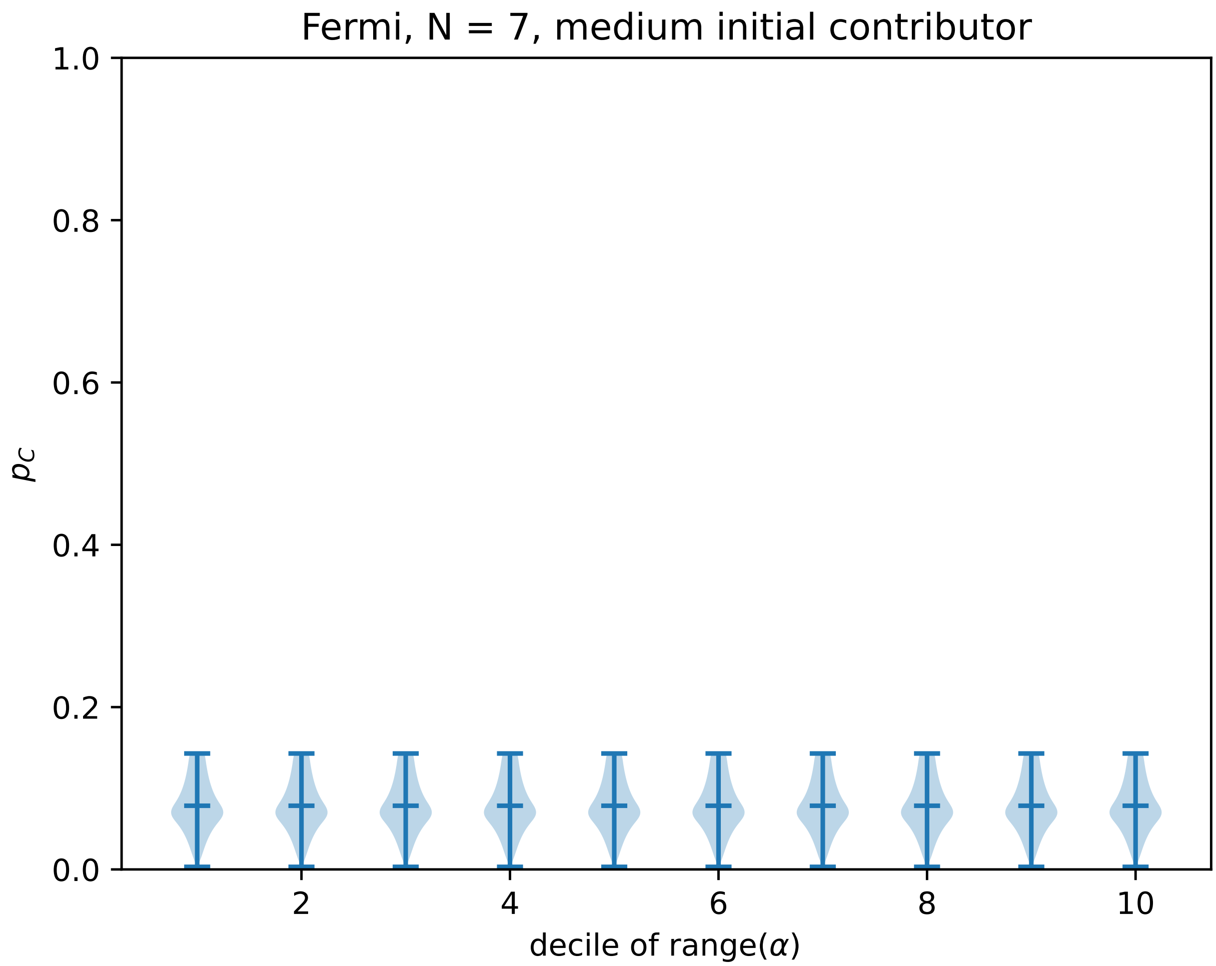
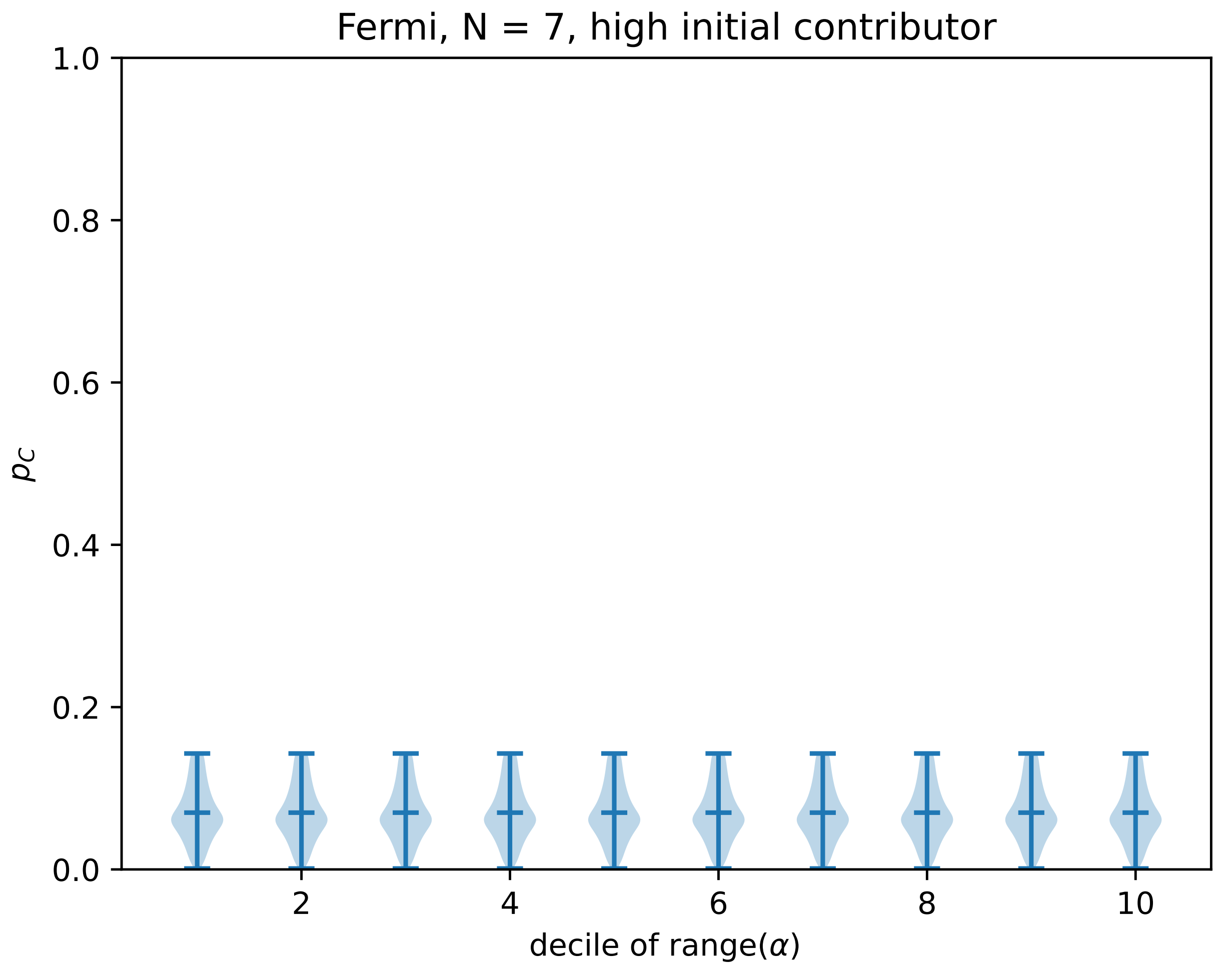
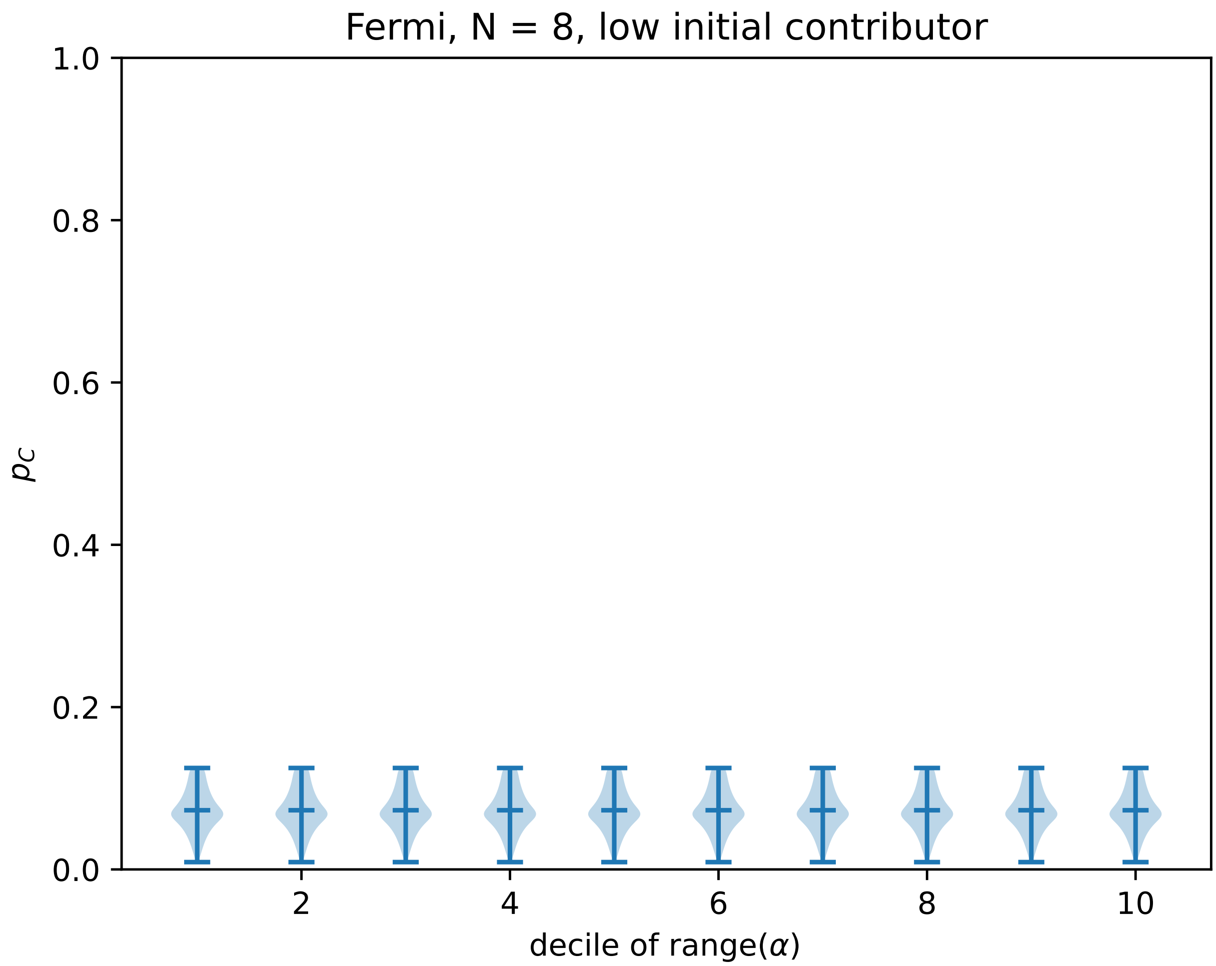
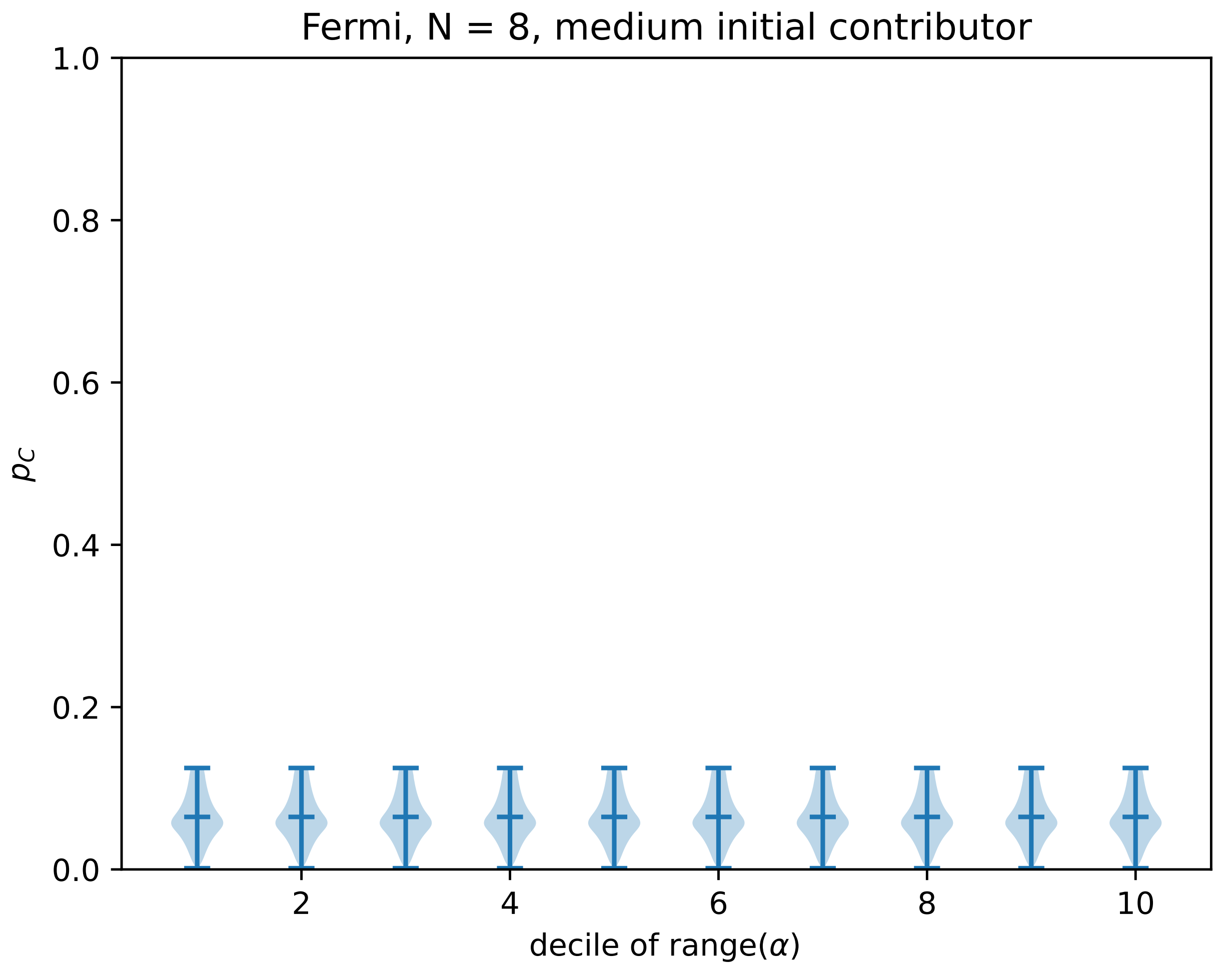
0.0
p_C
2
4
6
8
10
decile of range(α)

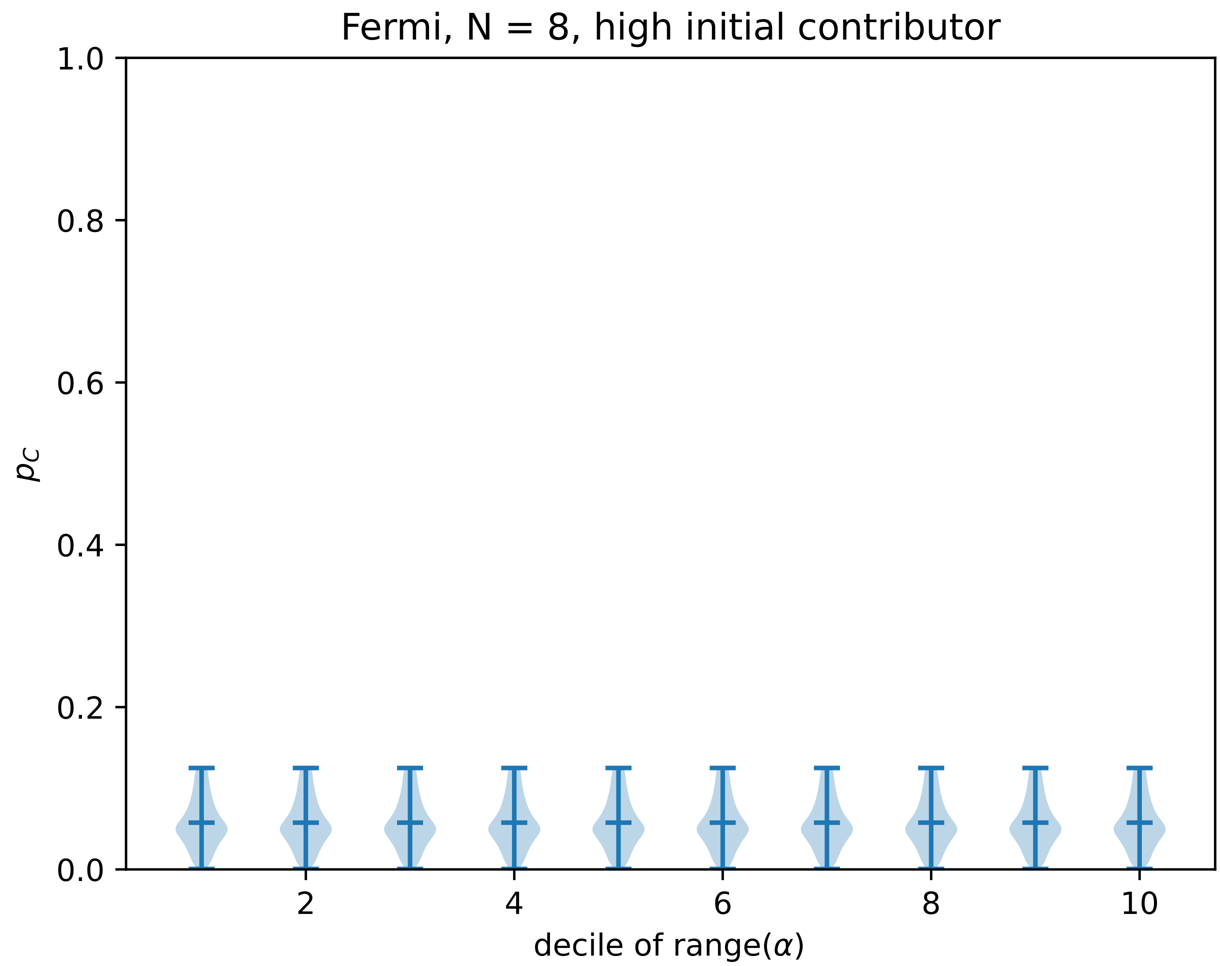

Fermi, N = 7, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 7, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

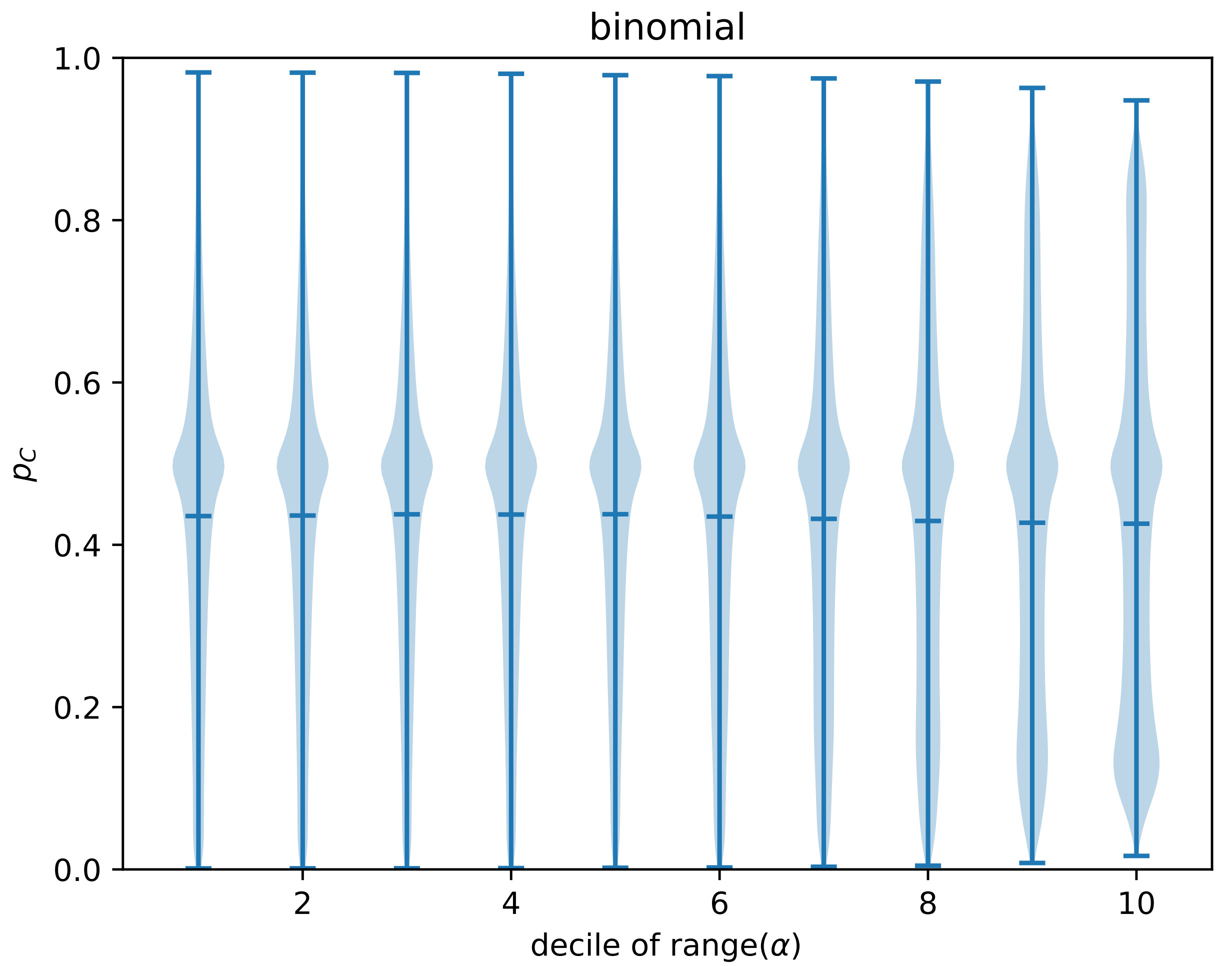


Fermi, N = 8, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 8, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



Fermi, N = 8, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


binomial
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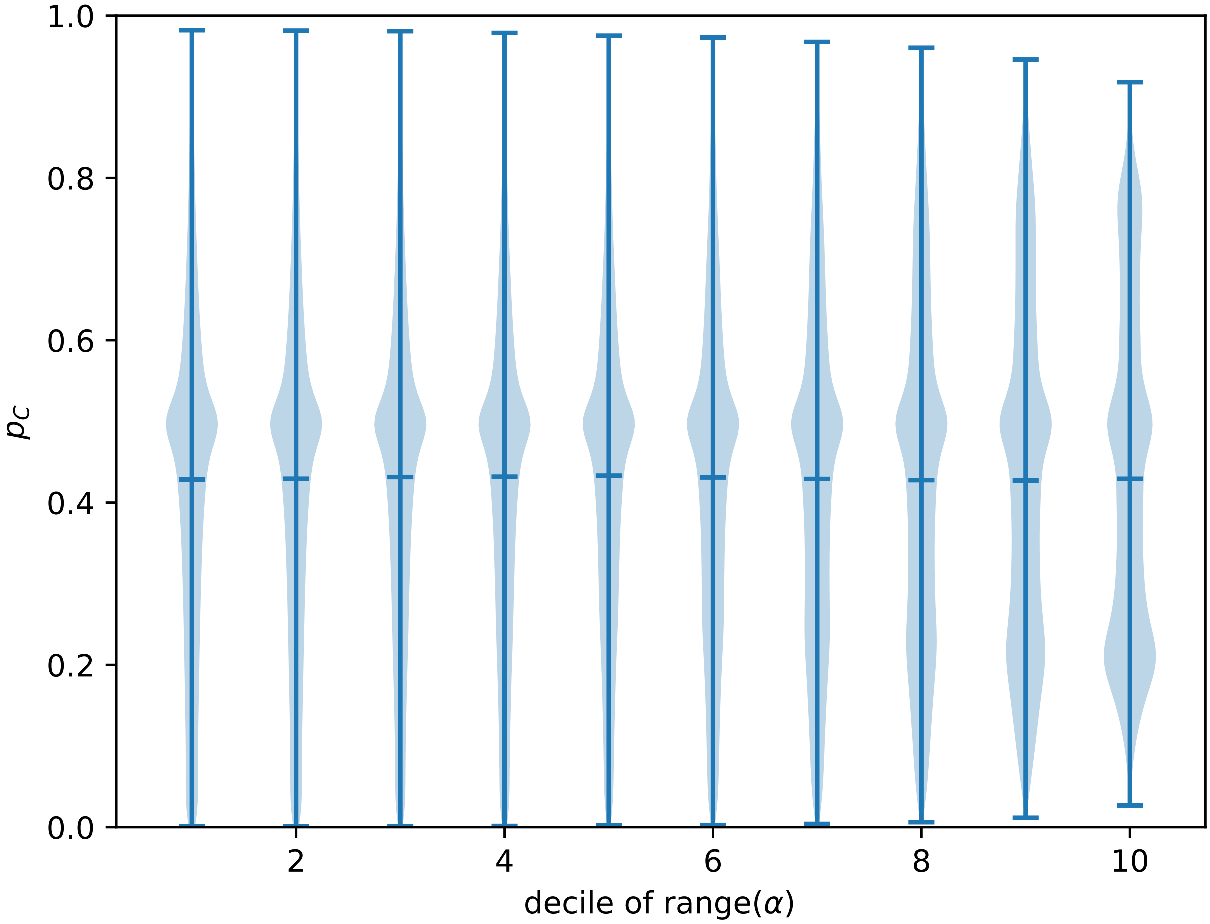
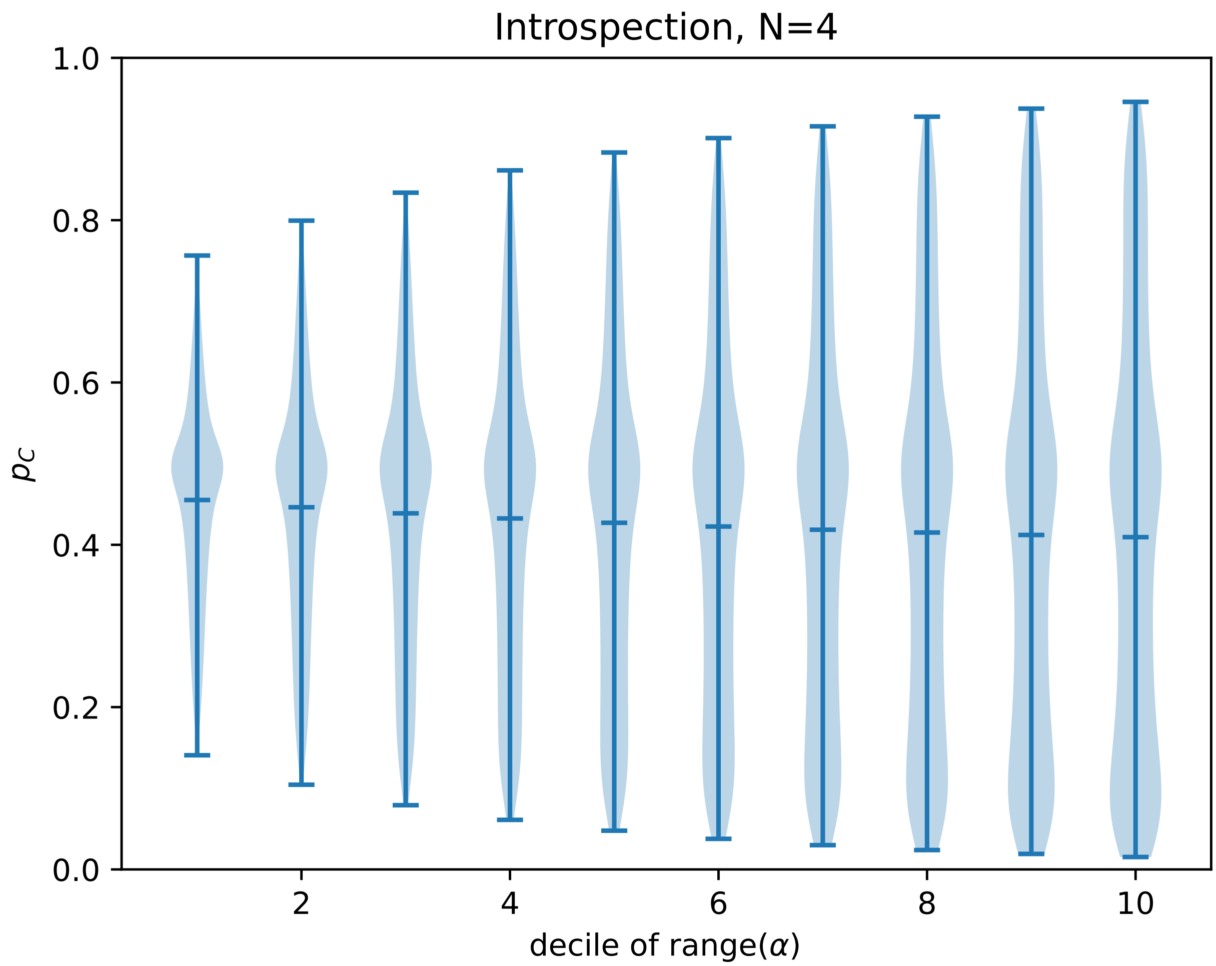
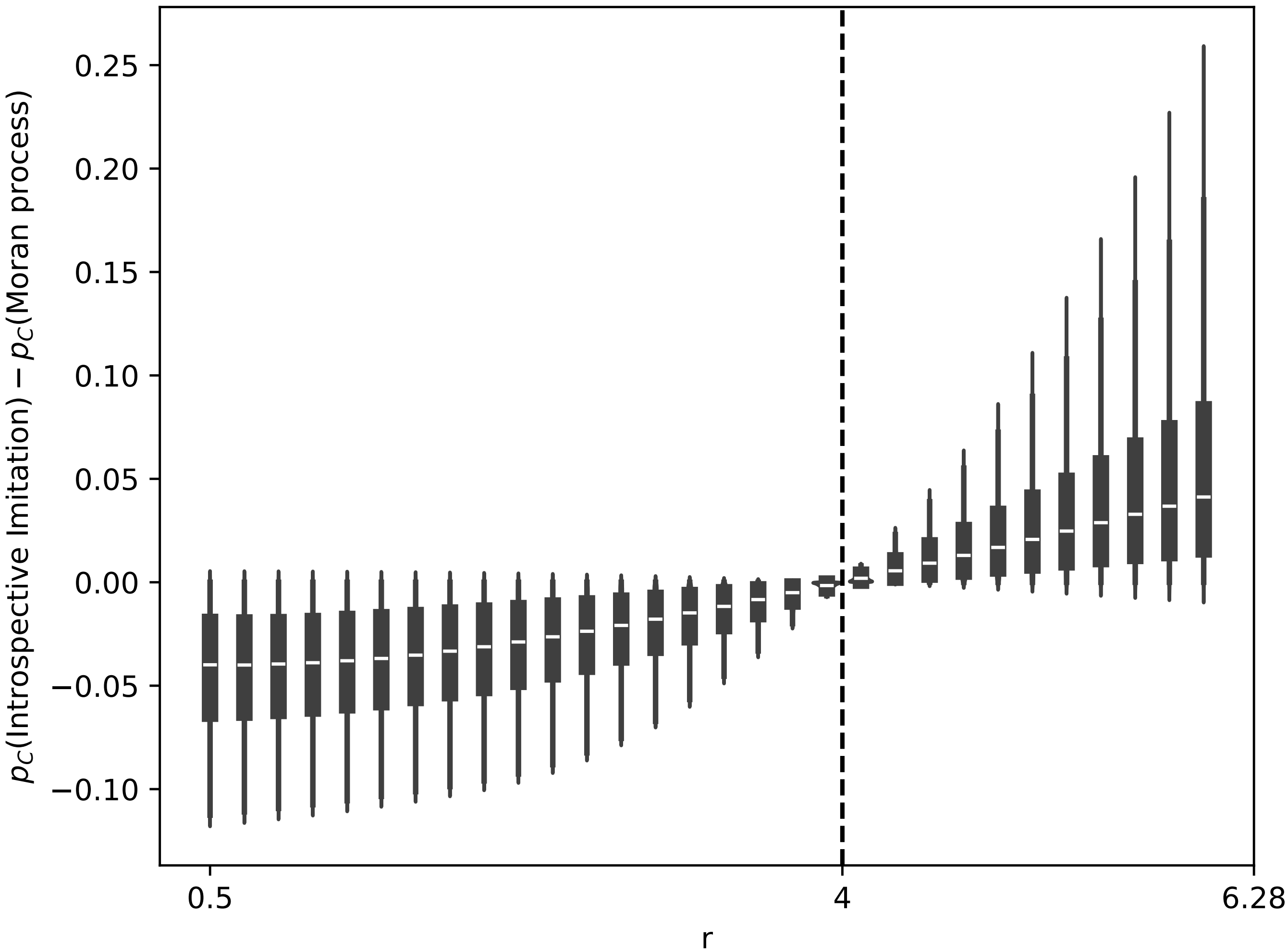
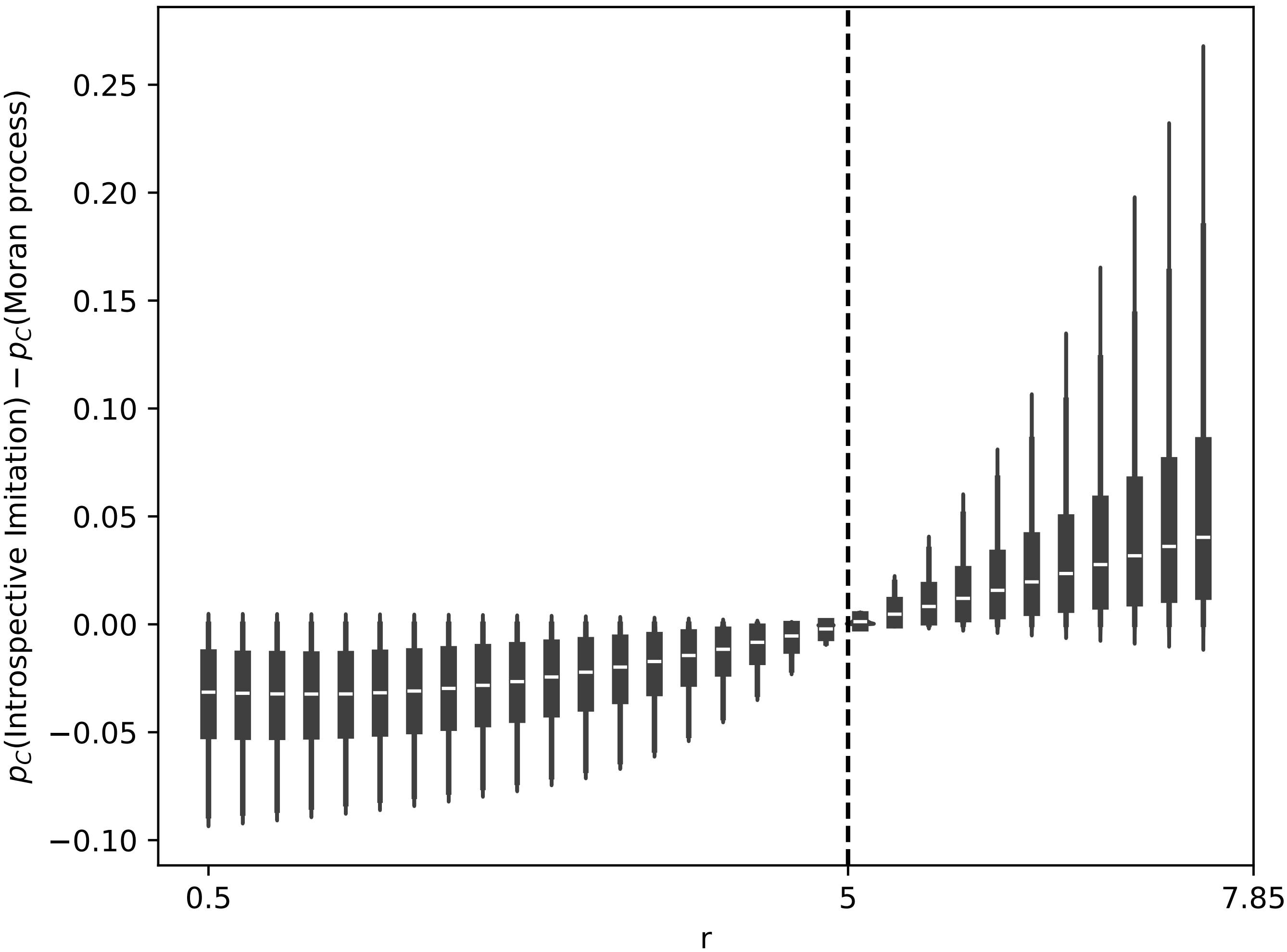
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



Introspection, N=4
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


p_C(Introspective Imitation) − p_C(Moran process)
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
0.5
4
6.28
r



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